National Marine Fisheries Service Species' Draft Biological Assessment for the 2001 Roadless Area Conservation Rule Rescission

USDA Forest Service

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INTRODUCTION

The U.S. Department of Agriculture proposes to rescind the 2001 Roadless Area Conservation Rule (2001 Roadless Rule) (66 FR 3244, 36 CFR Subpart B (2001)), which prohibits road construction, road reconstruction, and timber harvesting in inventoried roadless areas on National Forest System lands, with limited exceptions. Rescinding the 2001 Roadless Rule is intended to return decision making for these activities to local officials, in conjunction with Forest-level land management planning. The 2001 Roadless Rule applies to approximately 44.7 million acres of National Forest System lands, including in Alaska.

Accompanying the Proposed Rule, which is published in the Federal Register, and an environmental impact statement, the Forest Service has prepared this draft 'framework programmatic' biological assessment in compliance with the Endangered Species Act. This draft biological assessment discloses the potential effects to ESA-listed resources under National Oceanic and Atmospheric Administration, National Marine Fisheries Services (NFMS) jurisdiction that would result from the proposed 2001 Roadless Rule rescission. An additional biological assessment has been prepared to assess effects to ESA-listed resources under Department of the Interior, U.S. Fish and Wildlife Service (USFWS) jurisdiction. The Forest Service is requesting framework consultation and conferencing and commits to consulting on any future projects in the action area that may affect ESA-listed species or designated critical habitats.

All documents, including the proposed rule and other supporting information, may be found on the following website: https://www.fs.usda.gov/managing-land/planning/roadless.

Roadless Rule History

Throughout its history, the Forest Service has managed undeveloped areas to conserve their unique characteristics. After passage of the Wilderness Act of 1964 (Pub. L. 88-577), management of these areas, commonly known as "roadless areas," became the focus of national attention when the Forest Service initiated a review to determine their suitability for inclusion in the National Wilderness Preservation System, known as the Roadless Area Review and Evaluation (RARE) I (1973) and RARE II (1977). In the decades following the inventory, roadless area management continued to be controversial in land management planning and project proposals to harvest timber, build roads, or otherwise develop these areas. In 1999, President Clinton directed the Forest Service to undertake a public process to "provide appropriate long-term protection for most or all of these currently inventoried 'roadless' areas...", initiating the rule-making process. In 2001, the Forest Service issued the Roadless Area Conservation Rule establishing a national-level rule to maintain the values of inventoried roadless areas, limit their development, and address the history of controversy.

The 2001 Roadless Rule prohibited road construction, road reconstruction, and timber harvesting on inventoried roadless areas (IRAs) on 58.5 million acres, or 31 percent, of National Forest System lands (defined in 36 CFR 294.11 (2001) identified in the set of maps contained in the 2001 Final EIS). The Rule provided for certain limited exceptions, including for forest stewardship purposes, public health and safety, reserved or outstanding rights, existing or

renewed mineral leases, and realignment to prevent resource damage or address safety hazards (36 CFR 294.12 and 294.13 (2001)).

In the promulgation of the 2001 Roadless Rule, the Forest Service found that the Rule's prohibitions on road construction and timber harvest were not likely to adversely affect threatened or endangered species or adversely modify designated or critical habitat; were not likely to jeopardize proposed species or adversely modify proposed critical habitat; and may beneficially affect threatened, endangered, and proposed species and critical habitat (USDA 2000). The Services concurred with the determination (66 FR 9 3271).

Inventoried Roadless Areas

Today, inventoried roadless areas total approximately 58.2 million[1] acres, or about 30 percent of all National Forest System lands. The 2001 Roadless Rule applies to approximately 44.7 million acres of National Forest System lands, including about 9.3 million acres within the Tongass National Forest. The 44.7 million acres exclude approximately 9.3 million acres in Idaho and 4.2 million acres in Colorado, which are under state specific roadless rules. Current areas are reflected on national maps at the Forest Service Roadless Area website (https://www.fs.usda.gov/managing-land/planning/roadless)[2].

More than 95 percent of IRAs managed under the 2001 Roadless Rule (outside of Colorado and Idaho) are in ten Western states (from largest inventoried roadless area acreage to smallest): Alaska, Montana, California, Utah, Wyoming, Nevada, Washington, Oregon, New Mexico, and Arizona. In the West, IRAs were typically 5,000 acres or greater. Some roadless areas are less than 5,000 acres—either because they are adjacent to Congressionally designated wilderness or in the eastern US (east of the 100th meridian). While they are referred to as "roadless", because of the history of their inventory and designation, there are thousands of miles of existing roads within lands designated under the 2001 Roadless Rule, including National Forest System roads as well as roads of other ownerships. Additionally decommissioned roads and unauthorized routes that are not part of the official system or records may exist.

PURPOSE AND NEED

The U.S. Department of Agriculture (USDA or Department) is proposing to rescind the 2001 Roadless Area Conservation Rule (2001 Roadless Rule), which established broad prohibitions on road construction, road reconstruction, and timber harvesting within inventoried roadless areas on National Forest System lands. The intent of this proposed action is to reduce regulatory burden and return decision-making for the management of inventoried roadless areas to the land management planning process at the individual national forest level. Rescission of the national-level prohibitions provides responsible officials with flexibility to better guide management of National Forest System lands and respond to changing local resource conditions. The Forest and Rangeland Renewable Resources Planning Act of 1974, as amended by the National Forest

[1] While the original rule listed 58.5 million acres, mapping modernization and minor corrections reduced that to 58.2 million acres.

[2] To view the national maps in an interactive map viewer select 'Map Service for Inventoried Roadless Areas designated by the 2001 Roadless Rule', then select ArcGIS Online Map Viewer.

Management Act of 1976, the 2012 National Forest System Land Management Planning Rule (2012 Planning Rule), and the associated land management planning processes and plans are the appropriate and effective mechanisms to guide sustainable, integrated management of the resources within the plan areas in the context of the broader landscape, giving due consideration to the relative values of the various resources in particular areas. For additional discussion on the purpose and need please refer to the Final Rule.

CONSULTATION HISTORY

July 8, 2025 – The ESA compliance team is established for the proposed Roadless Rule Rescission.

August 21, 2025 – The ESA compliance team decided to conduct a programmatic analysis for the rule rescission despite all future activities being required to conduct project-level ESA consultation.

August 29, 2025 –Forest Service confirms a full analysis is preferred for this action given the history of the 2001 Roadless Rule.

November 21, 2025 –Analytical data needed to conduct exposure analyses is completed.

December 24, 2025 – Preliminary draft BA submitted to the Services to begin technical assistance.

DESCRIPTION OF THE ACTION

The proposed action described below is a "framework programmatic action" as defined in 50 CFR 402.02, where the framework programmatic action only establishes a framework for the development of specific future action(s) but does not authorize any future action(s). The effects to listed species and designated critical habitat of future actions that are subsequently authorized, funded, or carried out would be addressed in subsequent section 7 consultations. While this is a 'framework programmatic action', it does not provide a new framework. Instead, the proposed rule would remove the Roadless Area Conservation Rule regulation/prohibitions from the broader existing legal framework in which the Forest Service plans, authorizes, and implements activities. The programmatic effects are evaluated based on the legal framework that would remain in place, focusing on the effects of the removal of the prohibitions on roads construction and timber harvest (see Legal Framework, below).

Under the proposed rulemaking, the 2001 Roadless Area Conservation Rule prohibiting activities in inventoried roadless areas would be rescinded (also referred to as 'repealed'). This action would eliminate the national prohibitions set forth in the 2001 Roadless Rule – road construction, reconstruction, and timber harvest may be authorized within inventoried roadless areas, where they are consistent with land management plans, or other statutory and regulatory requirements. In addition, the proposed rule would remove the national "inventoried roadless

area" designation from the 44.7 acres million acres of National Forest System lands currently designated as inventoried roadless areas under the 2001 Roadless Rule[3].

Management of lands formerly managed as inventoried roadless areas would be governed by applicable land management plan direction alongside other relevant law, regulation, and policy. Some land management plans include components that restrict timber harvest or road building to meet other management objectives or legal requirements, such as to protect sensitive areas, to provide for primitive recreation opportunities, or to provide wildlife habitat. The proposed rescission of the Roadless Rule would not alter this existing land management plan direction.

Rescinding the Roadless Rule does not authorize, fund, or carry out future forest management activities that could occur in these areas. All future site-specific project decisions, including those to authorize road construction, reconstruction, and timber harvesting, must be consistent with land management plans. Any proposed actions must undergo environmental review, including National Environmental Policy Act compliance prior to a decision. Project decisions must comply with applicable Federal and State laws, such as the Clean Water Act, the Clean Air Act, Endangered Species Act, National Historic Preservation Act, and the Wilderness Act, as well as provisions for government-to-government consultation with Tribes and Alaska Native Corporations.

In addition, the proposed rule would not compel the amendment or revision of any land management plan. While the rulemaking would not require such changes, plans may be amended in any manner whatsoever and are revised periodically (16 U.S.C. 1604(f)). Local responsible officials would continue to have discretion to initiate land management plan amendment or revise, as deemed necessary. A determination to amend or revise a land management plan is based on a variety of factors. Any future administrative changes, amendments, or revisions to land management plans would comply with NFMA and planning regulations (36 CFR 219 and the planning directives at FSM 1920 and FSH 1909.12), which includes substantive requirements to provide for species biodiversity, recovery of threatened and endangered species, and ecological conditions to provide viable populations of species of conservation concern. To the degree that a land management plan could change in the future, the effects of such a change would be attributed to and analyzed as part of the decision-making for that action.

Rescinding the Roadless Rule does not constrain or alter the statutory requirement of sections 7(a)(1) through section 7(a)(4) of the Endangered Species Act. All future activities in currently inventoried roadless areas will require compliance with the ESA prior to implementation.

[3] The 9.3 million acres of inventoried roadless area within the Tongass National Forest would be included in the 44.7 million acres no longer managed according to the 2001 Roadless Rule. Rescinding the 2001 Roadless Rule does not affect state-level roadless rules that exist in Colorado and Idaho, nor does it preclude the development of state-led roadless efforts in the future.

ANALYTICAL METHODS

Action Area Determination

There are 44.7 million acres of IRAs subject to the 2001 Roadless Rule, although only ~40 million acres are included in the action area where the proposed rulemaking could result in changes to future management activities. In this assessment, the term "action area" and "affected environment" are used interchangeably.

The action area for the proposed Roadless Rule Rescission was defined by removing non-National Forest System lands, wilderness, wilderness study areas, and wild portions of wild and scenic rivers from Inventoried Roadless Areas managed under the 2001 Roadless Rule (resulting in ~40 million acres of action area) as potential future management of these areas would not be affected by the proposed rulemaking.

More specifically, review of the geospatial data found that 44.3 million acres of these are on National Forest System lands, meaning about 0.4 million acres of lands have had land ownership change since the 2001 Roadless Rule or there may be vertical integration problems (mapping errors) between the inventoried roadless data and official land ownership boundary data[4].

In addition, over 4 million acres are within Congressionally designated wilderness[5], wilderness study areas[6], and wild segments of wild and scenic rivers[7]. These statutory designations have more restrictive and more permanent mandates than the 2001 Roadless Rule. Therefore, changes in management activities in these areas under any alternative are highly unlikely and not reasonably certain to occur. The Forest Service did not arbitrarily buffer the action area to account for the possibility that some stressors such as noise and dust may extend beyond the IRA boundary. At the national scale, it is not reasonably certain that most stressors associated with a single action may extend beyond the boundary of the IRAs, except for sedimentation being carried downstream. Stressors such as noise and dust are expected to be minimally impactful where they occur, whereas sedimentation from proposed activities is considered a major stressor

[4] Vertical integration is a term used in the Geographic Information System (GIS) community to describe the process of aligning datasets from different sources. This includes the process of aligning the 2001 IRA boundary lines with National Forest System (NFS) boundary lines. NFS boundaries include, but are not limited to NFS property boundaries, national forest/national grassland boundaries, and special designated area boundaries (such as wilderness). Misalignments occur over time due to new surveys, improved technologies, and datum shirts (USDA Forest Service 2024 – Memo Procedure for Aligning and Making Technical Corrections (https://usdagcc.sharepoint.com/sites/fs-wo-emcarlh/NEPA-EADM/Roadless/Forms/AllItems.aspx?id=%2Fsites%2Ffs%2Dwo%2Demcarlh%2FNEPA%2DEADM%2FRoadless%2FProcedure%20for%20Aligning%20and%20Making%20Technical%20Corrections%20to%202001%20IRA%20Boundary%20Lines%20%28eSign%29%5FCombined%2Epdf&parent=%2Fsites%2Ffs%2Dwo%2Demcarlh%2FNEPA%2DEADM%2FRoadless). Very slight misalignment of the coverages can result in polygon slivers between the coverages, which can produce acreage differences initially. These differences can amount to tens or hundreds of acres or more, especially because of the large area under analysis.

[5] 1.3 million acres have been designated wilderness. Congress has included approximately 1.3 million acres (or 2.8 percent) of inventoried roadless areas managed under the 2001 Roadless Rule in the National Wilderness Preservation System.

[6] 2.8 million acres are Congressionally-designated wilderness study areas. Agency policy requires that "[a]ny area recommended for wilderness or wilderness study designation is not available for any use or activity that may reduce the wilderness potential of an area" (FSM 1923.03(3)).

[7] 85,000 acres are Wild segments of designated Wild and Scenic Rivers. Agency policy states that "[g]enerally, a wild river is accessible only by trail" (FSM 2354.42g) and that we must "[e]nsure trees are not sold, cut, or otherwise harvested in a designated wild river area except under emergency conditions, such as for insect or disease control, fire, natural catastrophe, disaster, public safety, or under specified conditions on valid mining claims" (FSM 2354.42d).

to the aquatic environment. Outside of the action area (i.e., outside IRAs) there are no prohibitions to road construction or timber harvest such that these activities could have already occurred, suggesting a buffer to include these areas may not be prudent. The fourth circuit court has defended that the action area for a programmatic action need not include areas outside of the relevant administrative boundary because the programmatic action does not authorize any specific activity and extension beyond those borders would be largely speculative. (Case 1:24-cv-00118-MR-WC).

For impacts considerations for aquatic species, downstream portions of aquatic habitats were included in the action area for up to 2 miles, to capture the most likely locations where sedimentation may be an issue. Instream work has been shown to cause turbidity that is generally not detectable more than half a mile downstream (see the Sedimentation section of the Fish discussion for more details). While activities on land can lead to increased sedimentation entering surface waters, best management practices will reduce these sediment loads. Although sediment transport is dependent on streamflow and particle size, a 2-mile downstream buffer is expected to capture most of the reasonably certain effects from sedimentation that would potentially occur with future actions (indirect effects). Although storm events may cause sediment plumes that travel more than 2 miles downstream, these events are often short in duration, unpredictable, and not a result of forest management. This downstream buffer distance is similar or longer than most aquatic effect analyses have required from consultations on timber and road construction activities in the west. National grasslands and forests in Idaho and Colorado are not included in the action area because roadless areas in these states are managed under state-specific roadless rules, which are not affected by rescission of the national roadless rule.

Legal Framework for Future Projects[8]

As stated under the description of the action, the proposed rulemaking does not authorize any site-specific actions, however, it does remove prohibitions on certain actions such that they are reasonably certain to be proposed in some of these areas in the future (see also "Scope" below). Under the proposed rulemaking, all future projects that authorize road construction and timber harvest within the action area must still be consistent with all federal laws, including, Endangered Species Act (ESA), National Forest Management Act (NFMA) of 1976, National Environmental Policy Act, and, Clean Water Act, and must comply with the relevant land management plan. To ensure compliance with these laws and plans, during project-level environmental review and decision-making, the Forest Service includes project design measures that minimize[9] adverse effects to species and their habitats.

Endangered Species Act

The Endangered Species Act of 1973 (ESA) the primary U.S. law protecting imperiled species and their habitats from extinction. Under Section 7 of the ESA, federal agencies must ensure actions they authorize, fund, or carry out, such as permits, construction, or resource management,

[8] Not all inclusive.
[9] Apply pre-project planning, best available technology, management practices, and scientific knowledge to reduce the magnitude, extent and/or duration of impacts.

do not jeopardize listed species or adversely modify critical habitats. Agencies must consult with the U.S. Fish and Wildlife Service (USFWS) or National Marine Fisheries Service (NMFS) if a project "may affect" protected species. Typically, ESA consultations include design measures that minimize adverse effects to ESA listed species and their designated critical habitats. Per the ESA, federal agencies are not allowed to jeopardize the continued existence of ESA listed species and destroy or adversely modify critical habitat. Because there is no way to predict where projects will occur and all future projects that occur in the affected environment will undergo ESA Section 7 consultation, this analysis will be high level effects that may be expected for actions proposed under this Environmental Assessment. In some cases, previous consultation effects to species and critical habitats for similar actions were utilized to determine effects.

National Forest Management Act

NFMA requires "guidelines for land management plans developed to achieve the goals of the Program which…provide for diversity of plant and animal communities based on the suitability and capability of the specific land area in order to meet overall multiple-use objectives, and within the multiple-use objectives of a land management plan adopted pursuant to this section, provide, where appropriate, to the degree practicable, for steps to be taken to preserve the diversity of tree species similar to that existing in the region controlled by the plan." NFMA is implemented through either the 2012 planning rule or the 1982 planning rule.

A primary purpose of the 2012 Planning Rule (planning rule) is to promote the ecological integrity of national forests and grasslands and other National Forest System administrative units. Ecological integrity – a substantive requirement of the planning rule – is designed to support Ecological Sustainability (36 CFR 219.8), Diversity of Plant and Animal Communities (36 CFR 219.9), and Multiple Uses (36 CFR 219.10). Similarly, the 1982 planning rule dictates "Fish and wildlife habitat shall be managed to maintain viable populations of existing native and desired non-native vertebrate species in the planning area. For planning purposes, a viable population shall be regarded as one which has the estimated numbers and distribution of reproductive individuals to insure its continued existence is well distributed in the planning area. In order to insure that viable populations will be maintained, habitat must be provided to support, at least, a minimum number of reproductive individuals and that habitat must be well distributed so that those individuals can interact with others in the planning area" and that "Forest planning shall provide for diversity of plant and animal communities and tree species consistent with the overall multiple-use objectives of the planning area. Such diversity shall be considered throughout the planning process. Inventories shall include quantitative data making possible the evaluation of diversity in terms of its prior and present condition. For each planning alternative, the interdisciplinary team shall consider how diversity will be affected by various mixes of resource outputs and uses, including proposed management practices. (Refer to Sec. 219.27(g).)"

All forest and grassland management plans, regardless of which planning rule they follow, must be consulted on under the Endangered Species Act.

Aquatic Species Conservation Strategies

Specific to aquatic species habitat, NFMA mandates that U.S. Forest Service protect streams, streambanks, shorelines, lakes, and wetlands from detrimental changes in water temperature,

sediment deposits, and blocked watercourses during timber harvesting and management. It requires, through land management plans, that soil and water resources are not irreversibly damaged, often by establishing riparian management areas and utilizing Best Management Practices (BMPs). Further, individual land management plans contain standards and guides for aquatic management zones, some in the form of Aquatic Conservation Strategies (ACS) (such as the Northwest Forest Plan (USFS and USBLM 1994b), PacFish (USFS 1995a), and InFish (USFS 1995b)). Aquatic conservation strategies and land management plans standards and guides are designed to move the aquatic management zones to desired ecological conditions.

Aquatic conservation strategies use an ecosystem approach to management of riparian and aquatic habitats and were designed to (1) protect watersheds that had good-quality habitat and strong fish populations at the time they were drafted, and (2) halt further declines in watershed condition and restore ecological processes that create and maintain favorable conditions in aquatic ecosystems in degraded ecosystems (FEMAT 1993). The long-term goal (100+ years) is to develop a network of functioning watersheds that supports populations of fish and other aquatic and riparian-dependent organisms (such as the Northwest Forest Plan (USFS and USBLM 1994b), PacFish (USFS 1995a), and InFish (USFS 1995b)). Aquatic conservation strategies are based on preserving and restoring key ecological processes, including the natural disturbance regimes (USFS and USBLM 1994b) that create and maintain habitat for native aquatic and riparian-dependent organisms, and it recognizes that periodic natural disturbances may be required to sustain ecological productivity. As a result, aquatic conservation strategies do not expect all watersheds to be in favorable condition (highly productive for the same aquatic organisms) at any point in time, nor does it expect that any watershed will remain in a certain condition through time. The primary objective of aquatic conservation strategies is to maintain and restore the distribution, diversity, and complexity of watershed-level features and processes to which aquatic and riparian species are uniquely adapted. Aquatic conservation strategies were also designed to develop a strategy to guide management of aquatic ecosystems on federal lands, where amended, that would meet potential ESA requirements (Reeves et al. 2018). Aquatic conservation strategies were expected to make significant contributions to the recovery of the ESA-listed fish by increasing the quantity and quality of freshwater habitat for Pacific salmon and protecting and enhancing habitats of other species (Reeves et al. 2006, FEMAT 1993).

Best Management Practices (Clean Water Act)

Forest Service National Best Management Practices for Water Quality Management on National Forest System Lands (USFS 2012a) was developed to improve agency performance and accountability in managing water quality consistent with the Federal Clean Water Act (CWA) and State water quality programs. Current Forest Service policy directs compliance with required CWA permits and State regulations and requires the use of BMPs to control nonpoint source pollution to meet applicable water quality standards and other CWA requirements. The National BMP program document provides consistency among Forest Service administrative units to efficiently administer the program and demonstrate improvements in performance and accountability at multiple scales. The National BMP Program consists of four main components: (1) a set of National Core BMPs, (2) a set of standardized monitoring protocols to evaluate implementation and effectiveness of those BMPs, (3) a data management and reporting structure, and (4) corresponding national direction.

Relationship Between the 2001 Roadless Rule and Land Management Planning

Administration of each national forest and grassland is directed by a land management plan, along with numerous land management laws, regulations, policies, and agency directives. A land management plan provides a framework for future site-specific project analysis and decision making. Land management plans are developed, amended, and revised under the National Forest Management Act and the 2012 Planning Rule at 36 CFR 219. All land management plans must conform to existing laws and regulations. Laws and regulations take precedence over management direction in the land management plans. When new or changed laws or regulations create conflicts with land management plan content, plans are administratively changed as needed to conform to new statutory and regulatory requirements (36 CFR 219.1(f) and 219.13(c)).

The 2001 Roadless Rule did not amend, revise, or compel amendment or revision of any land management plans (36 CFR 294.14(b) (2001); 66 FR 3273); however, the Roadless Rule supersedes individual land management plans where the provision of the rule is more restrictive than a plan. Similarly, for some areas, the plans may contain plan components that are more restrictive than the 2001 Roadless Rule. Under the 2012 Planning Rule, inventoried roadless areas are treated as a type of designated area,[10] with the option to also identify these areas as unique geographic areas or management areas in the plans. Plans may also include additional plan components for inventoried roadless areas if they are compatible with the restrictions of the applicable roadless rule. Within land management plans, inventoried roadless areas currently overlap other management areas or geographic areas, such as core habitats, research natural areas, alpine areas, or areas managed for backcountry recreation where there are expected to be low levels of development, including timber harvest and road construction.

Current land management-plans often include plan direction and components that identify where timber management and road construction may or may not be allowed. Example management areas where timber management or road construction may be limited include: recommended wilderness, research natural areas, or backcountry management areas. Some plans include plan components with direction specific to IRAs.

A national review of plan components in management areas overlapping the potentially affected environment was completed to identify where plans do or do not further restrict timber harvest and road construction activities. There is much variation between individual land management

[10] 36 CFR 219.7(d) Management areas or geographic areas. Every plan must have management areas or geographic areas or both. The plan may identify designated or recommended designated areas as management areas or geographic areas.
36 CFR 219.9 Designated area. An area or feature identified and managed to maintain its unique special character or purpose. Some categories of designated areas may be designated only by statute and some categories may be established administratively in the land management planning process or by other administrative processes of the Federal executive branch. Examples of statutorily designated areas are national heritage areas, national recreational areas, national scenic trails, wild and scenic rivers, wilderness areas, and wilderness study areas. Examples of administratively designated areas are experimental forests, research natural areas, scenic byways, botanical areas, and significant caves.
Geographic area. A spatially contiguous land area identified within the planning area. A geographic area may overlap with a management area.
Management area. A land area identified within the planning area that has the same set of applicable plan components. A management area does not have to be spatially contiguous

plans and Regions reflecting differences in local conditions, age of the plans, and changes in planning policies over time. Overall, within potentially affected IRAs, land management plan components:

- Further restrict (do not allow or generally do not allow) road construction on 21.8 million acres (54.5 percent of the potentially affected IRAs), and
- Further restrict (do not allow or generally do not allow) timber harvest on 26.7 million acres (66.5 percent of the potentially affected IRAs).

Note that this information is intended for programmatic analysis purposes, actual implementation of any activities would be subject to local interpretation of the land management plans based on unique local conditions, which would also include numerous considerations of forest-wide or habitat related forest plan direction also to likely be limiting on where timber management and road construction would be allowed or proposed.

Consideration of Existing Protections

Species-specific conservation measures and design criteria were considered on an individual basis for certain species. Conservation measures largely come from forest plans but might also originate from regional strategies and previous consultations. These measures were reviewed and analyzed to determine if they affected the standard determinations made based on the procedures described above. Species-specific measures associated with forest plans typically apply to any location on that forest that is inhabited by the species unless otherwise stated. Species-specific measures from forest plans and other approved strategy documents are mandatory in nature unless otherwise stated.

Scope

The following describes how the agency drew a reasonable and manageable line in consideration of effects of the proposed action that are reasonably certain to occur, and those that may extend outside the geographical territory of the proposal or might materialize later in time.

In general, it is reasonably likely activities with potential for beneficial or adverse environmental effects could occur within or near inventoried roadless areas following the proposed rulemaking. However, the rulemaking would not compel, require, or authorize any site-specific activities. Therefore, site-specific locations and amounts of activities that could occur in the future under the alternatives are uncertain. Any future decisions on these activities are greatly influenced by factors beyond the proposed rulemaking such as local decision-making through land management plans and projects, local industry capacity, agency budget allocations and resources, and other regulations and policies. Site-specific effects can be meaningfully measured and evaluated at the project and activity stage and are evaluated during the project-level ESA Section 7 consultation process. In addition, some of the potential adverse effects described could be reduced or mitigated at the project level, through complying with policy, following land management plan direction, implementing best management practices, or developing project-specific design criteria. Protections/requirements that reduce impacts are noted throughout this analysis.

The analysis focuses on the difference where (what locations) activities, particularly road construction and timber harvest, are reasonably likely to occur within inventoried roadless areas, and the resulting effects to listed species. Estimates of areas where activities could occur, were prepared for effects analysis purposes – they do not reflect proposed or planned activity areas. The stressors associated with these activities were assumed to potentially occur when species would be present and affected since the timing of future activities is not known.

The geographic extent of the environmental effects analysis focuses on activities with a close causal relationship to the rulemaking. Changes to regulations remain in place until additional rulemaking or Congressional action. Short- and long-term reasonably likely to occur environmental impacts of the rulemaking are considered.

The short-term potential for road construction and timber harvest or related activities and associated effects is more predictable and reasonably likely to occur. Within the potentially affected environment the following areas are more likely to have activities or impacts, in the short term (5 years):

- areas where activities are allowed under existing land management plans (see environmental baseline below);
- areas with a greater management priority or 'need' (ex. high risk firesheds and areas affected by insect and disease or other disturbances);
- areas that are more feasible or economical for equipment and workers to access (more operable); and,
- areas with high product potential/economic viability for timber production to support management objectives.

In the longer term (up to 20 years and beyond), the extent of activities is more uncertain. Still, it is reasonable to expect a greater proportion of the potentially effected environment could be impacted by activities over time. Factors affecting long term change include natural events such as storms and wildfires, local decision-making through projects and land management plan amendments/revisions, national policy changes, technological changes, industry and economic changes, etc. Areas that are least likely for development in the short term (currently restricted, less developed, etc.) could be impacted by change, in the longer term. However, given the long causal chain and distance from the action proposed, predicting the specific amounts is too speculative.

The Roadless Rule prohibited road construction and timber harvest (with exemptions) but did not directly restrict other activities. Rescinding the Roadless Rule increases the likelihood of timber and road construction activities occurring within the boundary of previously designated IRAs. However, removing the prohibitions on road construction in IRAs may increase accessibility and therefore, increase use of the land for purposes other than timber harvest. These activities are speculative, not reasonably certain to occur, and would require a lengthy causal chain to tie to the proposed rescission of the Roadless Rule. As such, only timber and road construction activities are considered in this analysis.

Action Deconstruction

Although rescinding the Roadless Rule does not dictate or prescribe forest management in previously inventoried roadless areas, it does remove the current prohibitions on timber harvest and road construction. Therefore, it is reasonably certain that additional active forest management and road construction activities could occur in these areas under the proposed roadless rule rescission (authorized under future decisions) for a variety of purposes. A deconstruction of potential future road construction and timber harvest activities is provided in Table 1 and the analysis assumptions for these activities that follow. The deconstruction table below does not include the impacts (stressors) associated with each tool or the expected response of listed species as those differed between aquatic and terrestrial organisms. Each taxonomic section has a deconstruction table that includes the stressors and responses relevant to those species.

Table 1. Deconstruction of the Roadless Rule Rescission.

Activity	*Management Element*	*Task/Method*	*Tool*
Timber Harvest	Vegetation Management	Timber Felling	Heavy Equipment
		Tree and Unit Marking	Walking
			ATV
		Landings - Vegetation Clearing and Removal	Heavy Equipment
		Site Restoration - Seeding and Noxious Weed Treatments	Hand and Vehicle
		Timber Yarding - Skidding, Skyline, Helicopter, Tethered, Forwarders	Heavy Equipment
		Timber Haul	Heavy Equipment
	Temporary Access Control	Road Gates Installation	Heavy Equipment
Roads - Maintenance or Roads - Reconstruction	Drainage	Maintenance (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment
		Installation (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment
		Replacement or repair (e.g., ditch-lines, water dips, cross-drain culverts, culvert replacement, or water bars)	Heavy Equipment
		Temporary stream flow divergence; Bypass	Heavy Equipment; Pumps; Screens
	Vegetation Management	Brushing, limbing, seeding, mowing, and mulching, hazard tree removal	Heavy Equipment/Hand Tools

Activity	*Management Element*	*Task/Method*	*Tool*
	Surface Repair/Replacement	Road cut and fill repair/stabilization	Heavy Equipment
		Asphalt repair, repaving, chip-sealing, skin patching, and rocking	Heavy Equipment
		Small slide removal (easily handled with maintenance equipment)	Heavy Equipment
		Dust abatement (water, lignin sulfate, magnesium chloride)	Heavy Equipment; Pumps; Screens
	Structure Maintenance, Repair, and Installation	Guardrails, signs, gates, etc.	Heavy Equipment, ATV
	Bridge Maintenance	Deck cleaning, painting, replacing decking, and minor structural repairs outside of the wetted perimeter or above the bank full	Heavy Equipment
	Snow	Plowing	Heavy Equipment
Permanent and Temporary Road Construction	Vegetation Clearance	Cut Above-Ground Vegetation	Heavy Equipment/Motorized Hand Tools
		Snags - Hand Felling	Motorized Hand Tools
		Snags - Mechanical Felling	Heavy Equipment
		Tree Removal - Log Decks	Heavy Equipment
		Tree Removal - Hauling	Heavy Equipment
	Roadway template	Soil Excavation	Heavy Equipment, Blasting
		Soil Removal	Heavy Equipment
		Fill Placement	Heavy Equipment
	Drainage	Ditch lines, Relief Culverts, Water bars, Riprap	Heavy Equipment
		Stream Crossings - Culvert Installation	Heavy Equipment
	Surfacing	Hauling of surfacing materials	Heavy Equipment
		Aggregate Surfacing Placement - Rolling and Blading	Heavy Equipment
Temporary Road Removal	Recontouring	blading/recontouring	Heavy equipment
	scarifying	Material removal	Heavy equipment
	culvert/drainage structure removal	culvert/drainage structure removal	Heavy equipment
	Revegetation	Seeding	hand tools; ground equipment; aerial equipment

Road Construction

Road construction[11] **or road reconstruction**[12] **or road maintenance**[13] may be associated with timber harvest or other forest vegetation management (fuels reduction, thinning, etc.), private or special uses, existing rights, minerals and energy exploration[14], watershed restoration, habitat improvement, and recreation development. Aside from timber harvest, these activities are currently allowed under the 2001 Roadless Rule (see environmental baseline section).

The objective of the NFS transportation system is to provide the minimal mileage road system needed for safe and efficient travel, and for administration, utilization and protection of NFS lands. Any new roads would be designed and maintained in a way that would minimize negative effects on water quality, soil stability, and wildlife habitat. The decision to construct a new, permanent road that would become part of the NFS road system would be made as part of transportation planning associated with planned resource management activities and use of NFS lands. Proposals for new roads related to development and the associated environmental effects would be considered through project-level planning. If future decisions authorize it, modifications to the transportation network would follow law, regulation and policy found at 16 USC 1608, FSM/FSH 7700, relevant State and Federal statutes such as the Clean Water Act, NEPA, and Endangered Species Act, and national forest land management plans. Implementation of these policies can minimize, but not eliminate, some adverse environmental effects.

Timber harvest activities would likely generate the highest need for permanent and temporary roads on the landscape. New roads will not be required for some timber activities. For example, it's assumed timber related activities within 0.5 miles of existing roads where terrain slopes are 40 percent or less, would less likely require the development of new, permanent roads (11.3 million acres or 28.3 percent of the affected environment). Conversely, where timber activities are occurring beyond 0.5 miles of existing roads, some construction of new temporary and permanent roads would be more likely[15]. New projects would not automatically be approved as Land Management Plan consistency and other environmental factors would need to be in alignment with the proposal. Any new permanent road construction[16] would be limited to areas

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[11] Activity that results in the addition of forest classified or temporary road miles.

[12] Road reconstruction. Activity that results in improvement or realignment of an existing classified road defined as follows:
(1) Road improvement. Activity that results in an increase of an existing road's traffic service level, expansion of its capacity, or a change in its original design function.
(2) Road realignment. Activity that results in a new location of an existing road or portions of an existing road, and treatment of the old roadway.

[13] Road maintenance. The ongoing upkeep of a road necessary to retain or restore the road to the approved road management objective.

[14] General Mining Act of 1872 and the 2001 Roadless Rule's exceptions (36 CFR 294.12(b)(3); 36 CFR 294.13(b)(2))

[15] A recent review of timber related projects in Forest Service Northern Region (1) revealed that approximately 25 percent of roads constructed in connection with timber contracts or other permits/leases were considered temporary and would be decommissioned within 10 years of contract, permit, or lease termination (16 USC 1608). For example, the Sawmill-Petty Project on the Lolo National Forest in Montana estimated road construction needs of approximately 19 miles of permanent road and 9 miles of temporary road to treat just under 17,000 acres. Another project, the Cyclone Bill Project on the Flathead National Forest estimated road construction needs of approximately 12 miles of permanent road and 2.5 miles of temporary road to treat approximately 12,000 acres.

[16] New road construction would be accordance with 16 U.S.C. 1608 which states "Unless the necessity for a permanent road is set forth in the forest development road system plan, any road constructed on land of the National Forest System in connection with a timber contract or other permit or lease shall be designed with the goal of reestablishing vegetative cover on the roadway

where it is currently allowed by land management plans (18.2 million acres or 45.5 percent of the affected environment).

Timber Harvest

Vegetation and Forest Management include commercial timber harvest[17] and non-commercial timber treatments[18] and are completed for multiple resource objectives, including but not limited to timber production, fuels reduction, public health and safety, forest health improvement, watershed improvement, and wildlife habitat improvement. Land management plans must ensure: (1) Timber harvest is only allowed in areas that will not be irreversibly damaged, where cutting units can be visually blended into the terrain, where streams, wetlands, riparian areas, and water bodies are protected; (2) Timber harvest does not exceed sustained yield and is not designated solely to maximize outputs; and, (3) Harvested areas must be regenerated within 5 years. Any project activity, including timber harvest, must be consistent with land management plans. The 2012 Planning Rule provides further direction for forest management activities including timber harvest and non-commercial treatments. These plans include plan components (desired conditions, goals, objectives, standards, and guidelines) and identify areas suitable and not suitable for timber production to maintain sustained yield levels, identify maximum harvest levels on suitable lands (allowable sale quantity or sustained yield limit), and allow, prohibit, or limit timber harvest to protect resources.

Timber harvest effects are the same and not dependent on the resource objective; therefore, the deconstruction table (Table 1) only includes a basic timber harvest activity. Silvicultural prescriptions are utilized to meet direction within individual land management plans. Timber sales and other removals of forest products support agency strategic objectives to foster resilient, adaptive ecosystems, mitigate wildfire risk, and strengthen communities (USDA 2015). Table 1 was developed to cover all types of timber harvest and silvicultural prescriptions covered by the Roadless Rule Rescission. On average per region, NFS harvests less than 1 percent of operable areas annually (Forest Service Cut and Sold Reports 2014-2024). About 4.8 million acres (16 percent of forested areas in the potentially affected IRAs) are in areas where current land management plans allow timber harvest and are more likely to be operable.

Other Activities Potentially Involving Road Construction or Tree Cutting

Access to National Forest System Lands and Special Use Authorizations

Access to the national forest system lands come in many forms from general usage by members of the public that do not require permits in most cases, to lands and special use authorization uses

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and areas where the vegetative cover has been disturbed by the construction of the road, within ten years after the termination of the contract, permit, or lease either through artificial or natural means. Such action shall be taken unless it is later determined that the road is needed for use as a part of the National Forest Transportation System."

[17] Timber Harvest – Forest management activities that include the removal of sawtimber. Timber sales are often used as a least-cost method (revenue is returned to the Federal treasury to offset the costs of preparing and carrying out the timber harvest while portions of receipts are used for post-sale activities) of managing vegetation to meet multiple resource objectives (Forest Service Manual 2400, Forest Service Handbook 2409.14).

[18] Non-commercial forest management – Forest management activities that do not include the removal of sawtimber, non-commercial uses of timber, or any forest products removed for personal or incidental use. These activities may be completed with mechanized systems or by hand. Non-commercial forest management are higher cost options for managing vegetation to meet multiple resource objectives than timber harvest (Forest Service Manual 2460, Forest Service Handbook 2409.18).

that are allowed through Land Management Plans and require a special use permit or similar legal instrument. Special use authorizations are needed for activities on or occupancy of National Forest System Lands including water transmission/storage, utilities, research, military training, communication uses, agriculture, pipelines, organized recreation events, ski areas, road access and much more. Pre-existing uses and those related to reserved or outstanding rights for access, occupancy, and use of NFS lands were not affected by the 2001 Roadless Rule. The continuation, re-issuance, or renewal of existing special use authorizations for activities in inventoried roadless areas is not prohibited by the 2001 Roadless Rule. Any future or additional special use authorizations would be reactive in nature to respond to public and customer requests. Under the proposed rulemaking, it is reasonably certain to occur that there would be an increase in new proposals for uses involving road construction or reconstruction that benefit the public or provide access to non-Federal lands within or adjacent to IRAs, as well as requests for expansion of existing uses (for example, additional access routes or improved road quality). An authorization is issued only after being evaluated and having passed through screening criteria (36 CFR 251.54), including compliance with the applicable land management plan and environmental analysis processes. Under the proposed action, new special use proposals would be evaluated on their own merits and no longer be rejected outright due to prohibitions for road construction or reconstruction. Requests for and approval of road construction-related authorizations are most likely to occur in areas that are 'operable' and where current land management plans allow these activities. For the purposes of this assessment, road construction is considered but future activities requiring a special use permit are outside the scope of this analysis and will be consulted on at the time of permit application.

Minerals Exploration and Development

Mineral and energy exploration and development occur within inventoried roadless areas (though to a lesser degree than other NFS lands) and would continue where there are existing rights or permits/leases/authorizations. Road construction, reconstruction, and tree cutting for locatable (gold, silver, copper, lead, zinc, etc.) mineral exploration and development is considered a right of access as provided by the General Mining Law and the 2001 Roadless Rule's exceptions (36 CFR 294.12(b)(3); 36 CFR 294.13(b)(2)). Changes may occur for leasable minerals exploration and development (primarily oil and gas and coal). Any new, modified or renewed leases would continue to be managed under existing laws and regulations governing oil and gas and coal, including project specific environmental review documents and compliance with other relevant law, regulation, and policy prior to leasing. For new leases or modifications of leases, construction or reconstruction of roads that are reasonable and necessary for development of existing energy or mineral leases, for access to new or existing energy or mineral leases, and for access to associated product conveyance lines could be allowed as necessary in accordance with the terms of the lease, where they are consistent with land management plans or other laws and policy requirements. In addition, the 2001 Roadless Rule and IRA boundary would not apply or be considered in any future analyses for the modification of existing leases. However, in a site-specific leasing analysis, a no surface occupancy stipulation may still be applied to protect surface resources. For example, to prevent excessive erosion on steep slopes, a no surface occupancy may still be applied to the Federal lease. The Forest Service is not proposing to extend this consultation to include future minerals exploration and development.

Future mineral exploration and development will require standalone consultation to comply with the Endangered Species Act.

Species Lists

Threatened or Endangered species and their designated critical habitats considered in this consultation are listed in Table 2. These species and critical habitat lists were compiled from existing consultation documents for species under NMFS's jurisdiction. Species lists and some critical habitat layers were compiled from other consultations and species lists from National Forests where NMFS species are known to occur (Table 3). This list was reviewed by regional Forest Service staff for consistency.

Species critical habitat shapefiles were uploaded from NOAA Fisheries National ESA Critical Habitat Mapper (https://www.fisheries.noaa.gov/resource/map/national-esa-critical-habitat-mapper), overlaid with the Forest Service administrative boundaries for each National Forest, then overlaid with the affected environment. For the Southern Oregon/Northern California coho salmon critical habitat, mapping information used by the Rogue River-Siskiyou National Forest during ESA consultation was used in this process as critical habitat maps are not included in the NOAA's Critical Habitat Mapper. The Forest Service included a 2-mile buffer to account for any downstream effects. Species lists and critical habitat were reviewed by the National Marine Fisheries Service in September and October 2025. All maps are in Appendix A.

Table 2. ESA listed species considered in this consultation, by ESU or DPS, listing status, federal register notice information, and protective regulations. T=Threatened, E=Endangered, XN=non-essential population, FR=Federal Register.

Species	**Scientific Name**	**Evolutionarily Significant Unit or Distinct Population Segment**	**Listing Status**	**Listing date and Federal Register notice of listing status**	**Critical Habitat listing date and Federal Register notice of listing status**	**Protective Regulations**
Chinook salmon	*Oncorhynchus tshawytscha*	California Coastal	T	09/16/1999; 64 FR 50394 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52488	06/28/2005; 70 FR 37160
Chinook salmon	*Oncorhynchus tshawytscha*	Central Valley Spring	T + XN	09/16/1999; 64 FR 50394 06/28/2005; 70 FR 37160	09/02/2005; 70 FR 52488	01/09/2002; 67 FR 1116
Chinook salmon	*Oncorhynchus tshawytscha*	Lower Columbia River	T	03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160
Chinook salmon	*Oncorhynchus tshawytscha*	Puget Sound	T	03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160
Chinook salmon	*Oncorhynchus tshawytscha*	Sacramento River Winter	E	08/04/1989; 54 FR 32085 11/05/1990; 55 FR 46515 01/04/1994; 59 FR 440	06/16/1993 58 FR 33212	11/05/1990; 55 FR 46515 (T) 01/04/1994; 59 FR 440 (E)
Chinook salmon	*Oncorhynchus tshawytscha*	Snake River Fall	T	04/22/1992; 57 FR 14653 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	12/28/1993; 58 FR 68543	06/28/2005; 70 FR 37160
Chinook salmon	*Oncorhynchus tshawytscha*	Snake River Spring/Summer	T	04/22/1992; 57 FR 14653 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	10/25/1999; 64 FR 57399	06/28/2005; 70 FR 37160
Chinook salmon	*Oncorhynchus tshawytscha*	Upper Columbia River Spring	E	03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	ESA section 9 applies
Chinook salmon	*Oncorhynchus tshawytscha*	Upper Willamette River	T	03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	9/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160

Species	Scientific Name	Evolutionarily Significant Unit or Distinct Population Segment	Listing Status	Listing date and Federal Register notice of listing status	Critical Habitat listing date and Federal Register notice of listing status	Protective Regulations
Chum salmon	*Oncorhynchus keta*	Columbia River	T	03/24/1999; 64 FR 14508 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	06/28/2005; 70 FR 37160	06/28/2005; 70 FR 37160
Chum salmon	*Oncorhynchus keta*	Hood Canal Summer	T	03/24/1999; 64 FR 14508 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	06/28/2005; 70 FR 37160	06/28/2005; 70 FR 37160
Coho salmon	*Oncorhynchus kisutch*	Central California Coast	E	10/31/1996; 61 FR 56138 06/28/2005; 70 FR 37159 04/14/2014; 79 FR 20802	05/05/1999; 64 FR 24049	ESA section 9 applies
Coho salmon	*Oncorhynchus kisutch*	Lower Columbia River	T	06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	02/24/2016; 81 FR 9252	06/28/2005; 70 FR 37160
Coho salmon	*Oncorhynchus kisutch*	Oregon Coast	T	08/10/1998; 63 FR 42587 06/20/2011; 76 FR 35755 04/14/2014; 79 FR 20802	02/11/2008; 73 FR 7816	02/11/2008; 73 FR 7816
Coho salmon	*Oncorhynchus kisutch*	Southern Oregon/Northern California Coast	T	05/06/1997; 62 FR 24588 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	05/05/1999; 64 FR 24049	06/28/2005; 70 FR 37160
Sockeye Salmon	*Oncorhynchus nerka*	Snake River	T	11/20/1991; 56 FR 58619 04/14/2014; 79 FR 20802 06/28/2005; 70 FR 37160	12/28/1993; 58 FR 68543	ESA section 9 applies
Steelhead	*Oncorhynchus mykiss*	California Central Valley	T	03/19/1998; 63 FR 13347 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52488	07/10/2000; 65 FR 42421
Steelhead	*Oncorhynchus mykiss*	Central California Coast	T	08/18/1997; 62 FR 43937 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52488	06/28/2005; 70 FR 37160
Steelhead	*Oncorhynchus mykiss*	Lower Columbia River	T	03/19/1998; 63 FR 13347 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160
Steelhead	*Oncorhynchus mykiss*	Northern California	T	06/07/2000; 65 FR 36074 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52488	06/28/2005; 70 FR 37160

Species	Scientific Name	Evolutionarily Significant Unit or Distinct Population Segment	Listing Status	Listing date and Federal Register notice of listing status	Critical Habitat listing date and Federal Register notice of listing status	Protective Regulations
Steelhead	*Oncorhynchus mykiss*	Middle Columbia River	T	03/25/1999; 64 FR 14517 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160
Steelhead	*Oncorhynchus mykiss*	Puget Sound	T	05/11/2007; 72 FR 26722 04/14/2014; 79 FR 20802	02/24/2016; 81 FR 9252	10/27/2008; 73 FR 55451
Steelhead	*Oncorhynchus mykiss*	Snake River	T	08/18/1997; 62 FR 43937 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160
Steelhead	*Oncorhynchus mykiss*	South-Central California Coast	T	08/18/1997; 62 FR 43937 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52488	06/28/2005; 70 FR 37160
Steelhead	*Oncorhynchus mykiss*	Southern California	E	08/18/1997; 62 FR 43937 05/01/2002; 62 FR 21586 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52488 07/11/2025; 90 FR 30833 (90-day finding to revise CH)	ESA section 9 applies
Steelhead	*Oncorhynchus mykiss*	Upper Columbia River	T	08/18/1997; 62 FR 43937 (E) 01/05/2006; 71 FR 834 (T) 08/24/2009; 74 FR 42605 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	02/01/2006; 71 FR 5178
Steelhead	*Oncorhynchus mykiss*	Upper Willamette River	T	03/25/1999; 64 FR 14517 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	09/02/2005; 70 FR 52630	06/28/2005; 70 FR 37160
Green sturgeon	*Acipenser medirostris*	Southern	T	04/07/2006; 71 FR 17757	10/09/2009; 74 FR 52300	06/02/2010; 75 FR 30714 ESA 4(d) Rule
Eulachon	*Thaleichthys pacificus*	Southern	T	03/18/2010; 75 FR 13012	10/20/2011; 76 FR 65324	None
Killer whale	*Orcinus orca*	Southern Resident	E	11/18/2005; 70 FR 69903 04/14/2014; 79 FR 20802	11/29/2006; 71 FR 69054 08/02/2021; 86 FR 41668	ESA section 9 applies

Table 3. National Forests that consult on NMFS regulatory species and/or critical habitat by DPS or ESU, miles of critical habitat located on the National Forest (NF), miles of critical habitat (CH) in the Inventoried Roadless Area, miles of critical habitat within 2-mile buffer from IRA, miles of critical habitat potentially affected by proposed action, and miles of critical habitat (CH) in the operable and operable but complex portions of IRAs.

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Cleveland	Pacific Southwest	Steelhead	Southern California	3.78	0.00	0.00	0.00	0.00
Klamath	Pacific Southwest	Coho salmon	Southern Oregon/Northern California Coast	0.00	0.00	0.00	0.00	0.00
Lassen	Pacific Southwest	Chinook salmon	Central Valley Spring	53.25	19.63	22.05	41.68	12.62
Lassen	Pacific Southwest	Steelhead	California Central Valley	65.13	25.38	25.61	50.99	17.85
Los Padres	Pacific Southwest	Steelhead	Southern California	279.56	47.57	111.93	159.50	24.11
Los Padres	Pacific Southwest	Steelhead	South-Central California Coast	109.41	0.67	32.62	33.29	0.09
Mendocino	Pacific Southwest	Chinook salmon	California Coastal	0.31	0.00	0.00	0.00	0.00
Mendocino	Pacific Southwest	Chinook salmon	Central Valley Spring	0.51	0.00	0.00	0.00	0.00
Mendocino	Pacific Southwest	Chinook salmon	Sacramento River Winter	0.56	0.00	0.00	0.00	0.00
Mendocino	Pacific Southwest	Coho salmon	Southern Oregon/Northern California Coast	0.00	0.00	0.00	0.00	0.00
Mendocino	Pacific Southwest	Steelhead	California Central Valley	20.77	8.68	3.62	12.30	2.87
Mendocino	Pacific Southwest	Steelhead	Northern California	70.79	1.55	39.57	41.12	0.73

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Mendocino	Pacific Southwest	Green Sturgeon	Southern	0.52	0.00	0.00	0.00	0.00
Shasta-Trinity	Pacific Southwest	Chinook salmon	Central Valley Spring	1.26	1.26	0.00	1.26	0.89
Shasta-Trinity	Pacific Southwest	Chinook salmon	Sacramento River Winter	0.00	0.00	0.00	0.00	0.00
Shasta-Trinity	Pacific Southwest	Coho salmon	Southern Oregon/Northern California Coast	0.00	0.00	0.00	0.00	0.00
Shasta-Trinity	Pacific Southwest	Steelhead	California Central Valley	12.54	4.89	2.22	7.11	3.35
Shasta-Trinity	Pacific Southwest	Steelhead	Northern California	0.00	0.00	0.00	0.00	0.00
Six Rivers	Pacific Southwest	Chinook salmon	California Coastal	1.87	0.00	0.00	0.00	0.00
Six Rivers	Pacific Southwest	Coho salmon	Southern Oregon/Northern California Coast	16.02	1.09	14.93	16.02	0.00
Six Rivers	Pacific Southwest	Steelhead	Northern California	24.21	0.66	13.42	14.08	0.61
Columbia River Gorge NSA	Pacific Northwest	Chinook salmon	Lower Columbia River	6.56	0.00	3.60	3.60	0.15
Columbia River Gorge NSA	Pacific Northwest	Chinook salmon	Snake River Fall	0.00	0.00	0.00	0.00	0.00
Columbia River Gorge NSA	Pacific Northwest	Chinook salmon	Snake River Spring/Summer	0.00	0.00	0.00	0.00	0.00

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Columbia River Gorge NSA	Pacific Northwest	Chinook salmon	Upper Columbia River Spring	0.00	0.00	0.00	0.00	0.00
Columbia River Gorge NSA	Pacific Northwest	Chum salmon	Columbia River	1.68	0.00	0.93	0.93	0.00
Columbia River Gorge NSA	Pacific Northwest	Coho salmon	Lower Columbia River	21.03	0.00	10.14	10.14	1.61
Columbia River Gorge NSA	Pacific Northwest	Sockeye Salmon	Snake River	0.00	0.00	0.00	0.00	0.00
Columbia River Gorge NSA	Pacific Northwest	Steelhead	Snake River	0.00	0.00	0.00	0.00	0.00
Columbia River Gorge NSA	Pacific Northwest	Steelhead	Lower Columbia River	24.13	0.00	13.17	13.17	1.54
Columbia River Gorge NSA	Pacific Northwest	Steelhead	Middle Columbia River	0.92	0.00	0.00	0.00	0.00
Columbia River Gorge NSA	Pacific Northwest	Steelhead	Upper Columbia River	0.00	0.00	0.00	0.00	0.00
Columbia River Gorge NSA	Pacific Northwest	Eulachon	Southern	0.22	0.00	0.00	0.00	0.00
Gifford Pinchot	Pacific Northwest	Chinook salmon	Lower Columbia River	77.24	6.42	55.06	61.47	6.26
Gifford Pinchot	Pacific Northwest	Chinook salmon	Puget Sound	0.00	0.00	0.63	0.63	0.00

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Gifford Pinchot	Pacific Northwest	Chum salmon	Columbia River	0.00	0.00	0.00	0.00	0.00
Gifford Pinchot	Pacific Northwest	Coho salmon	Lower Columbia River	146.25	11.55	73.42	84.96	10.72
Gifford Pinchot	Pacific Northwest	Steelhead	Lower Columbia River	222.85	21.03	93.88	114.92	19.57
Gifford Pinchot	Pacific Northwest	Steelhead	Middle Columbia River	0.03	0.00	0.00	0.00	0.00
Gifford Pinchot	Pacific Northwest	Eulachon	Southern	0.00	0.00	0.00	0.00	0.00
Gifford Pinchot	Pacific Northwest	Green Sturgeon	Southern	0.00	0.00	0.00	0.00	0.00
Deschutes	Pacific Northwest	Steelhead	Middle Columbia River	0.00	0.00	0.00	0.00	0.00
Malheur	Pacific Northwest	Steelhead	Middle Columbia River	418.32	57.43	129.58	187.01	44.17
Mt. Baker Snoqualmie	Pacific Northwest	Chinook salmon	Puget Sound	204.15	23.18	143.88	167.05	18.98
Mt. Baker Snoqualmie	Pacific Northwest	Steelhead	Puget Sound	367.20	47.17	245.99	293.15	39.39
Mt. Hood	Pacific Northwest	Chinook salmon	Lower Columbia River	50.34	2.25	37.53	39.78	2.09
Mt. Hood	Pacific Northwest	Chinook salmon	Snake River Fall	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Chinook salmon	Snake River Spring/Summer	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Chinook salmon	Upper Columbia River Spring	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Chinook salmon	Upper Willamette River	75.76	0.15	22.19	22.34	0.18

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Mt. Hood	Pacific Northwest	Chum salmon	Columbia River	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Coho salmon	Lower Columbia River	189.16	5.54	86.60	92.14	3.93
Mt. Hood	Pacific Northwest	Sockeye Salmon	Snake River	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Steelhead	Middle Columbia River	32.81	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Steelhead	Lower Columbia River	265.47	10.49	149.75	160.24	8.89
Mt. Hood	Pacific Northwest	Steelhead	Snake River	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Steelhead	Upper Columbia River	0.00	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Steelhead	Upper Willamette River	0.19	0.00	0.00	0.00	0.00
Mt. Hood	Pacific Northwest	Eulachon	Southern	0.00	0.00	0.00	0.00	0.00
Ochoco	Pacific Northwest	Steelhead	Middle Columbia River	99.24	29.12	15.76	44.88	25.95
Okanogan-Wenatchee	Pacific Northwest	Chinook salmon	Upper Columbia River Spring	188.40	7.61	135.71	143.32	13.45
Okanogan-Wenatchee	Pacific Northwest	Steelhead	Middle Columbia River	180.69	0.00	125.03	125.03	4.92
Okanogan-Wenatchee	Pacific Northwest	Steelhead	Upper Columbia River	236.61	2.46	166.18	168.64	16.07
Olympic	Pacific Northwest	Chinook salmon	Puget Sound	33.15	2.94	20.88	23.81	2.75
Olympic	Pacific Northwest	Chum salmon	Hood Canal Summer	0.04	0.00	0.04	0.04	0.00

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Olympic	Pacific Northwest	Steelhead	Puget Sound	56.75	3.36	32.69	36.06	3.16
Olympic	Pacific Northwest	Eulachon	Southern	0.00	0.00	0.00	0.00	0.00
Olympic	Pacific Northwest	Green Sturgeon	Southern	0.00	0.00	0.00	0.00	0.00
Rogue River-Siskiyou	Pacific Northwest	Coho salmon	Oregon Coast	7.66	0.00	7.23	7.23	0.00
Rogue River-Siskiyou	Pacific Northwest	Coho salmon	Southern Oregon/Northern California Coast	579.34	2.74	271.16	273.90	0.00
Rogue River-Siskiyou	Pacific Northwest	Eulachon	Southern	0.00	0.00	0.00	0.00	0.00
Rogue River-Siskiyou	Pacific Northwest	Green Sturgeon	Southern	0.00	0.00	0.00	0.00	0.00
Siuslaw	Pacific Northwest	Coho salmon	Oregon Coast	565.75	22.36	53.70	76.06	17.22
Siuslaw	Pacific Northwest	Steelhead	Upper Willamette River	0.47	0.00	0.44	0.44	0.00
Siuslaw	Pacific Northwest	Eulachon	Southern	0.00	0.00	0.00	0.00	0.00
Siuslaw	Pacific Northwest	Green Sturgeon	Southern	0.68	0.00	0.40	0.40	0.00
Umatilla	Pacific Northwest	Chinook salmon	Snake River Fall	4.73	0.00	4.73	4.73	0.00
Umatilla	Pacific Northwest	Chinook salmon	Snake River Spring/Summer	132.64	21.37	55.72	77.09	0.00

National Forest Service Unit	National Forest Service Region	Species	DPS or ESU	Total Miles of CH on NF	Miles of CH in Roadless only	Miles of CH in Roadless within 2-mile buffer only	Miles of CH potentially affected	Miles of CH in Operable/Operable but Complex Roadless Areas and 2-mile buffer
Umatilla	Pacific Northwest	Steelhead	Middle Columbia River	665.10	143.12	144.64	287.76	101.93
Umatilla	Pacific Northwest	Steelhead	Snake River	195.14	31.96	88.57	120.53	26.38
Umpqua	Pacific Northwest	Coho salmon	Oregon Coast	85.90	0.40	37.74	38.14	0.25
Wallowa-Whitman	Pacific Northwest	Chinook salmon	Snake River Fall	58.59	1.59	41.15	42.75	0.87
Wallowa-Whitman	Pacific Northwest	Chinook salmon	Snake River Spring/Summer	426.29	37.19	162.03	199.22	0.00
Wallowa-Whitman	Pacific Northwest	Sockeye Salmon	Snake River	12.95	0.00	12.95	12.95	0.00
Wallowa-Whitman	Pacific Northwest	Steelhead	Middle Columbia River	76.36	2.71	41.42	44.13	2.61
Wallowa-Whitman	Pacific Northwest	Steelhead	Snake River	883.29	144.49	255.98	400.46	92.72
Willamette	Pacific Northwest	Chinook salmon	Upper Willamette River	245.90	5.89	95.32	101.21	5.67
Willamette	Pacific Northwest	Steelhead	Upper Willamette River	30.40	4.78	17.11	21.89	4.41

In Alaska, all Endangered Species Act listed species are associated with the off-shore marine environment adjacent to the National Forest. There are no listed threatened or endangered species that occupy the terrestrial or aquatic ecosystems systems in Alaska except for a few isolated haul-out locations for the Stellar sea lions associated with the western distinct population segment (DPS). The review of listed species and habitats for the action alternatives associated with this analysis were found to have 'no effect' to any threatened or endangered species or designated critical habitat and therefore there is no need for consultation with either the U.S. Fish and Wildlife Service or National Marine Fisheries Service. The National Marine Fisheries Service reviewed the biological assessment for threatened and endangered species under their regulatory jurisdiction and concluded that the 2016 Tongass Forest Plan Amendment EIS was "not likely to adversely affect" threatened or endangered species occurring on the Tongass (the 2020 Alaska FEIS summarizes the analysis on wildlife species and their habitat from the 2016 Forest Plan Amendment EIS). Steller sea lion will not be discussed further in this biological assessment.

Data Collection and Use

We acquired species occurrence data from the Global Biodiversity Information Facility (GBIF) and Natural Resource Manager (NRM) using R and ArcGIS Pro, respectively. The databases do not differentiate between DPSes of vertebrate species listed under the ESA, which causes potential inflation of exposure to these entities. The data is managed in such a way that occasionally species occurrences occur in locations where the entity is not listed or protected by the ESA; however, the FS does not believe these discrepancies have altered the conclusions provided.

A custom exposure analysis is performed for some species known to be impacted by roads and timber activities and where national data could support such analyses. Where custom analyses are performed, they will be documented in either the administrative record or in the taxonomic sections below. While the potential for inflated exposure is possible the FS does not believe it would significantly affect the determinations made. These data discrepancies will also be addressed during project-level consultation. We processed GBIF data using three Python scripts to unzip the downloaded GBIF folders and join relevant data (e.g., NLCD, basic ownership, Roadless Rule Affected Environments), filter to ESA sensitive species, and summarize the data into region-wide csv files. For Forest Service Alaska Region (Region 10), we separated the GBIF data from the conterminous United States (CONUS) data and ran a separate Python script due to available NLCD data (2024 Annual NLCD data for CONUS and the 2016 NLCD Alaska data for Region 10). We designated NRM data features using ArcGIS Pro's ArcPy as either inside or outside of Roadless Rule Affected Environments in National Forest land. We then determined each feature's NLCD value and class using pixel count majorities in R, joined relevant data (e.g., basic ownership and Roadless Rule Affected Environments), created Forest Service region-wide summaries, and exported them as an excel workbook. Species occurrence data are opportunistic and do not represent the results of exhaustive structured species surveys nor do they include negative occurrence data. Additionally, for mobile species, an observation of an individual in a specific location does not mean that that location is occupied by resident individuals- some of these observations could have been of transient individuals.

Exposure was determined by comparing species occurrence data with roadless area operability (as defined in the Environmental Baseline section below) and categorized as high, moderate, and low likelihood of exposure. Low exposure was determined for species that had no occurrences in IRAs. Medium exposure was determined for species with less than 50% of occurrences in IRAs. High exposure was determined for species that had greater than 50% of occurrences in IRAs. Where there was no occurrence data the exposure was classified as unknown. Exposure estimates were performed in a manner to err on the side of likely exposure. Existing protections afforded by land management plans and other documents/decisions were also reviewed to determine how they may affect the determinations for each species, particularly if effects to the species were determined to be highly impactful. The response of species was derived from published scientific literature and previous consultations. Together, exposure and response were used to make the determinations: No effect (NE), Not likely to adversely affect (NLAA), and likely to adversely affect (LAA).

	Exposure		
	Low or unknown	Medium or unknown	High
Response is impactful	NLAA or LAA	LAA	LAA
Response is minimal	NE or NLAA	NLAA or LAA	LAA

Where available information indicated a less impactful determination was warranted compared to the general determinations a less impactful determination was made. Where information is lacking the Forest Service erred on the side of making the more impactful determination.

The exposure of critical habitat to stressors associated with road construction and timber activities was performed by comparing the overlap of critical habitat with the action area categorized by operability. Where overlap occurred, the physical and biological features of the critical habitat were assessed to determine whether the stressors associated with road construction and timber harvest could be expected to adversely affect those PBFs and the functionality of the critical habitat. Determinations for critical habitat were either no effect, not likely to adversely affect, or likely to adversely affect. Where information was lacking the Forest Service erred on the side of making the more impactful determination.

ENVIRONMENTAL BASELINE

This section briefly describes the status of inventoried roadless areas that may be affected (Roadless Affected Environment or Environmental Baseline). There are approximately 40 million acres of IRAs managed under 2001 Roadless Rule[19] within the action area. Additional information on the status of the action area is available in the environmental impact statement for the roadless rule rescission.

[19] Current areas are reflected on national maps at the Forest Service Roadless Area website: https://www.fs.usda.gov/managing-land/planning/roadless

Land Management Plans and Operability

A national review of plan components in management areas overlapping the potentially affected environment was completed to identify where plans do or do not further restrict timber harvest and road construction activities[20]. There is much variation between individual land management plans and Regions reflecting differences in local conditions, age of the plans, and changes in planning policies over time. Current land management-plans include plan direction and components that identify where timber management and road construction uses may or may not be allowed (Table 3). Example management area types where timber management or road construction may be limited include: recommended wilderness, research natural areas, or backcountry management areas are administrative land management planning designations that limit active management.

- Across the Roadless Affected Environment, land management plan components generally allow timber harvest across 13.4 million acres (33.4% of NFS land within the Roadless Affected Environment). [36% not allowed, 26.8 generally not allowed (with some exceptions)].
- Across the Roadless Affected Environment, land management plan components generally allow road construction across 18.2 million acres (45.5% of NFS land within the Roadless Affected Environment).

This analysis is directed at actions that are reasonably likely to occur in the remaining acres of inventoried roadless areas where road construction and reconstruction or timber harvest would be most likely to occur under current land management plans. A nationwide geospatial analysis was used to estimate potential operability, or the capability of mechanized treatment or activities, in each area based on slope, distance from roads, site productivity, and statutory restrictions (Table 2):

- Likely Operable: Less than 0.5 miles from a road, on slopes less than 40%, are productive for restocking (greater than 20 cubic feet/acre/year), are potentially forested, and have no statutory restrictions from timber harvest.
- Likely Operable but Complex: Less than 0.5 miles from a road and on slopes between 40% and 70% or between 0.5 miles and one mile from a road and on slopes less than 70%. These areas have the same productivity for restocking as likely operable areas, are potentially forested, and no statutory restrictions from timber harvest.
- Likely Not Operable: Include slopes over 70%, further than one mile from a road or areas over one mile from a road and less than 70% percent slope. These areas have low productivity, are potentially forested, and no statutory restrictions from timber harvest.

The affected environment's overlap with hydrologic unit codes (HUC) is presented in Table 4. Within the Tongass National Forest 9.3 million acres of inventoried roadless area managed under

[20] Note that this national level review is currently undergoing quality control and Regional/local vetting and the data presented will be updated in the final BA.

the 2001 Roadless Rule, or about 56 percent of the Forest. Approximately 9.2 million acres of these are in the affected environment (on National Forest System lands and outside of wilderness, wilderness study area and wild river segments). Operability analysis excluded portions of the action area that were not forested.

Table 4. Operability by region within the action area (Roadless Rule affected environment).

Region	Roadless Affected Environment Acres	Likely Operable		Likely Operable, but Complex		Likely Not Operable	
		Acres	*Percent of Affected Environment*	*Acres*	*Percent of Affected Environment*	*Acres*	*Percent of Affected Environment*
1	5,951,089	990,944	16.7%	2,020,381	33.9%	2,218,591	37.3%
2	1,733,284	411,357	23.7%	452,877	26.1%	505,389	29.2%
3	2,528,791	156,219	6.2%	262,988	10.4%	1,689,352	66.8%
4	8,474,301	1,030,112	12.2%	1,295,380	15.3%	4,219,049	49.8%
5	3,634,708	553,856	15.2%	1,216,690	33.5%	1,278,172	35.2%
6	3,729,946	787,299	21.1%	1,512,164	40.5%	1,182,516	31.7%
8	839,339	337,357	40.2%	384,545	45.8%	103,114	12.3%
9	605,812	270,002	44.6%	193,661	32.0%	137,154	22.6%
10	12,565,353	396,547	3.2%	651,833	5.2%	5,574,698	44.4%
Total	**40,062,623**	**4,933,693**	**12.3%**	**7,990,519**	**19.9%**	**16,908,035**	**42.2%**

Table 5. Acres of the affected environment (action area) where land management plans currently allow road construction and timber harvest.

Region	Total National Forest System Lands (thousands of acres)	2001 Roadless Rule IRA not including Co or ID (thousands of acres)	Potentially affected environment (thousands of acres)	Potentially affected environment where current LMPs allow/generally allow road construction (thousands of acres)	Potentially affected environment where current LMPs allow/generally allow timber harvest (thousands of acres)
1	25,624	6,707	5,951	3,635	1,253
2	22,059	1,761	1,733	109	443
3	20,539	2,724	2,529	1,085	1,216
4	31,856	8,946	8,474	6,187	5,621
5	20,253	4,200	3,635	1,085	1,578
6	25,012	3,967	3,730	617	611
8	13,457	954	839	113	158
9	12,180	664	606	264	271
10	22,132	14,779	12,565	5,125	2,255
Total	193,112	44,701	40,063	18,221	13,406

Table 6. HUC4s within the affected environment (action area) and their operability.

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
0104	Androscoggin	2,257,874	27,166	1.2%	13,740	4,682	8,575
0106	Saco	3,105,641	38,808	1.2%	17,898	6,344	14,315
0107	Merrimack	3,203,363	59,281	1.9%	29,321	9,699	19,819
0108	Connecticut	7,208,942	90,744	1.3%	41,564	19,026	29,739
0202	Upper Hudson	8,105,456	977	0.01%		692	221

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
0207	Potomac	9,394,592	159,330	1.7%	23,653	49,566	85,465
0208	Lower Chesapeake	13,237,535	160,810	1.2%	20,666	49,374	89,928
0301	Chowan-Roanoke	12,199,819	237	0.002%	0	200	29
0302	Neuse-Pamlico	9,139,348	20,742	0.2%	11,702	3,823	5,125
0305	Edisto-Santee	15,166,690	37,403	0.2%	10,388	8,533	18,238
0306	Ogeechee-Savannah	10,913,203	18,757	0.2%	614	8,522	9,489
0307	Altamaha-St. Marys	13,260,203	4,187	0.0%	525	1,675	1,969
0308	St. Johns	7,798,794	4,820	0.1%	0	4,171	268
0311	Suwannee	9,035,995	13,888	0.2%	1,324	7,420	5,076
0312	Ochlockonee	2,430,245	3,907	0.2%	0	3,617	221
0313	Apalachicola	13,203,758	20,578	0.2%	41	13,977	6,467
0315	Alabama	14,555,603	33,302	0.2%	2,764	14,981	15,352
0316	Mobile-Tombigbee	13,970,558	621	0.004%	0	598	17
0401	Western Lake Superior	6,225,236	56,712	0.9%	5,164	35,322	14,911
0402	Southern Lake Superior	4,606,993	4,336	0.1%	8	3,335	905
0403	Northwestern Lake Michigan	13,259,458	17,503	0.1%	0	16,393	876
0406	Northeastern Lake Michigan	5,903,166	3,854	0.1%	0	3,184	502
0420	Northern Lake Huron	8,713,390	7,615	0.1%	19	4,526	2,406
0424	Lake Huron	16,085,424	223	0.001%	4	115	94
0430	Middle Saint Lawrence River	8,593,027	14,115	0.2%	3,525	4,535	5,875
0501	Allegheny	7,506,369	24,767	0.3%	4,572	10,512	9,534
0502	Monongahela	4,720,782	45,408	1.0%	4,502	21,059	19,527
0505	Kanawha	7,838,989	139,350	1.8%	9,602	70,451	58,414
0507	Big Sandy-Guyandotte	3,822,713	4,744	0.1%	851	892	2,976

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
0510	Kentucky-Licking	6,829,672	2,847	0.0%	185	1,548	1,084
0514	Lower Ohio	8,063,733	14,833	0.2%	0	14,575	121
0601	Upper Tennessee	11,067,332	157,075	1.4%	25,561	43,597	87,009
0602	Middle Tennessee-Hiwassee	3,302,727	38,239	1.2%	6,571	9,579	21,855
0701	Mississippi Headwaters	12,870,704	8	0.0%	0	1	5
0703	St. Croix	4,940,765	948	0.0%		842	99
0705	Chippewa	6,105,019	35,519	0.6%	563	28,152	5,706
0707	Wisconsin	7,636,945	282	0.004%	0	257	0
0714	Upper Mississippi-Kaskaskia-Meramec	10,938,157	3,747	0.03%	3	2,764	958
0802	Lower Mississippi-St. Francis	10,351,746	2,676	0.03%	87	1,330	1,240
0804	Lower Red-Ouachita	13,124,880	8,087	0.1%	1,567	2,779	3,611
0806	Lower Mississippi-Big Black	4,530,136	2,620	0.1%	0	2,592	0
0902	Red	30,126,196	46,051	0.2%	2	3,073	250
0903	Rainy	17,195,988	17,931	0.1%	789	10,798	5,879
1002	Missouri Headwaters	8,982,371	1,652,068	18.4%	579,631	343,656	460,122
1003	Missouri-Marias	12,706,687	819,632	6.5%	364,078	129,948	258,897
1004	Missouri-Musselshell	15,031,372	232,225	1.5%	89,065	47,002	78,355
1007	Upper Yellowstone	9,254,806	568,818	6.1%	233,379	68,768	169,390
1008	Big Horn	14,641,079	805,533	5.5%	270,407	167,482	207,159
1009	Powder-Tongue	12,033,003	332,777	2.8%	94,538	100,594	91,394
1010	Lower Yellowstone	8,891,206	25,101	0.3%	2	447	46
1011	Missouri-Little Missouri	11,029,150	195,615	1.8%	2,310	31,277	13,407
1012	Cheyenne	15,501,512	116,830	0.8%	1,278	8,274	7,357
1013	Missouri-Oahe	23,734,929	14,165	0.1%	0	24	16

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
1018	North Platte	19,885,105	291,618	1.5%	74,813	92,947	92,862
1101	Upper White	14,302,523	25,260	0.2%	1,004	18,617	5,407
1108	Upper Canadian	7,983,248	11,869	0.1%	3,955	2,550	3,455
1111	Lower Arkansas	10,112,696	87,361	0.9%	981	72,048	13,945
1114	Red-Sulphur	17,697,863	16,835	0.1%	6	13,301	3,357
1203	Trinity	11,486,295	1,339	0.0%	0	1,323	10
1204	Galveston Bay-San Jacinto	5,323,577	2,754	0.1%	0	2,670	49
1301	Rio Grande Headwaters	4,871,558	7,575	0.2%	366	3,260	2,739
1302	Rio Grande-Elephant Butte	17,358,505	508,029	2.9%	309,272	40,678	68,261
1303	Rio Grande-Mimbres	13,916,028	200,847	1.4%	181,513	5,608	10,448
1305	Rio Grande Closed Basins	11,246,551	108,924	1.0%	59,917	2,433	4,520
1306	Upper Pecos	15,058,643	88,505	0.6%	40,071	14,108	22,133
1403	Upper Colorado-Dolores	5,352,777	77,699	1.5%	62,734	2,835	3,364
1404	Great Divide-Upper Green	13,283,979	700,232	5.3%	231,148	189,288	150,924
1405	White-Yampa	8,486,235	68,265	0.8%	7,287	33,741	22,634
1406	Lower Green	9,316,233	897,863	9.6%	296,324	197,988	218,344
1407	Upper Colorado-Dirty Devil	8,714,077	546,145	6.3%	305,736	73,368	64,362
1408	San Juan	15,955,361	49,434	0.3%	42,319	219	843
1501	Lower Colorado-Lake Mead	19,373,477	271,873	1.4%	199,422	15,403	16,340
1502	Little Colorado	17,316,510	57,712	0.3%	36,597	6,938	6,514
1503	Lower Colorado	13,280,094	45,500	0.3%	41,583	389	876
1504	Upper Gila	9,722,929	865,873	8.9%	650,032	53,057	78,942
1505	Middle Gila	11,508,412	235,202	2.0%	126,359	7,181	32,979
1506	Salt	8,624,188	256,342	3.0%	138,469	10,256	18,130

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
1507	Lower Gila	9,648,506	54,314	0.6%	39,299	2,444	5,650
1508	Sonora	15,386,447	42,206	0.3%	22,314	1,817	5,707
1601	Bear	4,809,334	355,039	7.4%	134,408	77,651	99,687
1602	Great Salt Lake	18,358,652	794,991	4.3%	434,506	108,199	171,935
1603	Escalante Desert-Sevier Lake	10,543,993	996,473	9.5%	664,234	94,409	133,741
1604	Black Rock Desert-Humboldt	18,329,433	475,723	2.6%	160,320	26,199	51,779
1605	Central Lahontan	8,115,069	574,982	7.1%	279,617	58,824	77,923
1606	Central Nevada Desert Basins	30,543,216	1,649,766	5.4%	1,006,726	17,294	33,295
1701	Kootenai-Pend Oreille-Spokane	33,298,571	2,610,131	7.8%	1,037,377	386,911	1,101,449
1702	Upper Columbia	29,592,139	827,984	2.8%	371,475	107,337	288,076
1703	Yakima	3,935,933	204,042	5.2%	57,517	46,419	91,620
1704	Upper Snake	22,958,517	944,925	4.1%	394,121	147,133	252,815
1705	Middle Snake	23,671,623	364,179	1.5%	74,047	68,957	100,804
1706	Lower Snake	22,428,309	425,288	1.9%	114,622	71,907	160,448
1707	Middle Columbia	19,138,642	553,310	2.9%	120,331	159,731	232,331
1708	Lower Columbia	3,985,028	180,740	4.5%	49,310	41,127	87,870
1709	Willamette	7,336,249	175,545	2.4%	31,119	56,721	85,324
1710	Oregon-Washington Coastal	16,121,566	490,054	3.0%	107,803	124,677	241,186
1711	Puget Sound	13,085,895	467,199	3.6%	237,227	42,819	164,847
1712	Oregon Closed Basins	11,155,153	86,239	0.8%	14,855	38,158	25,541
1801	Klamath-Northern California Coastal	16,744,271	774,800	4.6%	239,956	123,855	324,032
1802	Sacramento	17,794,824	715,873	4.0%	219,382	155,657	286,475
1803	Tulare-Buena Vista Lakes	10,501,344	493,058	4.7%	229,044	52,982	154,083
1804	San Joaquin	10,127,905	231,575	2.3%	75,488	59,843	75,879

HUC4 Number	*HUC4 Name*	*Total HUC4 Acres*	*Total Acres of HUC4 in the Affected Environment*	*Percent of Total HUC4 Acres in the Affected Environment*	*Acres Not Operable*	*Acres Operable*	*Acres Operable but Complex*
1806	Central California Coastal	8,472,485	408,848	4.8%	130,027	61,749	145,265
1807	Southern California Coastal	8,861,368	437,036	4.9%	141,749	76,724	187,761
1808	North Lahontan	2,913,621	36,488	1.3%	7,118	9,919	14,513
1809	Northern Mojave-Mono Lake	18,105,223	559,848	3.1%	225,083	25,081	32,199
1810	Southern Mojave-Salton Sea	10,934,158	23,140	0.2%	8,025	4,320	5,116
1901	Southeast Alaska	69,591,689	9,473,879	13.6%	4,626,703	357,067	589,753
1902	South Central Alaska	65,799,951	3,089,520	4.7%	942,450	36,276	61,108
1907	Upper Yukon River	76,785,649	1,897	0.002%	13	0	0
2101	Puerto Rico	3,373,641	12,607	0.4%	n/a	n/a	n/a

General Characteristics of the Affected Environment

Roadless Rule prohibitions do not mean no activities occur in Inventoried Roadless Areas. Activities that are ongoing and permissible, if not restricted by land management plans or other law, regulation, and policy include, but are not limited to:

- Access for reserved or outstanding rights/existing rights for minerals
- Continuation of existing land use authorizations (ex. utilities, recreation special uses, private access, easements, etc.)
- Road maintenance on existing roads
- Motorized and non-motorized trail construction or maintenance;
- Hunting, fishing, hiking, camping, and other recreational uses;
- Use of a motorized vehicle on a trail open to motorized use;
- Mountain biking on a trail open to mechanized use;
- Prescribed burning; and
- Livestock grazing.

The 2001 Roadless Rule prohibition on road construction and reconstruction in IRAs did not affect already existing access, occupancy, and use rights. These rights include those that exist by law, treaty, or by other authority. They include, but are not limited to, the right to provide reasonable access across NFS lands to private property, mining claims for locatable minerals under the 1872 Mining Law, and land uses protected by Native American treaty rights. These rights include, but are not limited to, rights of access provided in the ANILCA (Public Law 96-487), and recognized highway rights-of-way granted over NFS lands under Revised Statute 2477 (Public Law 94-579). The most common type of access pursued in conjunction with these two prominent statues is motorized road access. Land use authorizations and rights-of-way existing prior to the 2001 Roadless Rule are ongoing and part of baseline conditions.

While they are referred to as “roadless”, because of the decades between inventory and designation, there are thousands of miles of existing roads within lands designated under the 2001 Roadless Rule, including National Forest System roads as well as roads of other ownerships. Additionally decommissioned roads and unauthorized routes that are not part of the official system or records may exist.

Between 1984 and 2024, wildfire burned 20.5% of potentially affected IRAs (or 29.9% of potentially affected IRAs outside of Alaska). Of these, 4.2% (1.7 million acres) were high severity wildfire and 5.2% (2.1 million acres) were moderate severity wildfire. Nationally, wildfire hazards are commonly characterized using the Wildfire Hazard Potential (WHP) index, which integrates burn probability and potential fire intensity. Across potentially affected IRAs outside of Alaska, 42 percent of lands are classified as high or very high Wildfire Hazard

Potential, indicating a high potential for intense wildfire under severe weather conditions. Regional values range from 5 percent in the Eastern Region to 60 percent in the Pacific Southwest Region. Within potentially affected IRAs 9.8 million acres (24 percent) overlap with Healthy Forests Restoration Act Wildland Urban Interface. Within the potentially affected environment 5.9 million acres (14.8 percent) overlap with Community Wildfire Risk Reduction Zones (CWRRZ). Nearly all CWRRZ within the potentially affected environment occurs within the transition zone (within the 1.5-mile buffer surrounding building clusters).

2001 Roadless Area Conservation Rule Implementation

The 2001 Roadless Rule provided that the Responsible Official may grant certain limited exceptions to the road construction and timber harvest prohibitions, including for forest stewardship purposes, public health and safety, reserved or outstanding rights, existing or renewed mineral leases, and realignment of roads to prevent resource damage or address safety hazards (36 CFR 294.12(b) and 294.13(b) (2001)). It also provides exceptions to the timber harvest prohibitions for generally small diameter tree cutting to improve habitat for threatened, endangered, or sensitive species or to maintain or restore ecosystem characteristics and where it maintains or enhances one or more of the roadless characteristics, or when the activity is incidental to activities not otherwise prohibited (36 CFR 294.13(b) (2001)). The rule states that use of the timber harvest exceptions would be expected to be infrequent. Amid legal challenges to the 2001 Roadless Rule and to provide for consistent application, exceptions for specific activities in IRAs were reviewed and authorized by the Secretary and Forest Service Chief until 2012. In 2012, certain exceptions were delegated to the applicable regional forester. In 2018 all exceptions were delegated to the regional foresters, and some exceptions for public health and safety or small diameter tree cutting[21] may be further delegated to responsible officials. Approval of exceptions generally requires review documentation, including maps and justification.

Regional forester approval of exceptions under the 2001 Roadless Rule has been limited, and the majority have been for stewardship purposes. There is no national-level tracking of the review, approval, or implementation of 2001 Roadless Rule exceptions, though the Washington Office has recommended that Responsible Officials document reviews and approvals. Regional level tracking of review and approvals between 2018 and 2025 shows about 188 documented projects were granted exceptions. About 10 percent involved road related activities and 96 percent involved tree cutting (some involved both road and tree cutting activities). Since 2018, the majority were related to maintaining and restoring ecosystem characteristics. Eight exceptions approved tree cutting for the explicit purpose of "improving threatened, endangered, or sensitive species habitat," although some of the exceptions outside of this category may have served multiple purposes, such as providing benefits to species habitat. This information does not include responsible official reviews or approvals below the regional forester level[22], so it is not possible to list all exceptions that were granted with existing data sources.

The best available information on activities that have been implemented within IRAs over the last decade comes from the Forest Service Activity Tracking System (FACTS). This data shows

[21] Regional foresters and deputy regional foresters must retain sole authority for exceptions under §294.12(b)(3)-(7) and §294.13(b)(4) (2001).
[22] Similarly, the Agency does not track when requests for exceptions were abandoned or not approved.

an average of 500 acres per year of timber harvest and a total of 1.8 million acres of hazardous fuels reduction within IRAs managed under the 2001 Roadless Rule. More than half of the hazardous fuels reduction was from wildfire use, which is reported when acres burned by wildfire are determined to meet land management planning objectives in the post-fire assessment. An additional 25 percent of the hazardous fuels reduction was from prescribed burning activities for reducing fuel hazards and as resource management treatments. Best available records of permanent road construction indicate that approximately 60 miles of roads were built within inventoried roadless areas after 2013.

Use of the available exceptions has been limited in scope and inconsistent. This is due in part to rigorous project-level analysis requirements and associated reviews and approvals needed to demonstrate consistency with Roadless Rule requirements, as well the rule's intent for infrequent use.

STATUS OF SPECIES AND CRITICAL HABITAT

This section defines the biological requirements of each ESA listed species considered in this consultation (Table 7) that could be affected by future proposed actions within the Roadless Rule Rescission affected environment, and the status of each designated critical habitat (Table 8) relative to those requirements. Listed species facing a high risk of extinction and critical habitats with degraded conservation value are more vulnerable to the aggregation of effects considered under the environmental baseline, the effects by future proposed actions within the Roadless Rule Rescission affected environment. Maps showing designated critical habitat in relation to the Roadless Rule Rescission affected environment are in Appendix A.

Table 7. NMFS species listing classification and date from the federal register; recovery plan reference; most recent status review reference, including viability assessments; status summary; and limiting factors for each species considered in this opinion. All information is collected from federal register listings, species recovery plans, 5-year reviews and viability analyses. ESU = Evolutionarily Significant Unit; DPS = Distinct Population Segment; FR=Federal Register.

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
Chinook Salmon	California Coastal	Threatened 09/16/1999; 64 FR 50394 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	NMFS 2016g	NMFS 2024a	This ESU historically comprised of 38 populations which included 32 fall-run populations and 6 spring-run populations. All six of the spring-run populations were classified as functionally independent but have apparently been extirpated. Of the 32 fall-run populations, 15 populations were considered either functionally independent or potentially independent, while the remaining populations were classified as dependent populations. Low survival of juveniles in freshwater, in combination with poor ocean conditions, has led to the precipitous declines of Chinook salmon populations throughout the central and northern California coastal areas. Estimates indicate Chinook salmon have declined substantially in coastal populations of central and northern California over the past 70 years. It is difficult to characterize the status of this ESU based on the available	• Inadequate flow regime during migration • Reduced amount and complexity of estuarine and nearshore marine habitat • Predation • Degraded water quality • Disease • Loss of water quantity • Lack of floodplain connectivity • Blocked/impaired fish passage for outmigrants • Lack of water flow in estuary during migration • Inadequate instream water quantity in lower watershed • Degraded estuarine conditions and loss of historic estuarine habitat • Loss of outmigrant cover due to eelgrass wasting disease in Humboldt Bay • Reduced amount and complexity of freshwater habitat for migration, spawning, and rearing

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					data, but overall, did not find evidence of a substantial change in conditions since the last status reviews and maintained the ESU is threatened and likely to become endangered in the foreseeable future. NMFS estimates recovery of the California Coastal Chinook salmon ESU would take 50 to 100 years. Thirteen independent Essential populations attained low extinction risk criteria; three Supporting Independent populations attained moderate extinction risk criteria; one Supporting Dependent population contributed to redundancy and occupancy.	
Chinook Salmon	Central Valley Spring	Threatened 09/16/1999; 64 FR 50394 06/28/2005; 70 FR 37160	NMFS 2014b	NMFS 2016d	This ESU includes spring-run Chinook salmon populations spawning in the Sacramento River and its tributaries and spring-run Chinook salmon in the Feather River Hatchery. The San Joaquin River watershed and Delta are excluded as critical habitat and San Joaquin basin populations are considered extirpated. In 2014, Feather River Hatchery brood stock was used to actively reintroduce Central Valley spring-run Chinook salmon into the mainstem San Joaquin River as an ESA 10(j) experimental population. Several	• Present or threated destruction, modification, or curtailment of habitat or range • Overutilization for commercial, recreational, scientific, or educational purposes • Disease or predation • Inadequacy of existing regulatory mechanisms • Other natural or manmade factors • Hatchery-related effects

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					juveniles successfully survived to adulthood and returned to spawn in 2019. The status of this ESU in 2016 was improved compared to the previous assessment in 2010. This improved status was attributed to extensive habitat restoration and improved spatial structure, with historically extirpated populations (Battle and Clear Creeks) trending in a positive direction. However, recent declines of many of the dependent populations, high pre-spawn and egg mortality during the 2012 to 2016 drought, and uncertain juvenile survival during the drought likely increased the ESU's overall extinction risk. Monitoring data showed sharp declines in adult returns from 2014 through 2020. The most recent viability assessment concluded that the viability of the ESU has declined since the 2015 assessment with an increased risk of extinction for all independent populations, and overall, the species is at moderate to high risk of extinction. In fact, Mill, Deer, and Battle creeks changed from low/moderate to a high risk of extinction using one or more viability	

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					criteria. The total abundance of this ESU for the Sacramento River watershed in 2019 was approximately half of the population size in 2014, and close to the decadal lows which occurred as recently as the last two years. The Central Valley-wide abundance was driven largely by the annual variation in returns to Butte Creek. The Butte Creek population remains at low extinction risk, yet all viability metrics for the ESU (except hatchery influence) are trending in a negative direction relative to 2015.	
Chinook Salmon	Lower Columbia River	Threatened 03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	NMFS 2013a	NMFS 2022b; Ford 2022	This ESU comprises 32 independent populations. Relative to baseline VSP levels identified in the recovery plan, there has been an overall improvement in the status of a number of fall-run populations although most are still far from the recovery plan goals; Spring-run Chinook salmon populations in this ESU are generally unchanged; most of the populations are at a "high" or "very high" risk due to low abundances and the high proportion of hatchery-origin fish spawning naturally. Many of the populations in this ESU remain at "high risk," with low natural-origin abundance levels. Overall, the viability of the Lower Columbia River Chinook salmon	• Reduced access to spawning and rearing habitat • Hatchery-related effects • Harvest-related effects on fall Chinook salmon • An altered flow regime and Columbia River plume • Reduced access to off-channel rearing habitat • Reduced productivity resulting from sediment and nutrient-related changes in the estuary • Contaminant

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					ESU has increased somewhat since 2016, although the ESU remains at "moderate" risk of extinction.	
Chinook Salmon	Puget Sound	Threatened 03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	Shared Strategy for Puget Sound 2007 Ruckelshaus et al. 2006	NMFS 2016b; Ford 2022	This ESU comprises 22 populations distributed over five geographic areas. All population in this ESU continue to remain well below the Technical Recovery Team planning ranges for recovery escapement levels. Most of the populations remain consistently below the spawner–recruit levels identified by the Technical Recovery Team as necessary for recovery. Across the ESU, most populations have increased somewhat in abundance since the last status review in 2016 but have small negative trends over the past 15 years. Productivity remains low in most populations. This ESU remains at "moderate" risk of extinction.	• Degraded floodplain and in river channel structure • Degraded estuarine conditions and loss of estuarine habitat • Degraded riparian areas and loss of in-river large woody debris • Excessive fine-grained sediment in spawning gravel • Degraded water quality and temperature • Degraded nearshore conditions • Impaired passage for migrating fish • Severely altered flow regime
Chinook Salmon	Sacramento River Winter	Endangered 08/04/1989; 54 FR 32085 11/05/1990; 55 FR 46515 01/04/1994; 59 FR 440	NMFS 2014b	NMFS 2024d	This ESU consists of one population and has been completely displaced from its historical spawning habitat by the construction of Shasta and Keswick dams. Approximately 300 miles of tributary spawning habitat in the upper Sacramento River is now inaccessible to the ESU. Most components of this ESUs life history (for example, spawning, incubation, freshwater rearing) have been compromised by the habitat blockage	• Lack of access to historic spawning and rearing habitats due to dams (Shasta and Keswick dams). • Adverse effects related to dams and water diversions. • Barriers to migrations and spawning habitat. • Degraded freshwater habitat. • Elevated temperatures below Shasta and Keswick dams. • Highly variable water flows limiting downstream transport of juveniles

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					in the upper Sacramento River. The remaining spawning habitat in the upper Sacramento River is artificially maintained by cool water releases from Shasta and Keswick dams, and the spatial distribution of spawners is largely governed by the water year type and the ability of the Central Valley Project to manage water temperatures in the upper Sacramento River. This ESU includes winter-run Chinook salmon spawning in the mainstem Sacramento River downstream of Keswick Dam and the Livingston Stone National Fish Hatchery supplementation and captive broodstock programs (85 FR 81822). Within the established ESU delineations, new efforts were initiated in 2017 to establish a viable, self-sustaining, and locally adapted population in Battle Creek to add to the spatial diversity (such as spatial structure) and abundance of this ESU. The 2016 viability assessment concluded the extinction risk of this ESU is high. Several listing factors contributed to NMFS' recommendation that the species remain listed as endangered,	• Altered flow regime. • Degraded floodplain connectivity and function due to artificial hydrograph from intense water management. • Degraded estuarine and nearshore marine habitat in south Delta. • Altered flow regime in south Delta prohibiting successful rearing and migration. • Mortality due to entrainment within rearing and migratory corridor habitat due to export facilities operations in South Delta Water diversions are not properly screened along migration corridor entraining migrants. • Water diversions are not properly screened along migration corridors entraining migrants. • Contaminants. • Exposure to toxic chemicals. • Harmful algal blooms related to nutrient loading upstream.

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					including drought, poor ocean conditions, and hatchery influence. Until additional populations are established, this ESU will remain in the "High" biological extinction risk category. The overall viability of this ESU has continued to decline since the 2015 viability assessment, with the single spawning population on the mainstem Sacramento River now at high risk of extinction due to an increase in extinction risk from hatchery influence.	
Chinook Salmon	Snake River Fall	Threatened 04/22/1992; 57 FR 14653 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	NMFS 2017b	NMFS 2022f; Ford 2022	This ESU has one extant population The single extant population in the ESU is currently meeting the criteria for a rating of "viable" developed by the Interior Columbia Technical Recovery Team, but the ESU as a whole is not meeting the recovery goals described in the recovery plan for the species, which require the single population to be "highly viable with high certainty" and/or will require reintroduction of a viable population above the Hells Canyon Complex. This ESU is at a moderate- to- low risk of extinction.	• Degraded floodplain connectivity and function • Harvest-related effects • Loss of access to historical habitat above Hells Canyon and other Snake River dams • Impacts from mainstem Columbia River and Snake River hydropower systems • Hatchery-related effects • Degraded estuarine and nearshore habitat.
Chinook Salmon	Snake River Spring/Summer	Threatened 04/22/1992; 57 FR 14653 06/28/2005;	NMFS 2017c	NMFS 2022g; Ford 2022	This ESU comprises 28 extant and four extirpated populations. There have been improvements in abundance/productivity in several	• Degraded freshwater habitat • Effects related to the hydropower system in the mainstem Columbia River,

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
		70 FR 37160 04/14/2014; 79 FR 20802			populations relative to the time of listing, but most populations experienced sharp declines in abundance in the recent five-year period Overall, this ESU viability continues to be at moderate-to-high risk.	• Altered flows and degraded water quality • Harvest-related effects • Predation
Chinook Salmon	Upper Columbia River Spring	Endangered 03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	Upper Columbia Salmon Recovery Board 2007	NMFS 2022h; Ford 2022	This ESU comprises four independent populations. Current estimates of natural-origin spawner abundance decreased substantially relative to the levels observed in the prior review for all three extant populations. Productivities also continued to be very low, and both abundance and productivity remained well below the viable thresholds called for in the Upper Columbia Salmon Recovery Plan for all three populations. Based on the information available for this review, this ESU remains at high risk, with viability largely unchanged since 2016.	• Effects related to hydropower system in the mainstem Columbia River • Degraded freshwater habitat • Degraded estuarine and nearshore marine habitat • Hatchery-related effects • Persistence of non-native (exotic) fish species • Harvest in Columbia River fisheries
Chinook Salmon	Upper Willamette River	Threatened 03/24/1999; 64 FR 14308 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	ODFW & NMFS 2011	NMFS 2024e; Ford 2022	This ESU comprises seven populations. Abundance levels for all but Clackamas River DIP remain well below their recovery goals. Overall, there has likely been a declining trend in the viability of this ESU since the last review. The magnitude of this change is not sufficient to suggest a change in risk category, so this ESU	• Degraded freshwater habitat • Degraded water quality • Increased disease incidence • Altered stream flows • Reduced access to spawning and rearing habitats • Altered food web due to reduced inputs of microdetritus • Predation by native and non-native species, including hatchery fish

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					remains at "moderate" risk of extinction.	• Competition related to introduced salmon and steelhead • Altered population traits due to fisheries and bycatch
Chum Salmon	Columbia River	Threatened 03/24/1999; 64 FR 14508 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	NMFS 2013a	NMFS 2022b; Ford 2022	This ESU has 17 populations divided into three MPGs. Three population groups exceed the recovery goals established in the recovery plan. The remaining population groups have unknown abundances. Abundances for these population groups are assumed to be at or near zero. The viability of this ESU is relatively unchanged since the last review (moderate to high risk), and the improvements in some populations do not warrant a change in risk category, especially given the uncertainty regarding climatic effects in the near future.	• Degraded estuarine and nearshore marine habitat • Degraded freshwater habitat • Degraded stream flow because of hydropower and water supply operations • Reduced water quality • Current or potential predation • An altered flow regime and Columbia River plume • Reduced access to off-channel rearing habitat in the lower Columbia River • Reduced productivity resulting from sediment and nutrient-related changes in the estuary • Juvenile fish wake strandings • Contaminants
Chum Salmon	Hood Canal Summer Run	Threatened 03/24/1999; 64 FR 14508 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	Hood Canal Coordinating Council 2005; NMFS 2007b	NMFS 2016b; Ford 2022	The Puget Sound Technical Recovery Team identified two independent populations for Hood Canal summer chum, one which includes the spawning aggregations from rivers and creeks draining into the Strait of Juan de Fuca, and one which includes spawning aggregations within Hood Canal proper. Natural-origin spawner abundance has increased since ESA listing, and spawning abundance	• Reduced floodplain connectivity and function • Poor riparian condition • Loss of channel complexity • Sediment accumulation • Altered flows and water quality

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					targets in both populations have been met in some years. Productivity increased at the time of the last review but has been down for the last three years for the Hood Canal population, and for the last four years for the Strait of Juan de Fuca population. Productivity of individual spawning aggregates shows that only two of eight aggregates have viable performance. Spatial structure and diversity viability parameters, as originally determined by the Puget Sound Technical Recovery Team, have improved, and nearly meet the viability criteria for both populations. Despite substantive gains toward meeting viability criteria in the Strait of Juan de Fuca and Hood Canal summer chum salmon populations, the ESU still does not meet all the recovery criteria for population viability currently. Overall, this ESU remains at "moderate" risk of extinction.	
Coho Salmon	Central California Coast	Endangered 10/31/1996; 61 FR 56138 06/28/2005; 70 FR 37159 04/14/2014; 79 FR 20802	NMFS 2012a	NMFS 2023a	Historically, this ESU was comprised of 75 coho salmon populations, 11 functionally independent, one potentially independent and 64 dependent populations. Dependent populations needed immigration from other nearby populations to ensure their long-term survival. Eleven	• Lack of in-stream habitat complexity • Lack of high flow refugia habitat • Reduced access to off-channel rearing habitat • Degraded floodplain connectivity and function • Degraded estuarine conditions and

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					functionally independent populations and one potentially independent population of Central California Coast coho salmon once existed. Most of the populations in the Central California Coast coho salmon ESU are currently not viable, hampered by low abundance, range constriction, fragmentation, and loss of genetic diversity. None of the five Central California Coast coho salmon diversity strata currently support viable coho salmon populations. According to the 2016 viability report, surveys suggested Central California Coast coho salmon abundance had improved slightly since 2011 within several independent populations (including Lagunitas Creek), although all populations remained well below their high-risk dispensation thresholds. The Russian River and Lagunitas Creek populations are relative strongholds for the species compared to other Central California Coast coho salmon populations, the former predominantly due to out-planting of hatchery-reared juvenile fish. The 2016 viability assessment determined the overall risk of Central California Coast coho salmon extinction remained high, and the 2016 status	loss of estuarine habitat • Predation • Loss of freshwater channel complexity • Sediment accumulation • Reduced water quality • Increased stream temperatures

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					review recommended the ESU remain listed as an endangered species. Overall, new information available since 2016 indicates the extinction risk for Central California Coast coho salmon has not changed appreciably and remains high, with slight improvements in the two northernmost diversity strata, but little change in the Coastal Diversity Stratum and perhaps worsening conditions in the Santa Cruz Mountain Stratum. In its most recent status review, NMFS recommended that the Central California Coast coho salmon ESU remain listed as an endangered species.	
Coho Salmon	Lower Columbia River	Threatened 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	NMFS 2013a	NMFS 2022b; Ford 2022	Of the 24 populations that make up this ESU only six of the 23 populations for which we have data appear to be above their recovery goals. Overall abundance trends for the Lower Columbia River coho salmon ESU are generally negative. Natural spawner and total abundances have decreased in almost all DIPs, and Coastal and Gorge MPG populations are all at low levels, with significant numbers of hatchery-origin coho salmon on the spawning grounds. Improvements in spatial structure and diversity have been slight and overshadowed by declines	• Degraded estuarine and near-shore marine habitat • Fish passage barriers • Degraded freshwater habitat: • Hatchery-related effects • Harvest-related effects • An altered flow regime and Columbia River plume • Reduced access to off-channel rearing habitat in the lower Columbia River • Reduced productivity resulting from sediment and nutrient-related changes in the estuary • Juvenile fish wake strandings • Contaminants

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					in abundance and productivity. For individual populations, the risk of extinction spans the full range, from "low" to "very high." Overall, the Lower Columbia River coho salmon ESU remains at "moderate" risk, and viability is largely unchanged since 2016.	
Coho Salmon	Oregon Coast	Threatened 08/10/1998; 63 FR 42587 06/20/2011; 76 FR 35755 04/14/2014; 79 FR 20802	NMFS 2016h	NMFS 2022d; Ford 2022	This ESU comprises 56 populations including 21 independent and 35 dependent populations. The biological status of the ESU has decreased slightly since the 2015 review (high certainty of persistence, moderate certainty of sustainability), however, current ESU scores improved relative to the 2012 assessment (moderate certainty of persistence, low-to-moderate certainty of sustainability). The projected stream temperature and flow assessment indicated that Oregon Coast coho salmon will likely be negatively affected by increased temperatures by the end of the century at all stages of the life cycle. Overall, the Oregon Coast coho salmon ESU is therefore at "moderate-to-low" risk of extinction.	• Reduced amount and complexity of habitat including connected floodplain habitat • Degraded water quality • Blocked/impaired fish passage • Inadequate long-term habitat protection • Changes in ocean conditions

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
Coho Salmon	Southern Oregon/Northern California Coast	Threatened 05/06/1997; 62 FR 24588 06/28/2005; 70 FR 37160 04/14/2014; 79 FR 20802	NMFS 2014a	NMFS 2016c	This ESU comprises 31 independent, 9 independent, and 5 ephemeral populations all grouped into 7 diversity strata. Of the 31 independent populations, 24 are at high risk of extinction and 6 are at moderate risk of extinction. The extinction risk of an ESU depends upon the extinction risk of its constituent independent populations; because the population abundance of most independent populations is below their depensation threshold, the Southern Oregon/Northern California Coast coho salmon ESU is at high risk of extinction and is not viable.	• Lack of floodplain and channel structure • Impaired water quality • Altered hydrologic function • Impaired estuary/mainstem function • Degraded riparian forest conditions • Altered sediment supply • Increased disease/predation/competition • Barriers to migration • Fishery-related effects • Hatchery-related effects
Sockeye Salmon	Snake River	Threatened 11/20/1991; 56 FR 58619 04/14/2014; 79 FR 20802 06/28/2005; 70 FR 37160	NMFS 2015	NMFS 2022e; Ford 2022	This single population ESU is at remains at "extremely high risk", although there has been substantial progress on the first phase of the proposed recovery approach—developing a hatchery-based program to amplify and conserve the stock to facilitate reintroductions. Current environmental variation long-term temperature modeling supports the "extremely high risk" rating with the potential for extirpation in the near future. The viability of the Snake River sockeye salmon ESU therefore has likely declined since the time of the prior review, and the extinction risk category remains "high."	• Effects related to the hydropower system in the mainstem Columbia River • Reduced water quality and elevated temperatures in the Salmon River • Water quantity • Predation

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
Steelhead	California Central Valley	Threatened 03/19/1998; 63 FR 13347 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2014b	NMFS 2024b	This DPS includes steelhead populations spawning in the Sacramento and San Joaquin rivers and their tributaries. Populations upstream of migration barriers remain excluded from this DPS. Hatchery stocks within the DPS include Coleman National Fish Hatchery, Feather River Hatchery, and Mokelumne River Hatchery. In its most recent five-year review, NMFS determined the listing status of California Central Valley steelhead had not improved since the previous five-year review and recommended that the DPS remain listed as a threatened species. Most natural-origin California Central Valley populations are very small, are not monitored, and may lack the resiliency to persist for protracted periods if subjected to additional stressors, particularly widespread stressors such as climate change. The genetic diversity of California Central Valley steelhead has likely been impacted by low population sizes and high numbers of hatchery fish relative to natural-origin fish. The life-history diversity of the DPS is mostly unknown, as very few studies have been published on traits such as age	• Reduced access to spawning and rearing habitats due to impaired passage at dams (Shasta and Keswick dams). • Adverse effects related to dams and water diversions. • Barriers to migrations and spawning habitat. • Altered flow regime • Antiquated fish passage structures and screens • Degraded river channels • Reduction of spawning gravels • Loss of historic floodplain habitat • Loss of riparian functions • Loss of habitat complexity along corridors for rearing and migration • Wildfire intensity and effects

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					structure, size at age, or growth rates in California Central Valley steelhead. While updated data on steelhead in the American River is mostly based on hatchery returns, natural spawning populations within the Sacramento River tributaries had fluctuated, but showed a steady decline from 2008 to 2018. The most recent viability assessment determined that 11 of the 16 populations for which there are data available are at high risk of extinction based on abundance and/or hatchery influence, no populations are at low risk of extinction, and the overall viability of the ESU has not changed since 2016. Therefore, the species remains at moderate risk of extinction.	
Steelhead	Central California Coast	Threatened 08/18/1997; 62 FR 43937 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2016g	NMFS 2016a	This DPS historically consisted of five Diversity Strata with 38 independent populations of winter-run steelhead (12 functionally independent and 26 potentially independent) and 22 dependent populations. NMFS estimates recovery of this DPS would take 50 to 100 years. The biological recovery criteria for these populations show a low to moderate risk of extinction.	• Contaminants • Degraded water quality and quantity • Urbanization • Continued destruction and modification of habitat • Adverse effects related to dams and water diversions. • Degradation and loss of historic rearing habitat in estuary and wetlands • Predation by native and non-native species

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					A recent viability assessment concluded that populations draining to San Francisco Bay are "highly unlikely to be viable". Data from 2008-2010 showed a decline in returning adults compared to the previous decade, and the most recent status update finds that Central California Coast steelhead remain "likely to become endangered in the foreseeable future".	• Inadequate existing regulations for water quality • Increased water temperatures • Disrupted food web process due to non-native species
Steelhead	Lower Columbia River	Threatened 03/19/1998; 63 FR 13347 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2013a	NMFS 2022b; Ford 2022	This DPS comprises 23 historical populations, 17 winter-run populations and 6 summer-run populations. Ten populations are nominally at or above the goals set in the recovery plan; however, it should be noted that many of these abundance estimates do not distinguish between natural- and hatchery origin spawners. The majority of winter-run steelhead populations in this DPS continue to persist at low abundance levels (hundreds of fish), except for the Clackamas and Sandy River DIPs, which have abundances in the low 1,000s. Although the five-year geometric abundance means are near recovery plan goals for many populations, the recent trends are negative. Overall, the Lower Columbia River steelhead DPS is	• Degraded estuarine and nearshore marine habitats • Degraded freshwater habitat • Reduced access to spawning and rearing habitats • Avian and marine mammal predation • Hatchery-related effects • An altered flow regime and Columbia River plume • Reduced access to off-channel rearing habitat in the lower Columbia River • Reduced productivity resulting from sediment and nutrient-related changes in the estuary • Juvenile fish wake strandings • Contaminants

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					therefore considered to be at "moderate" risk.	
Steelhead	Middle Columbia River	Threatened 06/07/2000; 65 FR 36074 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2009b	NMFS 2022c; Ford 2022	This DPS comprises 17 extant populations. Recent (five-year) returns are declining across all populations, the declines are from relatively high returns in the previous five-to-ten-year interval, so the longer-term risk metrics that are meant to buffer against short-period changes in abundance and productivity remain unchanged. The Middle Columbia River steelhead DPS does not currently meet the viability criteria described in the Middle Columbia River steelhead recovery plan.	• Degraded freshwater habitat • Mainstem Columbia River hydropower-related impacts • Degraded estuarine and nearshore marine habitat • Hatchery-related effects • Harvest-related effects • Effects of predation, competition, and disease
Steelhead	Northern California	Threatened 03/25/1999; 64 FR 14517 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2016g	NMFS 2024c	This DPS historically consisted of 41 independent populations of winter-run steelhead (19 functionally independent and 22 potentially independent) and 10 populations of summer steelhead (all functionally independent). Overall, the available data for winter-run populations indicate all essential populations remain well below recovery targets. The viability of winter-run populations has not changed appreciably since the previous viability assessment. Summer-run populations remain a significant concern as they are well below	• Inadequate flow regime during migration and rearing • Degraded water quality • Loss of juvenile food supply • Lack of in-stream habitat complexity • Reduced amount and complexity of estuarine and nearshore marine habitat • Degraded water quality • Contaminants

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					recovery targets as some of the population are very small and little is known about other summer populations. NMFS estimates recovery of the Northern California steelhead DPS would take 50 to 100 years. The biological recovery criteria for these populations show a low to moderate risk of extinction.	
Steelhead	Puget Sound	Threatened 05/11/2007; 72 FR 26722 04/14/2014; 79 FR 20802	NMFS 2019	NMFS 2016b; Ford 2022; Hard et al. 2015	This DPS comprises 32 populations. Viability has improved somewhat since the Puget Sound Technical Recovery Team concluded that the DPS was at a very low level of viability, as were all three of its constituent Major Population Groups, and many of its 32 populations. Increases in spawner abundance were observed in a number of populations over the last five years within the Central & South Puget Sound and the Hood Canal & Strait of Juan de Fuca MPGs, primarily among smaller populations. There were also declines for summer- and winter-run populations in the Snohomish River basin. In fact, all summer- run steelhead populations in the Northern Cascades MPG are likely at a very high demographic risk.	• Continued destruction and modification of habitat • Widespread declines in adult abundance despite significant reductions in harvest • Threats to diversity posed by use of two hatchery steelhead stocks • Declining diversity in the DPS, including the uncertain but weak status of summer-run fish • A reduction in spatial structure • Reduced habitat quality • Urbanization • Dikes, hardening of banks with riprap, and channelization

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
Steelhead	Snake River	Threatened 08/18/1997; 62 FR 43937 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2017c	NMFS 2022e; Ford 2022	This DPS comprises 24 populations. Based on the updated viability information available for this review, all five Major Population Groups are not meeting the specific objectives in the draft recovery plan, and the viability of many individual populations remains uncertain. Of note, the updated, population-level abundance estimates have made very clear the recent (last five years) sharp declines that are extremely worrisome, were they to continue.	• Adverse effects related to the mainstem Columbia River hydropower system • Impaired tributary fish passage • Degraded freshwater habitat • Increased water temperature • Harvest-related effects, particularly for B-run steelhead • Predation • Genetic diversity effects from out-of-population hatchery releases
Steelhead	South-Central California Coast	Threatened 08/18/1997; 62 FR 43937 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	NMFS 2013b	NMFS 2023b	This DPS is composed of four Biogeographic Population Groups each group contains between 1 and 15 populations. Single Biogeographic Population Groups encompass multiple watersheds and multiple *O. mykiss* populations. However, many watersheds are relatively small and may be capable of supporting only small steelhead runs. This DPS includes all naturally spawned anadromous *O. mykiss* (steelhead) originating below natural and manmade impassable barriers from the Pajaro River to (but not including) the Santa Maria River. Recent drought conditions have reduced the successful recruitment of juvenile steelhead and size of adult	• Impaired fish passage in mainstems of rivers and tributaries. • Altered flow regime due to dams, water diversions, and groundwater extraction • Predation • Disease • Degraded quantity of estuary habitat for rearing and migration • Non-native species • Prolonged drought • Increased wildfire frequency and duration

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					runs of steelhead in all the core recovery populations within the South-Central California Steelhead Recovery Planning Area. The systemic anthropogenic threats identified at the time of the initial listing have remained essentially unchanged over the past 5 years, though there has been significant progress in removing small-scale fish passage barriers in a number of the core recovery watersheds. Threats to the South-Central California Coast Steelhead DPS posed by environmental variability resulting from projected climate change are likely to increase, further threatening the threatened South-Central California Coast Steelhead DPS. The most recent status review concludes that this DPS shows no appreciable change in viability metrics since the last status review and remains listed as threatened.	
Steelhead	Southern California	Endangered 08/18/1997; 62 FR 43937 05/01/2002; 62 FR 21586 01/05/2006; 71 FR 834	NMFS 2012b	NMFS 2023c	This DPS is composed of five Biogeographic Population Groups, each group contains between three and 29 populations. Recent monitoring shows persistent negative trends in adult abundance for most key populations, some population	• Urbanization • Habitat fragmentation • Decreased water quantity • Altered flow regime • Lack of access to historic upstream habitat due to dams and other barriers • Degrade habitat conditions due to

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
		04/14/2014; 79 FR 20802			experiencing a statistically significant decline. Densities of juvenile fish also plummeted during the recent multi-year drought. The extended drought, coupled with extensive wildfires, has exacerbated these declines, causing the adult steelhead life-history to have nearly disappeared and creating a high and increasing risk of its permanent loss. This DPS is at a high risk of extinction, driven by critically low abundance and productivity relative to viability objectives, a loss of native genetic diversity in its southernmost range, and the lack of access to critical upstream drought refugia, most of which are blocked by impassable dams.	prolonged droughts and extensive wildfires • Poor habitat conditions cause increase in disease • Predation • Degraded floodplain • Legacy effects due to non-native trout stocking
Steelhead	Upper Columbia River	Threatened 08/18/1997; 62 FR 43937 (E) 01/05/2006; 71 FR 834 (T) 08/24/2009; 74 FR 42605 04/14/2014; 79 FR 20802	Upper Columbia Salmon Recovery Board 2007	NMFS 2022h; Ford 2022	This DPS comprises four independent populations. The most recent estimates (five-year geometric mean) of total and natural-origin spawner abundance have declined since the last report, largely erasing gains observed over the past two decades for all four populations. Recent declines are persistent and large enough to result in small, but negative 15-year trends in abundance for all four populations. The overall viability of this DPS remains largely	• Adverse effects related to the mainstem Columbia River hydropower system • Impaired tributary fish passage • Degraded floodplain connectivity and function, channel structure and complexity, riparian areas, large woody debris recruitment, stream flow, and water quality • Hatchery-related effects • Predation and competition • Harvest-related effects

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					unchanged from the prior review, and the DPS is at high risk driven by low abundance and productivity relative to viability objectives and diversity concerns.	
Steelhead	Upper Willamette River	Threatened 03/25/1999; 64 FR 14517 01/05/2006; 71 FR 834 04/14/2014; 79 FR 20802	ODFW & NMFS 2011	NMFS 2024e; Ford 2022	This DPS has four demographically independent populations. Populations in this DPS have experienced long-term declines in spawner abundance. Although the recent magnitude of these declines is relatively moderate, continued declines would be a cause for concern. In the absence of substantial changes in accessibility to high-quality habitat, the DPS will remain at "moderate-to-high" risk. Overall, this DPS is therefore at "moderate-to-high" risk, with a declining viability trend.	• Degraded freshwater habitat • Degraded water quality • Increased disease incidence • Altered stream flows • Reduced access to spawning and rearing habitats due to impaired passage at dams • Altered food web due to changes in inputs of microdetritus • Predation by native and non-native species, including hatchery fish and pinnipeds • Competition related to introduced salmon and steelhead • Altered population traits due to interbreeding with hatchery origin fish
Green Sturgeon	Southern	Threatened 04/07/2006; 71 FR 17757	NMFS 2018	NMFS 2021c	The Sacramento River contains the only known green sturgeon spawning population in the Southern DPS. The current estimate of spawning adult abundance is between 824 - 1,872 individuals. Telemetry data and genetic analyses suggest that Southern DPS green sturgeon generally occur from Graves Harbor, Alaska to Monterey Bay, California and, within this range, most	• Reduction of its spawning area to a single known population • Lack of water quantity • Poor water quality • Poaching

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					frequently occur in coastal waters of Washington, Oregon, and Vancouver Island and near San Francisco and Monterey bays. Within the nearshore marine environment, tagging and fisheries data indicate that Northern and Southern DPS green sturgeon prefer marine waters of less than a depth of 110 meters.	
Eulachon	Southern	Threatened 03/18/2010; 75 FR 13012	NMFS 2017a	NMFS 2022a	This DPS includes all naturally spawned populations of eulachon that occur in rivers south of the Nass River in British Columbia to the Mad River in California. Sub populations for this species include the Fraser River, Columbia River, British Columbia, and the Klamath River. In the early 1990s, there was an abrupt decline in the abundance of eulachon returning to the Columbia River. Despite a brief period of improved returns in 2001-2003, the returns and associated commercial landings eventually declined to the low levels observed in the mid-1990s. Although eulachon abundance in monitored rivers has generally improved, especially in the 2013-2015 return years, recent poor ocean conditions and the likelihood that these conditions will persist into the near future suggest that population	• Changes in ocean conditions due to long term warming trends, particularly in the southern portion of the species' range where ocean warming trends may be the most pronounced and may alter prey, spawning, and rearing success. • Projected long-term temperature increases to freshwater habitats • Bycatch of eulachon in commercial fisheries • Adverse effects related to dams and water diversions • Poor water quality • Shoreline construction • Over harvest • Predation

Species	ESU/DPS	ESA Listing status, date, and FR notice of listing status	Recovery Plan Reference	Most Recent Status Review	Status Summary (From Listing Federal Registers, Recovery Plan, and Most Recent Status Review)	Limiting Factors
					declines may be widespread in the upcoming return years.	
Killer Whale	Southern Resident	Endangered 11/18/2005; 70 FR 69903 04/14/2014; 79 FR 20802	NMFS 2008	NMFS 2021b	This DPS is composed of a single population that ranges as far south as central California and as far north as southeast Alaska. While some of the downlisting and delisting criteria have been met, the biological downlisting and delisting of 63 criteria, including sustained growth over 14 and 28 years, respectively, have not been met. The Southern Resident killer whale DPS has not grown in numbers; the overall status of the population is not consistent with a healthy, recovered population. Considering the status and continuing threats, this DPS remains in danger of extinction.	• Quantity and quality of prey • Exposure to toxic chemicals • Disturbance from sound and vessels • Risk from oil spills

Designated Critical Habitat

This section describes the status of designated critical habitats that may be affected by the proposed action by examining the condition and trends of the essential physical and biological features (PBFs) of that habitat throughout the designated areas (Table 8). These features are essential to the conservation of the ESA-listed species because they support one or more of the species' life stages (for example, sites that support spawning, rearing, migration and foraging).

Table 8. NMFS species specific critical habitat status summaries collected from federal registers, recovery plans, and 5-year reviews. All information is collected from federal register listings, species recovery plans, 5-year reviews and viability analyses cited in Table 7. ESU = Evolutionarily Significant Unit; DPS = Distinct Population Segment; HSA=Hydrologic Sensitive Areas; HUC=Hydrologic Unit Code.

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
Chinook salmon	*Oncorhynchus tshawytscha*	California Coastal	09/02/2005 70 FR 52488	There are 45 occupied HSA watersheds within the freshwater and estuarine range of this ESU. Eight watersheds received a low rating, 10 received a medium rating, and 27 received a high rating of conservation value to the ESU. Two estuarine habitat areas used for rearing and migration (Humboldt Bay and the Eel River Estuary) also received a high conservation value rating. HSA watershed habitat areas for this ESU include approximately 1,634 mi (2,614 km) of stream habitat and approximately 25 mi^2 (65 km^2) of estuarine habitat (principally Humboldt Bay). Within these HSA watersheds, 1,475 mi (2,360 km) of streams and 25 mi2 (65 km2) of estuarine habitat were designated as critical habitat.
Chinook salmon	*Oncorhynchus tshawytscha*	Central Valley Spring	09/02/2005 70 FR 52488	There are 37 occupied watersheds within the freshwater and estuarine range of this ESU. Seven watersheds received a low rating, three received a medium rating, and 27 received a high rating of conservation value to the ESU. Four of these HSA watersheds comprise portions of the San Francisco-San Pablo-Suisun Bay estuarine complex which provides rearing and migratory habitat for this ESU. HSA Watershed habitat areas for this ESU include approximately 1,373 mi (2,197 km) of occupied stream habitat and approximately 427 mi^2 (1,102 km^2) of estuarine habitat in the San Francisco-San Pablo-Suisun Bay complex. Within these HSA watersheds, 1,158 mi (1,853 km) of streams and 254 mi^2 of estuary habitat were designated as critical habitat.
Chinook salmon	*Oncorhynchus tshawytscha*	Lower Columbia River	09/02/2005 70 FR 52630	Critical habitat for this ESU encompasses 10 subbasins in Oregon and Washington containing 47 occupied watersheds, as well as the lower Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some, or high potential for improvement. Conservation value was rated for HUC5 watersheds and found 30 watersheds rated high, 13 watersheds rated medium, and four watersheds rated low.
Chinook salmon	*Oncorhynchus tshawytscha*	Puget Sound	09/02/2005 70 FR 52630	Critical habitat for this ESU includes 1,683 mi of streams, 41 mi^2 of lakes, and 2,182 mi of nearshore marine habitat in Puget Sounds. This ESU has 61 freshwater and 19 marine areas within its range. Of the freshwater watersheds, 41 are rated high conservation value, 12 are rated low conservation value, and eight received a medium rating. All 19 marine areas are ranked with high conservation value.
Chinook salmon	*Oncorhynchus tshawytscha*	Sacramento River Winter	06/16/1993 58 FR 33212	Critical habitat for this ESU includes areas that are currently used by this ESU including the Sacramento River, all waterways and bays westward of Chipps Island to San Francisco Bay, and San Francisco Bay. Specific critical habitat includes: 1) the Sacramento River from Keswick Dam, Shasta County (River Mile 302) to Chipps Island (River Mile 0) at the westward margin of the Sacramento-San Joaquin Delta; 2) all waters from Chipps Island westward to Carquinez Bridge, including Honker Bay, Grizzly Bay, Suisun Bay, and Carquinex Strait; 3) all waters of San Pablo Bay westward of the Carquinez Bridge; and 4) all waters of San Francisco Bay (north of the San Francisco/Oakland Bay Bridge) from San Pablo Bay to the Golden Gate Bridge and north of the

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
				San Francisco-Oakland Bay Bridge. Within the Sacramento River, this CH designation includes the river water, river bottom (including those areas and associated gravel used by winter-run chinook salmon as spawning substrate), and adjacent riparian zone used by fry and juveniles for rearing. Critical Habitat is also included in the areas westward from Sherman Island to Chipps Island, it includes Kimball Island, Winter Island, and Browns Island. In the areas westward from Chipps Island, including San Francisco Bay to the Golden Gate Bridge, CH includes the estuarine water column and essential foraging habitat and food resources used by this ESU as part of their juvenile outmigration or adult spawning migration. This CH designation does not include any estuarine sloughs within San Francisco Bay or San Pablo Bay.
Chinook salmon	*Oncorhynchus tshawytscha*	Snake River Fall	12/28/1993 58 FR 68543	Critical habitat for this ESU consists of river reaches of the Columbia, Snake, and Salmon rivers, and all tributaries of the Snake and Salmon rivers presently or historically accessible to this ESU (except reaches above impassable natural falls, and Dworshak and Hells Canyon dams). Habitat quality in tributary streams varies from excellent in wilderness and roadless areas, to poor in areas subject to heavy agricultural and urban development. Reduced summer stream flows, impaired water quality, and reduced habitat complexity are common problems. Migratory habitat quality in this area has been severely affected by the development and operation of the dams and reservoirs of the Federal Columbia River Power System.
Chinook salmon	*Oncorhynchus tshawytscha*	Snake River Spring/Summer	10/25/1999 64 FR 57399	Critical habitat consists of river reaches of the Columbia, Snake, and Salmon rivers, and all tributaries of the Snake and Salmon rivers (except the Clearwater River) presently or historically accessible to this ESU (except reaches above impassable natural falls and Hells Canyon Dam). Habitat quality in tributary streams varies from excellent in wilderness and roadless areas, to poor in areas subject to heavy agricultural and urban development. Reduced summer stream flows, impaired water quality, and reduced habitat complexity are common problems. Migratory habitat quality in this area has been severely affected by the development and operation of the dams and reservoirs of the Federal Columbia River Power System.
Chinook salmon	*Oncorhynchus tshawytscha*	Upper Columbia River Spring	09/02/2005 70 FR 52630	Critical habitat for this ESU encompasses four subbasins in Washington containing 15 occupied watersheds, as well as the Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some, or high, potential for improvement. We rated conservation value of HUC5 watersheds as high for 10 watersheds, and medium for five watersheds. Migratory habitat quality in this area has been severely affected by the development and operation of the dams and reservoirs of the Federal Columbia River Power System.

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
Chinook salmon	*Oncorhynchus tshawytscha*	Upper Willamette River	09/02/2005 70 FR 52630	Critical habitat encompasses 10 subbasins in Oregon containing 56 occupied watersheds, as well as the lower Willamette/Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some, or high, potential for improvement. Watersheds are in good to excellent condition with no potential for improvement only in the upper McKenzie River and its tributaries. We rated conservation value of HUC5 watersheds as high for 22 watersheds, medium for 16 watersheds, and low for 18 watersheds.
Chum salmon	*Oncorhynchus keta*	Columbia River	09/02/2005 70 FR 52630	Critical habitat encompasses six subbasins in Oregon and Washington containing 19 occupied watersheds, as well as the lower Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some or a high potential for improvement. We rated conservation value of HUC5 watersheds as high for 16 watersheds, and medium for three watersheds.
Chum salmon	*Oncorhynchus keta*	Hood Canal Summer	09/02/2005 70 FR 52630	Critical habitat for this ESU includes 79 mi and 377 mi of near-shore marine habitat in Hood Canal. Primary constituent elements include: 1) Estuarine areas free of obstruction with water quality and aquatic vegetation to support juvenile transition and rearing; 2) Nearshore marine areas free of obstruction with water quality conditions, forage, submerged and overhanging large wood, and aquatic vegetation to support growth and maturation; 3) Offshore marine areas with water quality conditions and forage, including aquatic invertebrates and fishes, supporting growth and maturation.
Coho salmon	*Oncorhynchus kisutch*	Central California Coast	05/05/1999 64 FR 24049	Critical habitat for this ESU is designated to include all river reaches accessible to listed coho salmon from Punta Gorda in northern California south to the San Lorenzo River in central California, including Arroyo Corte Madera Del Presidio and Corte Madera Creek, tributaries to San Francisco Bay. Critical habitat consists of the water, substrate, and adjacent riparian zone of estuarine and riverine reaches (including off-channel habitats) in hydrologic units and counties identified in Table 5 of 64 FR 24049. Accessible reaches are those within the historical range of the ESU that can still be occupied by any life stage of coho salmon. Inaccessible reaches are those above specific dams identified in Table 5 of 64 FR 24049 or above longstanding, naturally impassable barriers (such as natural waterfalls in existence for at least several hundred years).
Coho salmon	*Oncorhynchus kisutch*	Lower Columbia River	02/24/2016 81 FR 9252	Critical habitat for this ESU encompasses 10 subbasins in Oregon and Washington containing 55 occupied watersheds, as well as the lower Columbia River and estuary rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some or a high potential for improvement. Conservation values of HUC5 watersheds were rated as high for 34 watersheds, medium for 18 watersheds, and low for three watersheds.
Coho salmon	*Oncorhynchus kisutch*	Oregon Coast	02/11/2008 73 FR 7816	Critical habitat for this ESU encompasses 13 subbasins in Oregon. The long-term decline in this ESUs productivity reflects deteriorating conditions in freshwater habitat as well as extensive loss of access to habitats in estuaries and tidal freshwater. Many of the habitat changes resulting from land use practices over the last 150 years that contributed to the ESA-listing of Oregon Coast coho salmon continue to hinder recovery of the populations; changes in the watersheds due to land use

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
				practices have weakened natural watershed processes and functions, including loss of connectivity to historical floodplains, wetlands and side channels; reduced riparian area functions (stream temperature regulation, wood recruitment, sediment and nutrient retention); and altered flow and sediment regimes. Several historical and ongoing land uses have reduced stream capacity and complexity in Oregon coastal streams and lakes through disturbance, road building, splash damming, stream cleaning, and other activities. Beaver removal, combined with loss of large wood in streams, has also led to degraded stream habitat conditions.
Coho salmon	*Oncorhynchus kisutch*	Southern Oregon/Northern California Coast (SONCC)	05/05/1999 64 FR 24049	Critical habitat for this ESU includes all areas accessible to any life-stage up to long-standing, natural barriers and adjacent riparian zones. Critical habitat within this geographic area has been degraded from historical conditions by ongoing land management activities. Habitat impairments recognized as factors leading to decline of this ESU were included in the original listing notice include: 1) Channel morphology changes; 2) substrate changes; 3) loss of in-stream roughness; 4) loss of estuarine habitat; 5) loss of wetlands; 6) loss/degradation of riparian areas; 7) declines in water quality; 8) altered stream flows; 9) fish passage impediments; and 10) elimination of habitat.
Sockeye Salmon	*Oncorhynchus nerka*	Snake River	12/28/1993 58 FR 68543	Critical habitat for this ESU consists of river reaches of the Columbia, Snake, and Salmon rivers; Alturas Lake Creek; Valley Creek; and Stanley, Redfish, Yellow Belly, Pettit and Alturas lakes (including their inlet and outlet creeks). Water quality in all five lakes generally is adequate for juveniles, although zooplankton numbers vary considerably. Some reaches of the Salmon River and tributaries exhibit temporary elevated water temperatures and sediment loads that could restrict sockeye salmon production and survival. Migratory habitat quality in this area has been severely affected by the development and operation of the dams and reservoirs of the Federal Columbia River Power System.
Steelhead	*Oncorhynchus mykiss*	California Central Valley	09/02/2005 70 FR 52488	There are 67 occupied HSA watersheds within the freshwater and estuarine range of this ESU. Twelve watersheds received a low rating, 18 received a medium rating, and 37 received a high rating for conservation value to the ESU. Four of these HSA watersheds comprise portions of the San Francisco-San Pablo-Suisun Bay estuarine complex which provides rearing and migratory habitat for this ESU. HSU watershed habitat areas for this ESU include approximately 2,604 mi (4,168 km) of stream habitat and approximately 427 mi^2 (1,102 km^2) of estuarine habitat. Within these HSA watersheds, 2,308 mi (3,693 km) of streams and 254 mi^2 (655 km^2) of estuarine habitat were designated as critical habitat.
Steelhead	*Oncorhynchus mykiss*	Central California Coast	09/02/2005 70 FR 52488	There are 46 occupied watersheds within the freshwater and estuarine range of this ESU. Fourteen watersheds received a low rating, 13 received a medium rating, and 19 received a high rating for conservation value to the ESU. Five of these HSA watersheds comprise portions of the San Francisco-San Pablo-Suisun Bay estuarine complex which provides rearing and migratory habitat for this ESU. HSA watershed habitat areas for this ESU include approximately 1,832 mi (2,931 km) of stream habitat and approximately 442 mi^2 (1,140 km^2) of estuarine habitat (principally San Francisco Bay-San Pablo Bay). Within these HSA watersheds, 1465 mi (2,344 km) of streams and 386 mi^2 (996 km^2) of estuarine habitat were designated as critical habitat.

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
Steelhead	*Oncorhynchus mykiss*	Lower Columbia River	09/02/2005 70 FR 52630	Critical habitat for this ESU encompasses nine subbasins in Oregon and Washington containing 41 occupied watersheds, as well as the lower Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some or a high potential for improvement. NMFS rated conservation value of HUC5 watersheds as high for 28 watersheds, medium for 11 watersheds, and low for two watersheds.
Steelhead	*Oncorhynchus mykiss*	Northern California	09/02/2005 70 FR 52488	There are 50 occupied HSA watersheds within the freshwater and estuarine range of this DPS. Nine watersheds received a low rating, 14 received a medium rating, and 27 received a high rating for conservation value to the DPS. Two estuarine habitat areas used for rearing and migration (Humboldt Bay and the Eel River Estuary) also received a high conservation value rating. HSA watershed habitat areas for this DPS include approximately 3,148 mi (5,037 km) of stream habitat and approximately 25 mi2 (65 km^2) of estuarine habitat (principally Humboldt Bay). Within these HSA watersheds, 3,028 mi (4,844 km) of streams and 25 mi^2 (65 km^2) of estuarine habitat were designated as critical habitat.
Steelhead	*Oncorhynchus mykiss*	Middle Columbia River	09/02/2005 70 FR 52630	Critical habitat for this DPS encompasses 15 subbasins in Oregon and Washington containing 111 occupied watersheds, as well as the Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some or a high potential for improvement. We rated conservation value of occupied HUC5 watersheds as high for 80 watersheds, medium for 24 watersheds, and low for 9 watersheds.
Steelhead	*Oncorhynchus mykiss*	Puget Sound	02/24/2016 81 FR 9252	Critical habitat for this DPS includes 2,031 stream miles. Nearshore and offshore marine waters were not designated for this species. There are 66 watersheds within the range of this DPS. Nine watersheds received a low conservation value rating, 16 received a medium rating, and 41 received a high rating to the DPS.
Steelhead	*Oncorhynchus mykiss*	Snake River	09/02/2005 70 FR 52630	Critical habitat for this DPS encompasses 25 subbasins in Oregon, Washington, and Idaho. Habitat quality in tributary streams varies from excellent in wilderness and roadless areas, to poor in areas subject to heavy agricultural and urban development. Reduced summer stream flows, impaired water quality, and reduced habitat complexity are common problems. Migratory habitat quality in this area has been severely affected by the development and operation of the dams and reservoirs of the Federal Columbia River Power System.
Steelhead	*Oncorhynchus mykiss*	South-Central California Coast	09/02/2005 70 FR 52488	There are 30 occupied watersheds within the freshwater and estuarine range of this DPS. Six watersheds received a low rating, 11 received a medium rating, and 13 received a high rating of conservation value to the DPS. One of these occupied watershed units is Morro Bay, which is used as rearing and migratory habitat for steelhead populations that spawn and rear in tributaries to the Bay. HSA watershed habitat areas for this DPS include approximately 1,251 mi (2,000 km) of stream habitat and approximately 3 mi^2 (8 km^2) of estuarine habitat (for example, Morro Bay). Within these HSA watersheds, 1,249 mi (2,000 km) of streams and 3 mi^2 (8 km^2) of estuarine habitat were designated as critical habitat.

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
Steelhead	*Oncorhynchus mykiss*	Southern California	09/02/2005 70 FR 52488	There are 32 occupied watersheds within the freshwater and estuarine range of this DPS. Five watersheds received a low rating, 6 received a medium rating, and 21 received a high rating of conservation value to the DPS. Watershed habitat areas for this DPS include approximately 741 mi (1,186 km) of stream habitat. Within these HSA watersheds, 708 mi (1,132 km) of streams were designated as critical habitat.
Steelhead	*Oncorhynchus mykiss*	Upper Columbia River	09/02/2005 70 FR 52630	Critical habitat for this DPS encompasses 10 subbasins in Washington containing 31 occupied watersheds, as well as the Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some or a high potential for improvement. Conservation values for HUC5 watersheds were rated as high for 20 watersheds, medium for eight watersheds, and low for three watersheds.
Steelhead	*Oncorhynchus mykiss*	Upper Willamette River	09/02/2005 70 FR 52630	Critical habitat for this DPS encompasses seven subbasins in Oregon containing 34 occupied watersheds, as well as the lower Willamette/Columbia River rearing/migration corridor. Most HUC5 watersheds with PCEs for salmon are in fair-to-poor or fair-to-good condition. However, most of these watersheds have some or a high potential for improvement. Watersheds are in good to excellent condition with no potential for improvement only in the upper McKenzie River and its tributaries. Conservation values of HUC5 watersheds were rated as high for 25 watersheds, medium for 6 watersheds, and low for 3 watersheds.
Green sturgeon	*Acipenser medirostris*	Southern	10/09/2009 74 FR 52300	Critical habitat for this DPS has been designated in coastal U.S. marine waters within 60 fathoms depth from Monterey Bay, California (including Monterey Bay), north to Cape Flattery, Washington, including the Strait of Juan de Fuca, Washington, to its United States boundary; the Sacramento River, lower Feather River, and lower Yuba River in California; the Sacramento-San Joaquin Delta and Suisun, San Pablo, and San Francisco bays in California; tidally influenced areas of the Columbia River estuary from the mouth upstream to river mile 46; and certain coastal bays and estuaries in California (Humboldt Bay), Oregon (Coos Bay, Winchester Bay, Yaquina Bay, and Nehalem Bay), and Washington (Willapa Bay and Grays Harbor), including, but not limited to, areas upstream to the head of tide in various streams that drain into the bays. Several activities threaten the PBFs in coastal bays and estuaries and need special management considerations or protection. The application of pesticides, activities that disturb bottom substrates/ adversely affect prey resources/ degrade water quality through re-suspension of contaminated sediments, commercial shipping and activities that discharge contaminants and result in bioaccumulation of contaminants in green sturgeon; disposal of dredged materials that bury prey resources; and bottom trawl fisheries that disturb the bottom/prey resources for green sturgeon.
Eulachon	*Thaleichthys pacificus*	Southern	10/20/2011 76 FR 65324	Critical habitat for this DPS includes portions of 16 rivers and streams in California, Oregon, and Washington. All of these areas are designated as migration and spawning habitat for this species. In Oregon, we designated 24.2 miles of the lower Umpqua River, 12.4 miles of the lower Sandy River, and 0.2 mi of Tenmile Creek. We also designated the mainstem Columbia River from the mouth to the base of Bonneville Dam, a distance of 143.2 mi. Dams and water diversions are moderate threats to this DPS in the Columbia and Klamath rivers where hydropower generation and flood control are major activities. Degraded water quality is common in some areas occupied

Species	Scientific Name	ESU or DPS	Designation Date and Federal Register Citation	Critical Habitat Status Summary
				by southern DPS eulachon. In the Columbia and Klamath river basins, large-scale impoundment of water has increased winter water temperatures, potentially altering the water temperature during spawning periods. Numerous chemical contaminants are also present in spawning rivers, but the exact effect these compounds have on spawning and egg development is unknown. Dredging is a low to moderate threat to eulachon in the Columbia River. Dredging during eulachon spawning would be particularly detrimental.
Killer whale	*Orcinus orca*	Southern Resident	08/02/2021 86 FR 41668	Critical habitat for this DPS includes approximately 2,560 mi^2 of marine inland waters of Washington: 1) the Summer Core Area in Haro Strait and waters around the San Juan Islands; 2) Puget Sound; and 3) the Strait of Juan de Fuca. Six additional areas include 15,910 mi^2 of marine waters between the 20-feet (ft) (6.1-meter (m)) depth contour and the 656.2-ft (200-m) depth contour from the U.S. international border with Canada south to Point Sur, California. The Quinault Range Site was excluded. Based on the natural history of this DPS and their habitat needs, NMFS identified three PCEs, or physical or biological features, essential for the conservation of this DPS: 1) Water quality to support growth and development; 2) prey species of sufficient quantity, quality, and availability to support individual growth, reproduction and development, as well as overall population growth; and 3) passage conditions to allow for migration, resting, and foraging Water quality in Puget Sound, in general, is degraded. Some pollutants in Puget Sound persist and build up in marine organisms including Southern Resident killer whales and their prey resources, despite bans in the 1970s of some harmful substances and cleanup efforts. The primary concern for direct effects on this DPS from water quality is oil spills, although oil spills can also have long-lasting impacts on other habitat features In regard to passage, human activities can interfere with movements of these killer whales and impact their passage. Vessels may present obstacles to whales' passage, causing them to swim further and change direction more often, which can increase energy expenditure which impacts foraging behavior. Reduced prey abundance, particularly Chinook salmon, is also a concern for critical habitat.

EFFECTS ANALYSES

Table 9 summarizes the effects determinations to species by ESU/DPS and its designated critical habitat for all activities combined. The amount of critical habitat within the action area is less than 0.34% regardless of species. However, critical habitat designations for many aquatic species under NMFS' jurisdiction are linear features and best analyzed by stream miles as shown in Table 2. Although overlap of the action area with designated critical habitat is low, exposure is possible. Analyses for salmon ESUs and steelhead DPSs are combined below due to the similarity of needs to the species and their critical habitat and similarity of effects to the species and their critical habitat.

Land management plans typically contain aquatic management strategies that have standards and guidelines for activities that can and cannot occur in the aquatic management zones. There are three examples of aquatic management strategies within the action area: the Northwest Forest Plan (NWFP) (USDA and USDI 1994); the interim strategies for managing anadromous fish-producing watersheds in eastern Oregon and Washington, Idaho, and Portions of California (PacFish) (USDA and USDI 1995); and the Inland native fish strategy (InFish) (USDA 1995). Within these aquatic strategies, riparian management objectives considered in each strategy need to be met before allowing certain activities to occur within the aquatic management zones. Aquatic management zones are generally identified within the three strategies as 1 site-potential-tree height (spth) or 150 feet, whichever is greater. Land Management Plans outside of these areas may have similar protections for the aquatic management zones. Forests where these aquatic management strategies do not apply, may have their own aquatic conservation strategy or utilize their land management plan standards and guides when determining what actions are allowable within the aquatic management zone. Aquatic management strategies and land management plans were considered when analyzing effects to species and their critical habitat.

Table 9. Threatened (T), endangered (E), NMFS regulatory species in the action area and the units they occur on. The presence of critical habitat (and percent overlap with the action area) is also provided. A summary of exposure and the effects determination is also shown. ESU=Evolutionarily Significant Unit; DPS=Distinct Population Segment; CH=Critical Habitat.

Species	Scientific Name	ESU or DPS	Status	CH Present, percent overlap	Species Effects	CH Effects
Chinook salmon	*Oncorhynchus tshawytscha*	California Coastal	T	Yes, 0.002	LAA	LAA
Chinook salmon	*Oncorhynchus tshawytscha*	Central Valley Spring	T	Yes, 0.05	LAA	LAA
Chinook salmon	*Oncorhynchus tshawytscha*	Lower Columbia River	T	Yes, 0.16	LAA	LAA
Chinook salmon	*Oncorhynchus tshawytscha*	Puget Sound	T	Yes, 0.18	LAA	LAA

Species	Scientific Name	ESU or DPS	Status	CH Present, percent overlap	Species Effects	CH Effects
Chinook salmon	*Oncorhynchus tshawytscha*	Sacramento River Winter	E	No, 0.0	LAA	NE
Chinook salmon	*Oncorhynchus tshawytscha*	Snake River Fall	T	Yes, 0.12	LAA	LAA
Chinook salmon	*Oncorhynchus tshawytscha*	Snake River Spring-Summer	T	Yes, 0.08	LAA	LAA
Chinook salmon	*Oncorhynchus tshawytscha*	Upper Columbia River Spring	E	Yes, 0.27	LAA	LAA
Chinook salmon	*Oncorhynchus tshawytscha*	Upper Willamette River	T	Yes, 0.10	LAA	LAA
Chum salmon	*Oncorhynchus keta*	Columbia River	T	Yes, 0.06	LAA	LAA
Chum salmon	*Oncorhynchus keta*	Hood Canal Summer	T	Yes, 0.03	LAA	LAA
Coho salmon	*Oncorhynchus kisutch*	Central California Coast	E	No, 0.0	LAA	NE
Coho salmon	*Oncorhynchus kisutch*	Lower Columbia River	T	Yes, 0.16	LAA	LAA
Coho salmon	*Oncorhynchus kisutch*	Oregon Coast	T	Yes, 0.05	LAA	LAA
Coho salmon	*Oncorhynchus kisutch*	Southern Oregon-Northern California Coast	T	Yes, 0.35	LAA	LAA
Sockeye Salmon	*Oncorhynchus nerka*	Snake River	T	Yes, 0.04	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	California Central Valley	T	Yes, 0.04	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Central California Coast	T	No, 0.0	LAA	NE
Steelhead	*Oncorhynchus mykiss*	Lower Columbia River	T	Yes, 0.19	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Middle Columbia River	T	Yes, 0.16	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Northern California	T	Yes, 0.03	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Puget Sound	T	Yes, 0.16	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Snake River	T	Yes, 0.10	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	South-Central California Coast	T	Yes, 0.04	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Southern California	E	Yes, 0.34	LAA	LAA

Species	Scientific Name	ESU or DPS	Status	CH Present, percent overlap	Species Effects	CH Effects
Steelhead	*Oncorhynchus mykiss*	Upper Columbia River	T	Yes, 0.25	LAA	LAA
Steelhead	*Oncorhynchus mykiss*	Upper Willamette River	T	Yes, 0.03	LAA	LAA
Green sturgeon	*Acipenser medirostris*	Southern	T	Yes, 0.02	NLAA	LAA
Eulachon	*Thaleichthys pacificus*	Southern	T	Yes, 0.07	LAA	LAA
Killer whale	*Orcinus orca*	Southern Resident	E	No, 0.0	NLAA	NE

Effects to Pacific Salmon and Steelhead and their Critical Habitat

Pacific salmon and steelhead are anadromous fish, meaning adults spend most of their life in the ocean but migrate into fresh water to spawn. The effects of Forest Service actions are almost always within the freshwater systems these salmonids utilize during their life cycles. Table 10 shows the effects pathway to species compared to the impacts/benefits described in the deconstruction table.

The proximity and probability of actions occurring near listed salmonid species or their designated critical habitat (sometimes referred to as listed fish habitat (LFH)) are assumed to occur at some point in the future. Specific ESUs and DPSs for the salmonids and which National Forest projects could cause effects to them are in Table 3. Maps of salmonid critical habitats are in Appendix A. Maps and shown in Figures 1 through 27. The magnitude of effects to species and critical habitat are assumed to be localized to the action area based on the aquatic conservation strategies listed above. Action distribution typically will not occur within an entire watershed during the same year but will be dependent on how timber sales are planned and the outcome of future NEPA and ESA consultation. This could mean several small effects in a watershed or one large effect for the watershed could occur. Frequency of actions would be based on many different factors, including the need for timber, but timber harvest in one area typically span decades since this timeframe is dependent on timber growth.

Table 10 shows the effects pathway for salmonid species. The impacts/benefits potential effects to salmonid species from the deconstruction table are summarized in Table 11, Table 12, Table 13, and Table 14. Potential effects are described as direct and/or indirect, neutral, and beneficial. The effects duration is further described as short-term or long-term effects; short-term effects subside immediately or up to one year, and long-term effects persist over a year, and potentially sets a new threshold for a species environment. There would be no direct effects from the proposed rulemaking as it would not authorize site-specific actions.

Table 10. Impact and/or benefit pathways for effects pathways for salmonids.

Impact and/or Benefit	Pathways for salmonid effects
Change in Riparian Vegetation (Temperature)	Water Quality
Chemical Contamination/Nutrients	Habitat Elements
Drainage Network	Channel Conditions and Dynamics; Flow/Hydrology; Habitat Elements
Large Woody Debris	Habitat Elements
Peak/Baseflows	Flow/Hydrology
Riparian Resilience	Habitat Elements
Road Density/Location	Habitat Access
Sediment/Turbidity	Water Quality
Spawning Substrate	Habitat Elements

Critical Habitat essential primary and biological features (PBFs) are the same for all pacific salmonids although conditions for each ESU or DPS may be different depending on their location on the landscape. Primary and biological features are outlined in Table 11 and crosswalked with the impact/benefit pathway found in the effects tables (Table 12, Table 13, Table 14, and Table 15).

Table 11. Crosswalk between critical habitat PBFs and the impacts and/or benefits for Pacific salmon and steelhead species in the aquatic deconstruction table.

Physical and Biological Features	Impact and or Benefit Crosswalk with PBFs
Spawning Habitat, as defined by water quality, water quantity, substrate	**Water Quality**: Change in Riparian Vegetation (Temperature), Sediment/Turbidity, Spawning Substrate **Flow/Hydrology**: Peak/Baseflows; Drainage Network **Habitat Elements**: Riparian Resilience, Substrate/Embeddedness
Rearing as defined by adequate water quantity and floodplain connectivity	**Habitat Access**: Road Density/Location **Channel Conditions and Dynamics**: Drainage Network **Flow/Hydrology**: Peak/Baseflows **Habitat Elements**: Riparian Resilience
Rearing as defined by adequate water quality and forage	**Water Quality**: Change in Riparian Vegetation (Temperature), Sediment/Turbidity, Spawning Substrate **Habitat Elements**: Riparian Resilience, Large Woody Debris, Chemical Contamination/Nutrients
Rearing as defined by adequate natural cover	**Habitat Elements**: Riparian Resilience, Large Woody Debris, Drainage Network
Migration as defined by habitat free of artificial obstructions, and adequate water quality, water quantity, and natural cover	**Habitat Access**: Road Density/Location **Water Quality**: Change in Riparian Vegetation (Temperature) **Flow/Hydrology**: Peak/Baseflows, Drainage Network **Habitat Elements**: Riparian Resilience, Large Woody Debris, Chemical Contamination/Nutrients

Table 12. Timber harvest task/methods effects summary activities that are likely to cause Indirect (D), Neutral (N), Beneficial (B), long-term (+), or short-term (-) effects to salmonid species and critical habitat.

Task/Method	Change in Riparian Vegetation (Temperature)	Chemical Contamination/ Nutrients	Drainage Network	Large Woody Debris	Peak/ Baseflows	Riparian Resilience	Road Density/ Location	Sediment/ Turbidity	Spawning Substrate
Timber Felling	D-	N	N	D+	N	D-, B+	N	D-	N
Tree and Unit Marking	N	D-	N	N	N	N	N	N	N
Landings - Vegetation Clearing and Removal	D-	N	N	D-	D-	D-	N	D-	D-
Site Restoration - Seeding and Noxious Weed Treatments	N	B+	N	N	N	D-, B+	N	N	N
Timber Yarding – for example, Skidding, Skyline, Helicopter, Tethered, Forwarders	D-	N	N	D-	D-	D-	N	D-	D-
Timber Haul	N	N	N	N	N	N	N	D-	D-
Road Gates Installation	N	N	N	N	N	N	N	D-	D-

Table 13. Road maintenance/repair task/methods task/methods effects summary activities that are likely to cause Indirect (D), Neutral (N), Beneficial (B), long-term (+), or short-term (-) effects to salmonid species and critical habitat.

Task/Method	Change in Riparian Vegetation (Temperature)	Chemical Contamination/ Nutrients	Drainage Network	Large Woody Debris	Peak/ Baseflows	Riparian Resilience	Road Density/ Location	Sediment/ Turbidity	Spawning Substrate
Drainage Maintenance	D-	D-, B+	N	N	D-	N	N	D-	D-
Drainage Installation	D-	D-, B+	N	N	D-	N	N	D-	D-
Drainage Replacement or Repair	D-	D-, B+	N	N	D-, B+	N	N	D-	D-, B+
Drainage Temporary Stream Flow Divergence; Bypass	D-	D-, B+	N	N	D-	N	N	D-	D-
Vegetation - Brushing, Limbing, Seeding, Mowing, Mulching, Hazard Tree Removal	D-	D-	N	D-	N	N	N	D-	D-
Surface Repair/Replacement-Road Cut and Fill Repair/Stabilization	N	N	N	N	N	N	N	D-	D-
Surface Repair/Replacement-Asphalt Repair, Repaving, Chip-Sealing, Skin Patching, and Rocking	N	D-	N	N	N	N	N	D-	D-
Surface Repair/Replacement-Small Slide Removal (easily handled with maintenance equipment)	N	D-	N	N	N	N	N	D-	D-
Surface Repair/Replacement-Dust Abatement	N	N	N	N	D-	N	N	D-	D-
Structure Maintenance, Repair, and Installation	N	N	N	N	N	N	N	D-	D-
Bridge Maintenance	N	D-	N	N	N	N	N	N	N
Snow Plowing	N	N	D-	N	N	N	N	N	N

Table 14. Permanent or temporary road construction task/methods effects summary activities that are likely to cause Indirect (D), Neutral (N), Beneficial (B), long-term (+), or short-term (-) effects to salmonid species and critical habitat.

Task/Method	Change in Riparian Vegetation (Temperature)	Chemical Contamination/ Nutrients	Drainage Network	Large Woody Debris	Peak/ Baseflows	Riparian Resilience	Road Density/ Location	Sediment/ Turbidity	Spawning Substrate
Vegetation Clearance- Cut Above-Ground Vegetation	D-	N	N	D+	D+	N	N	D+	D-
Vegetation Clearance- Snags - Hand Felling	N	N	N	D-	N	N	N	N	N
Vegetation Clearance- Snags - Mechanical Felling	D-	N	N	D-	N	N	N	D-	N
Vegetation Clearance- Tree Removal - Log Decks	N	N	N	N	N	N	N	D-	N
Vegetation Clearance- Tree Removal - Hauling	N	N	N	N	N	N	N	D-	N
Roadway Template- Soil Excavation	N	N	D-	N	N	N	N	D+	D-
Roadway Template- Soil Removal	N	N	N	N	N	N	N	D-	D+
Roadway Template- Fill Placement	N	N	D-	N	N	N	D+	D+	D-
Drainage- Ditchlines, Relief Culverts, Waterbars, Riprap	N	D-	D-	N	D+	N	N	D-	D-
Drainage- Stream Crossings - Culvert Installation	D-	D-, B+	D-	N	D-	N	N	D-	D-
Surfacing- Hauling of Surfacing Materials	N	D-	N	N	N	N	N	D-	D-
Surfacing- Aggregate Surfacing Placement - Rolling and Blading	N	D-	N	N	N	N	N	D-	D-

Table 15. Temporary road removal task/methods effects summary activities that are likely to cause Indirect (D), Neutral (N), Beneficial (B), long-term (+), or short-term (-) effects to salmonid species and critical habitat.

Task/Method	Change in Riparian Vegetation (Temperature)	Chemical Contamination/ Nutrients	Drainage Network	Large Woody Debris	Peak/ Baseflows	Riparian Resilience	Road Density/ Location	Sediment/ Turbidity	Spawning Substrate
Recontouring- Blading	N	N	B+	N	N	N	N	D-	D-
Scarifying- Material Removal	N	N	N	N	N	N	N	D-	D-
Culvert/Drainage Structure Removal	N	N	B+	N	N	N	N	D-	B+
Revegetation- Seeding	D-, B+	N	N	B+	N	N	N	D-	D-
Mulching- Weed free material	N	N	N	N	N	N	N	B+	B+

Change in Riparian Vegetation (Temperature)

Change in riparian vegetation is associated with stream temperature. Tree canopies disrupt the solar radiation from reaching the stream and increasing stream temperatures. Timber felling, landing creation and yarding corridors typically involve the loss of the tree canopy resulting in shade loss.

While stream shade correlates with the width of stream protection buffers, the relationship is quite variable, depending on site-specific factors such as stream size, substrate type, stream discharge, topography (Caissie 2006), channel aspect, and forest structure and species composition. Inputs of cold water from the streambed, seepage areas on the stream bank, and tributaries can help cool the stream on hot summer days if they are sufficiently large relative to the stream discharge (Wondzell 2012). The density of vegetation in riparian areas affects shade and thermal loading to a stream due to the penetration of solar radiation through gaps in the canopy and among the branches and stems (Brazier and Brown 1973; DeWalle 2010). In some instances (such as narrow streams with dense, overhanging streamside vegetation, or stands on the north sides of streams with an east-west orientation), no-harvest buffers as narrow as 30 feet adjacent to clearcuts can maintain stream shade (Brazier and Brown 1973). Wider buffers, in general, will provide increased protection of stream temperature (Anderson et al. 2007, Leinenbach et al. 2013, Groom et al. 2011a, Groom et al. 2011b).

Timber harvest adjacent to riparian buffers can influence the riparian microclimate adjacent to streams, which can, in turn, influence stream temperatures (Moore et al. 2005a). Gomi et al. (2006) reported that inclusion of a 30m (98') buffer in clear-cut harvesting units increased maximum daily temperatures by <2˚C in British Columbia. Wilkerson et al. (2006) reported maximum increases of 1.0-1.4 ˚C with 11 m buffers and negligible increases with 23m (75') or partially cut buffers in Maine. DeWalle (2010) theorized that at least 80% shade could be attained with 12m (39') wide buffers on small (<6m [20'] wide) streams. In western Oregon, Zwieniecki and Newton (1999) reported that the presence of near continuous shade from an uncut buffer positioned on the south side of water led to protection from the effect of solar radiation. On the Oregon Coast Range, Groom et al. (2011a) found that on post-harvest lands with a protection buffer (no-cut zone) of 6m (20') and limited harvest in the remainder of the riparian management area increased stream temperature by an average of 0.7 degrees C, but no change to maximum stream temperatures on lands managed to the more stringent state standard of 8m (26') protection buffer and limited harvest in the remainder of the riparian management zone. The high variability in buffer width and shade condition is a function of the many variables that influence the amount of shade produced by riparian vegetation. Low-density stands with limited vertical distribution of branches and foliage will require wider buffer widths to produce the same amount of shade as high-density stands.

Riparian treatments on perennial streams greater than 1,000' from LFH are not expected to increase stream temperatures in LFH when there is hyporheic flow and flow accumulation. Several papers have been published recently indicating that hyporheic flow and stream substrate composition has a significant influence on stream temperature. Janishch et al. (2012) found that that the canopy cover of "buffers" was not a strong predictor for temperature in small (<7 feet wide) headwater streams. Streams with coarse-textured streambeds tended to be thermally

unresponsive as compared with fine-textured streambeds or those with small, near-stream wetland areas. Moore et al. (2005b) found that heat exchange associated with hyporheic flow paths was a cooling effect during daytime, with a magnitude up to approximately 25% that of net radiation during the period of maximum daytime warming in clear cut harvest units on streams in British Columbia. Davis et al. (2015) found that in Western Oregon streams, pre-harvest to post-harvest temperature approximately 300m downstream of the harvest will be 56% of the temperature change that occurred within the harvest reach. Rapid cooling has been observed after an extensive exposure to the sun after water re-entered forest cover in Australia (Rutherford et al. 2004), and, in Idaho, warming of headwater streams with no buffers did not lead to warming of fish-bearing streams downstream (Gravelle and Link 2007). Zwieniecki and Newton (1999) indicated that stream temperature increases from harvesting activity were frequently lost within 150–300 m (492-984') beneath downstream cover. This re-emphasizes the important role gravel and large wood play in channels. Therefore, thinning on perennial channels 1,000' or greater from LFH is expected to have only minor negative effects to temperature in LFH.

Accurately predicting the magnitude of stream temperature increases on a highly variable landscape is extremely difficult. However, the research cited above supports the conclusion that while there may be a decrease in effective shade, localized increases in stream temperature are likely to be too small to differentiate from background variability. If localized increases in stream temperature occur, they will likely be less than 1 degree C and are unlikely to be measurable in LFH. The tributary streams that flow through the harvest units are small in size (<1-5 cfs in most cases) and research has shown that the magnitude of temperature effect is a function of the amount of water entering (or exchanged with) the stream relative to the stream discharge (Johnson and Jones 2000, Story et al. 2003, Wondzell 2006, Wondzell 2012) and the difference in temperature between the stream and the inflowing water. Therefore, any localized increase in stream temperature from commercial thinning activities would typically be minor negative effect on LFH streams.

Typically, timber harvest activities are associated with an aquatic conservation strategy or land management plan component that provides for a stream protection buffer to limit the solar radiation reaching the stream. It is probable future projects will include some type of minimization measure for stream buffers for LFH areas, but since specific buffers are unknown for now, this analysis assumes there could be projects that will have limited stream protection buffers and therefore have effects on stream temperature. As described above, most of the time timber harvest activities result in insignificant localized stream temperature increases because rapid cooling should occur as the stream flows into the uncut aquatic management zone buffer; therefore, effects should last less than an hour.

Road maintenance typically includes tree removal or brushing near streams and has the potential to create small minor changes to stream temperature. If there are drainage needs for road maintenance, such as culvert replacements, there may be a need to dewater the area and move it to a different location using pumps and screens. Depending on how this is accomplished, it could cause water temperatures to increase. If there are increases in the water temperature, the changes should be minor but could be measurable as future projects are unknown at this time. Past consultations have provided minimization measures that limit effects to temperature fluctuations.

While the specific future road maintenance and reconstruction, permanent and temporary road construction actions are uncertain, if those future actions are implemented within one site potential tree height there would be some shade loss from timber harvest, . Effects are expected to mostly be insignificant, unmeasurable effects, but could result in small localized measurable increases in stream temperatures. Temporary road removal would not create measurable effects to increases in stream water temperature and can regrow shrubs quickly in the area, which would be beneficial.

Overall, under the proposed rulemaking, where future road construction and timber harvest activities occur within the action area, they would have small measurable effects on salmonids and their critical habitat.

Chemical Contamination/Nutrients

Chemical contamination and nutrients for projects in this consultation are associated with equipment utilizing petroleum products and dust abatement chemicals. Timber vegetation management, road maintenance, road construction, and road removal activities all utilize heavy machinery, some utilize hand tools (for example, chainsaws), water pumps, and dust abatement chemicals have the potential to create chemical contamination reaching the streams.

Coarse woody debris is an important component of forest ecosystems because it can provide cover for many wildlife species, provide important structures for small mammals, reptiles, and amphibians, and the decomposition of logs on the forest floor that provides nutrients to plants, habitat for wildlife, and food for insects and microorganisms. This process of nutrient input is important for the food web processes for salmonids as they feed on insects and microorganisms during certain life stages. Where timber harvest has been implemented, replanting is important for this process to continue. Harvest areas are typically localized to units and timber harvest must meet certain criteria if it is to be implemented within the aquatic management zones. For many of the forests in the action area, future projects will maintain a no-cut buffer for the aquatic management zone. This will insulate some food-web negative effects.

Any work utilizing chemicals and petroleum products typically requires an approved Spill Prevention Control and Containment Plan (Spill Plan) as required by contract provisions G.3.4.1/BT6.341. The Spill Plan would include a description of the hazardous materials that would be used; and a spill containment kit would be located on-site. All heavy equipment used for refueling would carry a hazardous material recovery kit. All vehicles and machinery would be free of petroleum leaks. Any leaks that occur would be immediately repaired. Power equipment would be refueled at least 150 feet from water bodies or as far away as possible to prevent direct delivery of contaminants into a water body. If local site conditions do not allow for a 150-foot setback, then refueling would be as far away as possible from the water body (USDA 2012). For all immobile equipment, absorbent pads would be used. All petroleum products being transported or stored would be in approved containers meeting Occupational Safety and Health Administration and appropriate state standards. The Forest Service Contracting Officer would be notified of any spills. Any contaminated soil, vegetation or debris must be removed from National Forest System lands and disposed of in accordance with state laws.

Fuel spills of petroleum products from heavy equipment for all management activities have the potential to result in chemical contamination of streams. Hand tool fuel capacity is small (<1/4 gallon), and thus any potential spills during refueling would be unlikely to reach the stream channel resulting in a neutral effect for hand tools. Water pump fuel capacity is also small, and refueling is completed outside of the stream. It is extremely unlikely that heavy equipment in these activities will spill contaminants due to the contract clause requiring the Spill Plan. It is a standard contract clause that vehicles are checked regularly for leaks by purchasers and timber sale administrators and any leaks are promptly repaired before the vehicle is allowed to continue operating. Based on over 2 decades of experience that encompasses tens of thousands of log truck loads, as well as contractor supplied equipment for instream and road work, there is very low probability of chemical spills, and there have been no known spills from these activities that have impacted LFH. Forest Service best management practices require spill kits to be available if there is a spill and require fueling stations to be located 150 feet from streams when possible, and as far as practical when not possible.

The Forest Service expects limited applications for dust abatement. Dust abatement is typically only implemented during the dry season. Dust abatement can utilize water and chemicals (for example, lignosulfonate, Magnesium Chloride), depending on the conditions in the work area. The Forest Service has a guide on applications for dust suppression (USDA 1999). The application of dust abatement chemicals would reduce fine sediment delivery to streams typically from dry season haul. Several different chemicals are discussed in the Forest Service guide, with pros and cons of each including an environmental impact section. Most chemicals do have negative environmental impacts, and these will be taken into consideration when utilized.

Lignosulfonate is an organic, nonpetroleum product that has limited ability to migrate since it is a bonding agent that adheres to material in dry conditions. Lignosulfonate is the chemical expected to be utilized near listed fish habitat, but there may be other chemicals utilized as well. There is potential for the dust abatement product to directly enter waterways. However, the single application of lignosulfonate during the dry season would reduce the likelihood of it entering a stream channel. There would not be an effect to LFH fish from applying lignosulfonate on road surfaces. Lignosulfonate has been tested to determine its potential impact on water quality and fish species when introduced directly into surface waters and it is rather benign. A 48-hour LC50 of 7,300 parts per million (ppm) was observed by Roald (1977) in rainbow trout (Oncorhynchus mykiss). Compounds with values greater than a value of >1,000 are considered to have a low order of toxicity. Due to the limited use and application of lignosulfonate, the timing of application, the use of best management practices for application, and the known non-toxicity to fish at dust abatement concentrations, the probability of affecting listed fish is discountable.

Magnesium chloride can negatively impact aquatic systems by reducing macroinvertebrates (Kotalik et al. 2017) and directly affecting fish (Lewis 1998). If Magnesium chloride is utilized, there will be direct and indirect effects to fish through death (at high enough concentrations) and food web processes. The magnitude of effect is difficult to determine without having project specific information. Future project consultation will determine which chemical is utilized for dust abatement if it is needed but if utilized, we hypothesize the effects will be short-term but not discountable.

Timber harvest, road maintenance and reconstruction, and temporary road removal could cause short-term negative direct effects to the transport of nutrients downstream when implementing the projects, however, when projects utilize a water bypass system, effects would be limited to project implementation. These activities could provide for long-term beneficial effects when maintaining or repairing the current drainage system, installing appropriate sized drainage systems for the stream, and utilizing water bypass systems during the maintenance process.

Road construction could cause short-term negative direct effects to the transport of nutrients downstream when implementing the projects and fuel spills of petroleum products from machinery and equipment used in road work have the potential to result in chemical contamination of streams. Fuel powered equipment (hand tools and heavy machinery) will be used within the Riparian Reserves for road construction, including over stream crossings, where they are needed.

Overall, under the proposed rulemaking, where future road construction and timber harvest activities occur within the action area, they would have small measurable effects on salmonids and their critical habitat.

Drainage Network

The natural drainage networks of streams and rivers can be increased through road building and impervious road ditching and channeling that redirects surface flow and increase the active channel network. The primary management effect is associated with the road network, especially road compaction and road drainage systems. An increased drainage network can increase peak volume discharge and cause peak flows to occur earlier than unroaded drainages.

Wemple et al. (1996) studied the effect of channel network extension on two basins in the western Cascades of Oregon and found that 57% of the road network was connected to the stream network (road segments draining to streams, or road segments draining to culvert outlets with gullies incised below them), resulting in an increased drainage density of 21-50%. They concluded that this increase in road drainage density, and enhanced water routing efficiency was a possible explanatory mechanism for the changes in hydrograph shape (such as earlier peak, higher volume discharge) as presented in Jones and Grant (1996). Many other watershed factors (for example, soil percolation rates, soil stability, road construction type) may influence the magnitude or probability that road extension will affect flows.

System road reconstruction, road (system and non-system) maintenance, and permanent (system roads) and temporary (non-system roads) roads will be built, and decommissioning may occur in areas adjacent to streams with listed fish habitat (LFH) as well as in areas that may be several miles upstream from LFH.

Road reconstruction and maintenance have the potential for reducing the distance water travels along roadways, with the implementation of larger cross drain culverts, reduced spacing between culverts, and/or better placement of near-stream cross drains and their outflows. Road Decommissioning may occur near LFH streams, or on tributary streams that feed into LFH streams.

There are no causal effects of timber harvest on the drainage network, and the effects of timber harvest would be neutral to this element.

The only causal mechanism for effects for road maintenance and road re-construction would be through snow plowing. These would create short-term negative effects during the spring when snow melts. These would be localized effects to the area where snow plowing occurs. Snow plowing is not an action that is routinely implemented on National Forest System roads, so the probability of occurrence is low, but not zero. If effects do occur to drainage networks, these effects would be minor and localized.

Permanent and temporary road construction will have negative effects for drainage network lasting as long as the road is in place.

Temporary road removal will have long-term beneficial effects on drainage network. The process would be the opposite of road building.

Overall, activities will have small measurable effects on salmonids and their critical habitat.

Large Woody Debris

Large woody debris enters streams through chronic and episodic processes (Bisson et al. 1987). Chronic sources include streamside tree mortality and bank undercutting (Murphy and Koski 1989). Episodic sources include windthrow (for example, Harmon et al. 1986; Lienkaemper and Swanson 1986; May and Gresswell 2003), wildfire (Agee 1993), infrequent flooding, landslides/debris flows (Benda et al. 2002, 2003), and insect infestations or disease. The quality, complexity, and function of fish habitat is often affected by the level of instream large woody debris. Large woody debris influences the size and distribution of stream habitat units, provides cover from predators, shades the water surface, and increases the sorting of substrate, providing suitable spawning areas (Megahan 1982; Bilby and Ward 1991; Crispin et al. 1993; Montgomery et al.1995; May and Gresswell 2003). Large woody debris is instrumental in nutrient retention and provides a substrate for macroinvertebrates (Cederholm and Peterson 1985; Bilby and Ward 1989; Bilby and Bisson 1998; Gregory et al. 1991; Wallace et al. 1995). Large woody debris influences channel morphology (Swanson and Lienkaemper 1978; Heede 1985; Bisson et al. 1987; Swanston 1991). Fish densities are higher in streams with abundant large woody debris (Fausch and Northcote 1992; Roni and Quinn 2001b).

The quality, complexity, and function of fish habitat is often affected by the level of instream large woody debris. Large woody debris influences the size and distribution of stream habitat units, provides cover from predators, shades the water surface, and increases the sorting of substrate, providing suitable spawning areas (Megahan 1982; Bilby and Ward 1991; Crispin et al. 1993; Montgomery et al.1995; May and Gresswell 2003). LWD is instrumental in nutrient retention and provides a substrate for macroinvertebrates (Cederholm and Peterson 1985; Bilby and Ward 1989; Bilby and Bisson 1998; Gregory et al. 1991; Wallace et al. 1995). Large woody debris influences channel morphology (Swanson and Lienkaemper 1978; Heede 1985; Bisson et al. 1987; Swanston 1991). Fish densities are higher in streams with abundant Large woody debris (Fausch and Northcote 1992; Roni and Quinn 2001b).

Timber felling, landings creation, and yarding create pathways for effects to salmonids and critical habitat if they are done within the aquatic management zone. Projects where these activities are completed in aquatic management zones are typically implemented to move the aquatic management zone to a better desired condition, which are better for species and critical habitat. Where large woody debris is lacking, ESA consultations typically contain criteria for large woody debris to be left onsite where it can contribute to future in-stream wood. It is assumed that future proposed projects will contain the same requirements. There are direct short-term negative effects from removing large woody debris from aquatic management zone even with minimization measures.

Road maintenance could cause some minor short-term direct negative effects to large woody debris when vegetation management occurs next to the road and is within the aquatic management zone. Large woody debris is important to both salmonids and their critical habitat needs. These negative effects would be localized to road sections near LFH. Effects would still be localized and minor even where roads parallel LFH.

Road construction could cause some minor longer-term direct negative effects to large woody debris when roads are built in the aquatic management zone. Large woody debris is important to both salmonids and their critical habitat needs. These negative effects would be localized to roads built near LFH. Longer-term effects would still be localized if new roads are built parallel to LFH. Proposed future projects would need to complete ESA consultation prior to any new road building in the affected environment. Past ESA consultations that contain road building as an activity require minimization measures to reduce effects to salmonids and critical habitat. Any future projects would most likely also contain minimization measures as well. The magnitude of effects is difficult to predict. However, road building would be associated with timber harvest and roads for timber harvest are built once to use throughout the harvesting timeframe. This does mean that the effects of large woody debris may be years to re-establish in the area if it is cut. The effects would be localized to the stream segment where the road is built.

Temporary road removal includes removal of culverts, re-establishment of natural drainage patterns, and blocking vehicle access. The most likely point of entry for sediment will be during activities that occur next to streams, typically culvert removal or reducing the fill over culverts that are left in place. It is likely that temp road removal will result in short-term localized sediment delivery to streams. It is also expected that in the long-term, temporary road removal will have a positive effect on sediment delivery by putting to bed or removing roads, and stream crossings, where pulsed sediment delivery can potentially occur.

Overall, activities will have small measurable effects on salmonids and their critical habitat.

Peak/Baseflows

Peak/baseflows can be defined as quantity of water and the timing of input of water to an area. Loss of trees and vegetation may change the rate of evapotranspiration, increasing soil moisture and overall water yield. A short term (5-10 years) increase in discharge during the wet and the dry periods could occur from two mechanisms for the thinned stands: increased snow accumulation (wet period) will create small increases in peak flows (Jones and Grant 1996), and

reduced canopy (dry periods) will reduce transpiration rates which will account for small increases in summer flows.

Peak (and pulsed) stream flows occur regularly in the action area, normally from November to April, and carry the majority of the sediment load with high stream energies that may erode and change channel form (Cooper 2005).

Tree harvest and new permanent and temporary road construction do have potential to change the peak and base flows by decreasing the current canopy cover within the treated stands. Loss of trees and vegetation may change the rate of evapotranspiration, increasing soil moisture and overall water yield. A short term (5-10 years) increase in discharge during the wet and the dry periods could occur from two mechanisms for the thinned stands: increased snow accumulation (wet period) will create small increases in peak flows (Jones and Grant 1996), and reduced canopy (dry periods) will reduce transpiration rates which will account for small increases in summer flows.

Timber harvest has two causal mechanisms for peak/baseflows, these are timber yarding and timber landing construction. The pathway for effects for these two activities is based on compaction of soil. Soil compaction may cause water runoff to increase when there is heavy rainfall. These effects would be localized and small in magnitude.

The causal effect of road maintenance or reconstruction for peak/baseflows are due to water withdrawals, dewatering of a work site, or water pumping to a different location. These activities create localized short-term direct negative effects that last the duration of the implementation time for these activities but could have long-term beneficial effects if appropriately sized culverts are installed. These activities can last less than an hour for dust abatement to a few days for culvert replacement.

The causal effect of road building for peak/baseflows creates the largest effects to peak/baseflows of any other activity in this consultation. Depending on where roads are built and the associated water diversion systems used for the road, longer-term direct effects are expected since the roadbed can create a dam-like effect on water flow. These effects will last as long as the road is in place.

Temporary road removal will re-establish the natural drainage patterns to the streams resulting in a beneficial effect. Temporary road removal will re-vegetate over time resulting in an intact canopy and a reduction in openings. Temporary roads will be decommissioned at the end of the project. Although this increase in canopy cover on decommissioned roads is positive, at the watershed scale, the road area is small and the creation of temporary roads will offset the canopy closures, thus is discountable in terms of the potential to affect peak and base flows.

Water withdrawals have direct effects to peak/base flow. Drafting from streams may occur during spring, summer and fall, which encompasses the base flow period when the impact locally will be greatest, but will be isolated in space and time in that multiple drafting operations is unlikely to occur simultaneously in the action area. Stream flows may be temporarily reduced downstream from the withdrawal point. The amount of flow decrease from water withdrawals would depend on the amount of stream flow, how much water is withdrawn, and the duration of

water drafting. Water trucks commonly hold 500 gallons of water, with a withdrawal rate of up to approximately 7.5 gal/sec. This means the maximum withdrawal would last at least 67 sec, and the maximum withdrawal period commonly is less than 5 minutes at lower rates of withdrawal. The magnitude of flow reductions will be insignificant due to the high stream flow volume available and the short period of withdrawal.

Water withdrawal for road work typically occurs during the dry season (generally June to October). While water quantity to LFH may be temporarily reduced for short periods of time, the magnitude of flow reductions would be insignificant to affect the species or quality of their critical habitat because the periods of withdrawal will be short and would not affect inflow from adjacent tributary streams and hyporheic flow downstream from the point of withdrawal.

The implementation of future road construction or timber harvest activities would result mostly in insignificant overall effects to peak and base flows, mainly through drafting of surface water for fuels treatment (during peak flow from fall through spring), and road maintenance (during base flow in summer and early fall). However, future projects may create small significant effects to peak and base flows, it would depend on the specifics of the project.

Vegetation changes would result in short term reduced evapotranspiration, increasing soil water, and therefore increasing yield. Road work will reduce soil water infiltration affecting storage, delivery and timing of flow to stream channels.

Overall, activities will have small measurable effects on salmonids and their critical habitat.

Riparian Resilience

Riparian resilience is related to the effects and processes of aquatic management zone to move toward more desirable conditions. Typically, treatments within the aquatic management zone area designed to maintain or improve riparian conditions even though some short-term local impacts are possible. These short-term impacts include a potential slight loss of large wood recruitment from treated riparian stands, potential short-term sediment, and potential minor increase in stream temperature, resulting in a short-term negative effect to this indicator at the local scale. In the long-term the effect would be positive as riparian stand diversity and diameters increase. Utilizing a stream protection buffer determined by the Aquatic Conservation Strategy or the Land Management Plan minimizes short term effects. In the long term, riparian resilience may improve due to benefits expected with future project development.

Timber felling and site restoration can have direct short-term negative effects and long-term beneficial effects, depending on the prescribed treatment within the aquatic management zones. Timber thinning prescriptions within the aquatic management zones can provide long-term beneficial effects by reducing competition between trees allowing trees left to grow and provide future large woody debris. Effects from timber treatments in the aquatic management zone come with short-term negative effects as well just because it takes time for causal mechanisms to improve. Forest thinning remains one of many forest management tools employed to meet ecological, social, and economic objectives on Forests. Thinning within the aquatic management zones is expected to result in a closed canopy within 3-5 years. The temporal boundary for cumulative effects on aquatic species is 5 years. Timber harvest projects within aquatic

management zones are expected to be low and the objectives would be to improve the condition of the riparian areas, which would provide for long-term beneficial effects.

Timber yarding and landings have short-term negative effects through compaction of soil, which would take time to rebound. There are measures that are typically used during timber yarding to minimize soil compaction depending on what types of yarding will be utilized. This can range from utilizing slash for heavy equipment to drive over, to skyline yarding. There is no way to predict what type of yarding practices will be utilized for future projects, but we do know any future project will need ESA consultation, and all previous consultations contain project designs that minimize effects to species and critical habitats. The expectation for effects would still be measurable low-level effects to species and critical habitat for these actions.

There are no causal effects of road maintenance or reconstruction and permanent and temporary road construction for riparian resilience, and the effects of road maintenance or reconstruction would be neutral to this element.

There are no causal effects of permanent and temporary road construction for riparian resilience, and the effects of permanent and temporary road construction would be neutral to this element.

Temporary road removal will have long-term beneficial effects for riparian resilience by removing the road prism, soil decompaction, and recontouring the landscape within the aquatic management zone.

Overall, activities will have small measurable negative effects and small measurable beneficial effects to salmonids and their critical habitat.

Road Density/Location

Road density/location is based on where new permanent roads and temporary roads will be built. Almost all forest watersheds in the action area have existing primary and secondary road systems in place. Current road systems need maintenance and repairs over time to maintain safety standards. Any effects to salmonids and their critical habitat occur on roads that are hydrologically connected to LFH or where effects can reach LFH. The probability of new roads being built somewhere in the action area is high, it would be more likely to occur where the lands are operable and land management plans allow. The probability of the new road being a permanent road is relatively low, as new permanent roads need funding to build and maintain. Temporary roads are associated with timber harvest needs. If timber is harvested in an area, temporary road will be built.

There are no causal effects of timber harvest for road density/location, and the effects of timber harvest would be neutral to this element.

There are no causal effects of road maintenance or reconstruction for road density/location, and the effects of road maintenance or reconstruction would be neutral to this element.

New road building would be the largest contributor to increased road density of all actions for the affected environment. Permanent roads are considered system roads, while temporary roads are considered non-system roads. Temporary roads will increase road density for a short time during logging operations, but the increase will be minor over the existing condition as all

temporary roads will be fully decommissioned (temporary road removal) after sale contract harvest and yarding operations are complete. Increased road density affects salmonids and critical habitat typically through habitat fragmentation and sedimentation. Legacy roads on Forest Service lands are the largest contributor to habitat fragmentation of salmonids. New roads would need to meet certain Forest Service standards, which are designed to minimize effects on aquatic systems. Even so, all new roads would continue to have negative effects to salmonids for as long as they are in place.

Overall, activities will have small measurable effects on salmonids and their critical habitat.

Sediment/Turbidity

Sediment and turbidity tend to be the most frequent effects associated with timber harvest tasks, road maintenance and reconstruction and road construction. The potential for surface erosion is directly related to the amount of bare compacted soil exposed to rainfall and runoff and timber road haul; hence timber harvest activities may contribute large quantities of fine sediment to streams (Chamberlin et al. 1991, Swanston and Swanson 1976). Not all hillside erosion reaches stream channels (Chamberlin et al. 1991, Rice et al. 1972), but the potential exists for negative effects to the sedimentation/turbidity pathway. Heavy equipment can cause compaction and rutting of the ground which aids the movement of sediment- laden runoff. Excessive erosion may occur on poorly designed and placed skid trails, skid trails used during wet conditions, and trails that have connection to stream channels. Soil compaction from ground-based systems can be reduced when this equipment is restricted to dry soil conditions (Sidle and Drlica 1981). Fine sediment deposited in spawning gravels can reduce egg survival and developing alevins (Everest et al. 1987; Hicks et al. 1991) by reducing the availability of dissolved oxygen in the gravel.

The capacity of headwater streams to retain sediment is supported by many studies. The channels contain vegetation and woody debris of varying amounts, much of it small in diameter but in these small streams this material acts as an important obstruction to store sediment (Sullivan et al. 1987). Some studies indicate that annual sediment yields from small watershed make up less than ten percent of the sediment stored in channels (Swanson and Lienkaemper 1978), in other words, headwater stream channels with have a tremendous capability to store large amounts of fine sediment (Benda et al. 2005).

Chase et al. (2019) completed a soil and streamside effects study on 30 harvest units across Western Oregon and Washington on Weyerhaueser ownership on slopes up to 160%; with most 0-60%. Timing of clear-cut harvest was during the wet season (November – March). They observed that streams with riparian buffers >15m (49’) wide had little to no stream-adjacent disturbance.

Broderson (1973, cited in Castelle et al 1992) found that 50-foot buffers were sufficient for controlling most sedimentation on less than 50% slopes, while steeper slopes required wider buffers. Broderson noted that buffers have little or no effect on sediment removal if sediment-laden water flows across the buffers as channelized flow; buffers can be effective only if they resist channelization and maintain overland flow as sheetflow. A maximum buffer width of 200 feet was found to be effective even on extremely steep slopes.

The timing of turbidity delivery to non-LFH stream channels will coincide with the rainstorms (and snowmelt) from fall through spring that will also naturally increase flow volume and turbidity in LFH. Aquatic conservation plans tend to limit the usage of heavy equipment within 150 feet of a stream. Even when following the aquatic conservation strategy and utilizing BMPs, there is the possibility of sediment entering streams and creating direct and indirect effects to species. These effects are typically short-term effects lasting less than one year. The nature and timing of future projects, and inclusion of minimization measures are unknown which means the effects of sediment and turbidity magnitude is uncertain. It is typical to include minimization measures when there is LFH within the action area. Even with minimization measures, sedimentation and turbidity could reach measurable effects for any given project based on the incremental effects of almost every timber harvest activity creating sedimentation/turbidity. Even if measurable effects occur, these effects should be low level effects but would still reach a level of significance to LFH.

Road maintenance and reconstruction may create source areas for sediment input to streams, particularly during heavy rainfall events. Failures of road fill at stream crossings are common source areas for landslide and debris torrents. Vegetated buffer areas ranging in width from 40 to100 feet were successful in keeping sediment out of streams (Corbett and Lynch 1985, Lynch et al. 1985, Moring 1982, Newbold et al. 1980, and Gomi et al. 2005). Broderson (1973, cited in Castelle et al 1992) found that 50-foot buffers were sufficient for controlling most sedimentation on less than 50% slopes, while steeper slopes required wider buffers. A maximum buffer width of 200 feet was found to be effective even on extremely steep slopes. Previous consultations have design measures that minimize sediment effects to streams.

Surfacing roads with rock aggregate or pavement can significantly reduce the amount of erosion from the road surface (Burroughs and King 1989), particularly if used in the wet season. The type of rock used for surfacing can also influence sediment production. Hard, durable rock is more resistant to breakdown while softer rock is more easily crushed into particles small enough to be washed from the road surface (Bilby et al. 1989). Timber hauling may occur year-round, or seasonally depending on the road surface, design, and condition. The Forest Service Standards are described in several different Forest Service publications, including but not limited to the list in Appendix B. Forest Service Road Standards.

The turbidity generated by restoring flow at a culvert site is likely to affect the channel at the installation and removal sites and for approximately 1000 feet. Foltz and Yanosek (2005) found that turbidity decreased by an order of magnitude within 328 feet of the disturbance and that turbidity was not expected to be detectable more than 2,400 feet (800 meters) downstream from the project site.

Overall, road maintenance and reconstruction activities are ground disturbing actions and localized increase in sedimentation to streams is expected. However, future projects will most likely include minimization measures in their consultations which will help alleviate effects on species and critical habitats.

In general, modern road construction practices produce less sediment delivered to streams from forest roads than older road construction practices (Copstead 1998). Road construction and maintenance may create source areas for sediment input to streams, particularly during heavy

rainfall events. Failures of road fill at stream crossings are common source areas for landslide and debris torrents. Vegetated buffer areas ranging in width from 40 to100 feet were successful in keeping sediment out of streams (Corbett and Lynch 1985, Lynch et al. 1985, Moring 1982, Newbold et al. 1980, and Gomi et al. 2005). Broderson (1973, cited in Castelle et al 1992) found that 50-foot buffers were sufficient for controlling most sedimentation on less than 50% slopes, while steeper slopes required wider buffers. A maximum buffer width of 200 feet was found to be effective even on extremely steep slopes.

Some increases in sedimentation (and resulting increases in turbidity) can be expected during culvert and bridge replacement, as equipment is operating in the stream channel and some trees would need to be removed from the stream bank. However, such increases are likely to be of local extent and of short duration. From past experience, typically, only 1-2 yards, or less, of sediment enters a stream during culvert projects.

Forest roads are compacted surfaces and can potentially allow for accelerated erosion and potential sediment delivery to stream channels from a variety of sources, including small slumps and slides into the roadway from the cut bank, water channeling from the road or ditches if not properly directed and controlled, and blocked culverts and road fill washouts during floods. Road construction is a ground disturbing action, and the potential exists for some increased sedimentation to streams.

Past consultations include minimization measures to decrease the effects of new road construction to streams. There is reason to believe these minimization measures will be included in any potential future consultation in the action area. Even if minimization measures were included in future consultation, there is reason to believe the effects of sedimentation for new road building would be the largest contributor of sedimentation to stream systems.

Overall, activities will have small measurable effects on salmonids and their critical habitat.

Spawning Substrate

Timber harvest activity effects have similar mechanisms between spawning substrate and sediment/turbidity. Spawning substrate can be defined by water quality, water quantity, and substrate. Spawning substrates are necessary for successful reproduction of Pacific salmonids. Fine sediment deposited in spawning gravels can reduce egg survival and developing alevins (Everest et al. 1987; Hicks et al. 1991) by reducing the availability of dissolved oxygen in the gravel. Sedimentation/turbidity is discussed above as well as temperature. Water quality factors such as dissolved oxygen and water quantity are also important for spawning substrate.

Road reconstruction and new roads construction have the potential to have the largest negative effects to spawning substrate within the affected environment if they are built near the aquatic management zone. Habitat fragmentation would have negative effects on bedload movement. Effects would occur throughout the duration of the road.

Overall, activities will have small measurable effects on salmonids and their critical habitat.

Conclusion/Determinations

Species specific salmonid and their critical habitat effects are located in Table 9. For all the salmonid species, we determined there could be measurable effects with future projects with the action area, therefore, future projects **may affect, likely to adversely** affect salmonids and their critical habitat (Table 9), with exception. The critical habitat exception is for two critical habitats that are not located in the action area or within 2 miles of the action area. For these two critical habitat areas, the effects of any future action would be absent, therefore, a **no effect** determination was made (Table 9).

Eulachon

The eulachon is a smelt in the Family Osmeridae. The Genus Thaleichthys has only one species. Eulachon are an anadromous fish, meaning adults spend most of their life in the ocean but migrate into fresh water to spawn. Overall eulachon are endemic to the northeastern Pacific Ocean, ranging from northern California to the southeastern Bering Sea in Bristol Bay, Alaska (Willson et al. 2006). This distribution coincides closely with the distribution of the coastal temperate rain forest ecosystem on the west coast of North America (except for populations spawning west of Cook Inlet Alaska). On March 18, 2010, the southern Distinct Population Segment of Eulachon were listed as threatened under the Endangered Species Act (ESA) (75 FR 13012). The southern DPS consists of populations spawning in rivers south of the Nass River in British Columbia, Canada, to, and including, the Mad River in California (NMFS 2011a). Detailed information about the biology, habitat, and conservation status and trends of eulachon can be found in the listing regulations and critical habitat designations published in the Federal Register (75 FR 13012), as well as in recovery plans and other sources (NMFS 2011a; NMFS 2011b).

Eulachon spend 95 to 98 percent of their lives at sea (Hay and McCarter, 2000), little is known of their life history for their saltwater life stage (NMFS 2011a). Spawning generally occurs in January, February, and March in the Columbia River and the coastal rivers of Washington and Oregon (NMFS 2011b). Entry into the spawning rivers appears to be related to water temperature and the occurrence of high tides (NMFS 2011a). Eulachon lay their eggs in freshwater where the eggs stick to the substrate until they hatch 20-40 days later. The larvae are carried to the estuary/ocean by the flow of the natal creek or river. It is not known how long larval eulachon remain in the estuary before entering the ocean. Similar to salmon, juvenile eulachon are thought to imprint on the chemical signature of their natal river basins (NMFS 2011a). Once juvenile eulachon enter the ocean, they move from shallow nearshore areas to deeper areas over the continental shelf. Larvae and young juveniles become widely distributed in coastal waters, where they are typically found near the ocean bottom in waters 20-180 m deep. Little information is available about eulachon movements in nearshore marine areas and the open ocean (NMFS 2011a).

In the portion of the species' range that lies south of the U.S.–Canada border, most eulachon production originates in the Columbia River Basin. Within the Columbia River Basin, the major and most consistent spawning runs return to the mainstem of the Columbia River and the Cowlitz River (Gustafson et al. 2010). Spawning also occurs in other tributaries to the Columbia

River, including the Grays, Elochoman, Kalama, Lewis, and Sandy rivers (WDFW and ODFW 2001). Eulachon also spawn in the lower portions of the Elwha and Quinault rivers in Washington (NMFS 2011b). Little is known about southern DPS of eulachon compared to salmonids, although the distribution and biology of this species is similar to salmon and steelhead.

NMFS (2011c) identified the following Physical and Biological Features essential to conservation of eulachon as:

1. Freshwater spawning and incubation sites with water flow, quality and temperature conditions and substrate supporting spawning and incubation, and with migratory access for adults and juveniles. These features are essential to conservation because without them the species cannot successfully spawn and produce offspring.
2. Freshwater and estuarine migration corridors associated with spawning and incubation sites that are free of obstruction and with water flow, quality and temperature conditions supporting larval and adult mobility, and with abundant prey items supporting larval feeding after the yolk sac is depleted. These features are essential to conservation because they allow adult fish to swim upstream to reach spawning areas and they allow larval fish to proceed downstream and reach the ocean.
3. Nearshore and offshore marine foraging habitat with water quality and available prey, supporting juveniles and adult survival. Eulachon prey on a wide variety of species including crustaceans such as copepods and euphausiids, unidentified malacostracans, cumaceans, mysids, barnacle larvae, and worm larvae. These features are essential to conservation because they allow juvenile fish to survive, grow, and reach maturity, and they allow adult fish to survive and return to freshwater systems to spawn.

When NMFS listed eulachon as a threatened species, they identified a number of activities that may affect the physical and biological features essential to the southern DPS of eulachon such that special management considerations or protection may be required. This information is found in Table 7 and relates to dredging activities during eulachon spawning season. Future projects within the affected environment would not meet this consideration.

Effects on Eulachon

Table 3 shows which National Forests could have effects to Eulachon due to a low probability individual eulachon could be affected through downstream processes of future actions when eulachon are in freshwater systems or near estuaries during their spawning and rearing phases. It is reasonable to say effects to eulachon, if any, would be due to small measurable increases in sedimentation on the few streams and rivers upstream of where eulachon may occur. Sedimentation could be caused by projects involving timber harvest and road maintenance even when all pertinent Best Management Practices are implemented to minimize effects. Any sedimentation will have short-term effects to eulachon typically not lasting longer than one year.

Sedimentation effects will be minor for eulachon since they inhabit the downstream most portions of the rivers within the Columbia River system and typically stick to the intertidal zones in other rivers where they spawn and rear. The dispersed nature of the affected environments

combined with the fact that eulachon spend less than 5 percent of their lifespan in freshwater or estuaries and have limited distribution downstream of National Forest System lands. It is hard to predict where projects will be proposed within the affected environment. Since all proposed projects would need to go through the ESA consultation process, there is the ability to agree to project design criteria that minimize effects to species. However, given the unknowns of future proposed projects within the affected environment and unknown future consultation agreements, impacts to eulachon cannot be dismissed. Any future project effects to eulachon are expected to be minor effects on species level abundance or productivity.

Since future proposed activities could have measurable effects on eulachon, it is concluded the proposed action **may affect, likely to adversely affect** eulachon.

Effects on Eulachon Critical Habitat

Within the action area, future proposed actions have little opportunity to create effects to eulachon critical habitat. There is no critical habitat within the affected environment. Two forests have critical habitat within 2 miles of the affected environment, Mt. Hood National Forest and Siuslaw National Forest (Figure 28). The probability of projects occurring in these areas is possible. If projects were to occur in these areas, the causal effects to critical habitat would be through the sedimentation/turbidity pathway. These effects to PBFs would be short-term in nature lasting up to one year after projects are completed. The flows of both the Columbia River and the Umpqua River is high enough that any effects to species and critical habitat would be short lived as background levels would return quickly. Road and timber projects could create sedimentation effects to eulachon critical habitat. Since designated critical habitat is typically far downstream of proposed actions, the magnitude of effect for sedimentation would be minor but still may be measurable given the unknowns of future projects.

Since future proposed activities could have measurable effects to eulachon critical habitat, it is concluded the proposed action **may affect, likely to adversely affect** eulachon critical habitat.

Green Sturgeon, Southern DPS

Two DPSs have been defined for green sturgeon: a northern DPS (spawning populations in the Klamath and Rogue rivers) and a southern DPS (spawners in the Sacramento River). The southern DPS (sDPS) of green sturgeon was listed as threatened in 2006 and includes all naturally spawned populations of green sturgeon that occur south of the Eel River in Humboldt County, California (71 FR 17757). When not spawning, this anadromous species is broadly distributed in nearshore marine areas from Mexico to the Bering Sea. Although it is commonly observed in bays, estuaries, and sometimes the deep riverine mainstem in lower elevation reaches of non-natal rivers along the west coast of North America, the distribution and timing of estuarine use are poorly understood. The principal factor for the decline of southern green sturgeon is the reduction of its spawning area to a single known population limited to a small portion of the Sacramento River. It is currently at risk of extinction primarily because of human-induced ‘‘takes’’ involving elimination of freshwater spawning habitat, degradation of freshwater and estuarine habitat quality, water diversions, fishing, and other causes (NMFS 2018). Adequate water flow and temperature are issues of concern. Water diversions pose an unknown but potentially serious threat within the Sacramento and Feather Rivers and the

Sacramento River Delta. Poaching also poses an unknown but potentially serious threat because of high demand for sturgeon caviar. The effects of contaminants and nonnative species are also unknown but potentially serious threats. Retention of southern green sturgeon in both recreational and commercial fisheries is now prohibited within the western states, but the effect of capture/release in these fisheries is unknown. Climate change is likely to reduce the conservation value of designated critical habitats in the Pacific Northwest. Limiting factors outlined in the recovery plan (NMFS 2018) lists four factors: reduction in spawning area to a single population, lack of water quantity, poor water quality, and poaching (Table 7). Detailed information on the natural history of green sturgeon is provided in the proposed rule to designate critical habitat (73 FR 52084; September 8, 2008) and in the final biological report (NMFS 2009a) prepared in support of this final rule.

Since spawning occurs in the Eel River in Humboldt County, California, any future projects in the affected environment, adult and subadult southern green sturgeon would be the only individuals that may be affected by future projects. These subadult and adult southern green sturgeon would be distributed near shore marine areas or near designated critical habitat areas. Critical habitat is described in Table 8. The Lower Columbia River estuary to river mile 46 and the Strait of Juan de Fuca are at least 5 miles downstream of any activity occurring on National Forest System lands.

NMFS (2009) identified the following Physical and Biological Features essential to conservation of Southern green sturgeon as:

1. Pesticide application;
2. Activities affecting bottom substrates;
    a. Adversely affecting prey resources;
    b. Degrade water quality through resuspension of contaminated sediments;
3. Commercial shipping and other activities that discharge contaminates;
4. Disposal of dredge materials;
5. Bottom trawl fisheries.

Effects on Green Sturgeon

Project activities could occur adjacent to marine systems or where southern green sturgeon may be located. If southern green sturgeon are in an area when future projects are implemented, the only causal mechanism for effects to species would be a slight increase in sediment/turbidity plumes. Any sediment/turbidity increase would most likely cause insignificant effects to visibility of prey for subadult and adult southern green sturgeon. There are no other causal mechanisms for effects relating to the limiting factors for southern green sturgeon.

Since future proposed activities could have insignificant effects on southern green sturgeon, it is concluded the proposed action **may affect, not likely to adversely affect** southern green sturgeon.

Effects on Green Sturgeon Critical Habitat

There is a small section of critical habitat located in the mouth of the Umpqua River at Winchester Bay (Figure 29), near the affected environment on the Siuslaw National Forest. This area is the Oregon Dunes National Recreation Area, a special area designation where timber harvest and road building will most likely not occur. Even if there were a small vegetation project along within the Oregon Dunes National Recreation Area, the only causal mechanism for effects to species and critical habitat would be a slight increase in sediment/turbidity plumes. These plumes would be expected to return to background levels quickly due to the amount of water flowing down the Umpqua River at Winchester Bay. The only causal mechanism from the aquatic deconstructed action table that is similar to the PBFs for southern green sturgeon might be the sediment/turbidity effects; however, that is a stretch since the PBFs are related to contaminated sediments. The probability of this occurring is low, but not zero.

Since future proposed activities could have insignificant effects to southern green sturgeon critical habitat, it is concluded the proposed action **may affect, not likely to adversely affect** southern green sturgeon designated critical habitat.

Killer Whale

Southern resident killer whales spend several months of the summer and fall each year in the inland waters of the Salish Sea including Washington's Puget Sound. The southern resident killer whale population has fluctuated considerably since the 1970's. The population now numbers in the 70s. When in the Salish Sea, southern resident killer whale prey primarily on adult Chinook salmon returning to the Fraser River in Canada (NMFS 2016i). During summer forays to the outer coast the whales may also consume salmon from other stocks, such as from the Columbia River system. Southern resident killer whales have been known to travel to Southeast Alaska to Monterey Bay in California preying on salmonids (NMFS 2016i). Overall studies have shown that their prey base includes 22 species of fish and one species of squid (Ford et al. 1998; Ford and Ellis 2006). Their late summer and fall diet is more of a mystery, but data indicates that it includes coho salmon in late summer and Puget Sound Chinook and chum salmon in fall (NMFS 2021a). Detailed information about the biology, habitat, and conservation status and trends of southern resident killer whale can be found in the listing regulations and critical habitat designations published in the Federal Register (70 FR 69903), as well as in the recovery plans and other sources.

NMFS (2021a) identified the following physical and biological features essential to conservation of southern resident killer whale (essential features):

1. Water quality to support growth and development;
2. Prey species of sufficient quantity, quality, and availability to support individual growth, reproduction, and development, as well as overall population growth; and
3. Passage conditions to allow for migration, resting, and foraging.

Researchers are unsure about which threats are most significant to the southern resident killer whale population, and none of the threats have been identified as the cause of the recent decline

of the population (Krahn et al. 2002). There is limited information on how these factors or additional unknown factors may be affecting southern resident killer whale in coastal waters in winter. It is possible that two or more of these factors may act together to harm the whales. The small size of the population increases the level of concern for all these risks (NMFS 2008).

Water quality concerns are focused on biomagnification of pollutants, such as chemical pollution from industrial waste and oil spills. The contaminants are concentrated near areas of high human population and industrialization or sedimentation and affect southern resident killer whale typically through trophic relationships they have with their prey base.

Quantity and quality of prey availability is identified as a major threat to the survival of southern resident killer whale (PFMC 2020). Salmon, particularly Chinook salmon, are the preferred prey of southern resident killer whale in inland waters of Washington State during spring, summer, and early fall which is most likely due to the species' large size, high fat and energy content, and year-round occurrence in the area (Baird and Hanson 2004; Hauser 2006). Killer whales also captured older (larger) than average Chinook salmon (Ford and Ellis 2006). When prey abundance is low, southern resident killer whale may spend more time and energy foraging than when prey abundance is high; potential consequences for fitness include reduced reproductive rates and higher mortality rates. Ford and Ellis (2006) correlated coast-wide reduction in Chinook abundance (Alaska, British Columbia, and Washington) with decreased survival of resident whales (Northern and Southern Residents), but changes in killer whale abundance have not been linked to local areas or changes in salmon stock groups. No recent changes in salmon populations are obviously apparent that might be responsible for the recent decline in the killer whale population between 1996 and 2001 (NMFS 2008).

Past peer reviewed information indicated salmon are the preferred prey of southern resident killer whale year-round (Krahn et al. 2002, 2004), including while in coastal waters, and that Chinook are the preferred salmon species (Ford and Ellis 2006). Prey abundance is a concern for killer whales, both near-term and long-term. To survive in the near term, southern resident killer whale require regular supplies of adult Chinook prey in the ocean. To recover over the longer term, southern resident killer whale require abundant Chinook stocks coast-wide, likely including stocks from the Columbia River (Couture et al. 2022).

Although there is currently no quantitative model that identifies a low abundance that will definitively cause adverse effects to the whales or appreciably alter the value of their habitat, there is evidence southern resident killer whale and other killer whale populations that are known to consume Chinook salmon may have experienced adverse effects from low prey availability in the late 1990s (NMFS 2021a). In recent years, peer reviewed information found the relationship between Chinook salmon abundance and SRKW demographic rates has weakened, as southern resident killer whale status has declined despite varying levels of Chinook abundance (NMFS 2021a, 2021b; PFMC 2020). There are several challenges to quantitatively characterize the relationship between southern resident killer whale and Chinook salmon, including (1) there are multiple, interacting factors at play; (2) the strength of any one effect likely varies through time leading to a situation known as "non-stationarity"; and (3) the small population size of the southern resident killer whale population makes detection of any true effects difficult, using statistical analyses, due to a high degree of random variation (NMFS 2021a). Multiple threats

affect southern resident killer whale demographic performance through time, in addition to random chance, and these effects can confound the analysis of the effects of prey abundance (NMFS 2021a).

Effects on Southern Resident Killer Whale

The above effects analysis for Chinook salmon concluded the proposed action is likely to adversely affect Chinook salmon and Chinook salmon designated critical habitat for most ESUs, therefore because of the trophic relationship between Chinook salmon and southern resident killer whale will lead to an insignificant reduction to the prey base of southern resident killer whale. Southern resident killer whale are limited to marine water habitats and would not be directly exposed to any proposed activities within this project, but there is the potential projects implement within the affected environment could create indirect effects through the trophic web with relation to southern resident killer whale prey availability.

Indirect effects to southern resident killer whale would be expressed through effects to Chinook and other salmon species comprising the prey base. However, these indirect effects are likely to be insignificant. NMFS found that Pacific salmon fisheries (commercial, recreational, and tribal harvest in ocean and inland waters) conducted during the 2008-2009 regulatory cycle would cause a negligible reduction in prey resources, with no detectible change in the ratio of prey availability to needs for southern resident killer whale within their coastal range (NMFS 2008).

Proposed Forest Service project activities within the riparian area will mostly have short-term negative effects to salmonids and a few long-term benefits. Fixing the factors limiting the viability of natural-origin populations of salmon is essential for long-term viability of southern resident killer whale. These expected short-term effects would be indistinguishable from background levels compared to salmonid populations across the action area because few watersheds will be impacted annually. Therefore, indirect effects on southern resident killer whale prey, is likely to be discountable or insignificant in the short-term and potentially beneficial effects in the long-term. Any negative effects to salmonids including Chinook salmon would result in an insignificant reduction in adult equivalent prey resources for southern resident killer whale that may intercept these species within their range. For the reasons expressed immediately above, it is concluded that the proposed action **may affect, not likely to adversely affect** ESA-listed southern resident killer whale.

Effects on Southern Resident Killer Whale Critical Habitat

Project activities could occur adjacent to southern resident killer whale designated critical habitat but not within a two-mile buffer of the affected environment (Figure 30). Effects to southern resident killer whale designated critical habitat would be limited to the effects to the southern resident killer whale prey base. Since the proposed activities effects to salmonids would result in an insignificant reduction in adult equivalent prey resources as expressed immediate above it is concluded that the proposed action **may affect, not likely to adversely affect** southern resident killer whale designated critical habitat.

Determinations

The final effects for ESA-listed species and designated critical habitats under NMFS' jurisdiction are presented in Table 9. Exposure of most salmon ESUs and eulachon is likely and effects to these species and their critical habitat is expected to follow the effects pathways identified in Table 10, Table 11, Table 12, Table 13, Table 14, and Table 15.

Cumulative Effects

Under the ESA, cumulative effects are the consequences of future State, tribal, local, and private activities that are reasonably certain to occur within the action area. The action area for rescission of the Roadless Rule is owned and managed by the US Forest Service. As such, nearly all activities that will occur in the action area will require ESA consultation, with the FS as the lead agency for those consultations. Although private activities may occur within proximity to currently inventoried roadless areas, there are no reasonably certain private activities that overlap the time and scale of this consultation. Cumulative effects will be considered more thoroughly during project level consultation.

Project-level Consultation

The proposed action described below is a "framework programmatic action" as defined in 50 CFR 402.02, where the framework programmatic action only establishes a framework for the development of specific future action(s) but does not authorize any future action(s). The effects to listed species and designated critical habitat of future actions that are subsequently authorized, funded, or carried out under this program will be addressed in subsequent section 7 consultations, as appropriate. The Forest Service proposes to rescind the 2001 Roadless Rule. This action does not affect state-level roadless rules that may exist, nor does it preclude the development of state-led roadless efforts. Rescinding the Roadless Rule does not authorize, fund, or carry out future forest management activities that could occur in these areas. Rescinding the Roadless Rule does not constrain or alter the statutory requirement of sections 7(a)(1) through section 7(a)(4) of the Endangered Species Act. All future activities in currently inventoried roadless areas will require compliance with the ESA prior to implementation. Rescinding the Roadless Rule does not alter the protections afforded to designated wilderness areas or areas protected by other law, regulation, or policy.

Although rescinding the Roadless Rule does not dictate or prescribe forest management in previously inventoried roadless areas, it is reasonably certain that active forest management and road construction will increase in these areas. However, currently the Forest Service does not have a reliable method to identify, describe, or analyze the likelihood and location of any future projects that may be proposed for currently inventoried roadless areas.

Future projects in current inventoried roadless areas will require compliance environmental laws and regulations, including the Endangered Species Act. As is typical, future project designs in these areas will avoid and minimize adverse effects to ESA-listed resources to the greatest extent feasible. All projects must be consistent with the land management plan that governs management for their geographic location.

Adverse effects to ESA-listed resources from road construction and timber projects in roadless areas may include, but are not limited to: behavioral avoidance, increased energy expenditure, reduction in fitness, abandonment, degradation of unique critical habitat features, and even mortality. However, these effects will be fully explored during project-level compliance efforts, including consultation under section 7(a)(2) of the Endangered Species Act.

The Forest Service has determined that rescinding the Roadless Rule will have no direct effect on ESA-listed resources because it does not authorize, fund, or carry out future forest management activities in areas currently identified as inventoried roadless areas. However, after rescinding the Roadless Rule, future forest management may occur in inventoried roadless areas that result in road construction and timber harvest activities. These future activities may affect ESA-listed resources. Future activities in current inventoried roadless areas will comply with the ESA through project-level consultation, when details about the project are known. The Forest Service shall consult with the Services on any future projects in currently inventoried roadless areas that may affect ESA -listed resources.

The Magnuson–Stevens Fishery Conservation and Management Act (MSA) mandates that agencies initiate consultation with the National Marine Fisheries Service (NMFS) for any activities that could affect essential fish habitat (EFH). This consultation is completed for site-specific projects with ground-disturbing activity. The application of Forest-wide standards and guidelines and Best Management Practices (USDA 2012) developed to meet soil protection, water quality standards, and fish habitat protection will help protect EFH on National Forest Service lands and adjacent estuarine and marine waters. Under all alternatives any action that would be taken following the Roadless Rule Rescission that could affect EFH would undergo project-level consultation. All projects must be consistent with the forest land management plan.

CONCLUSION

Rescinding the 2001 Roadless Rule will increase the likelihood of road construction and timber harvests occurring in locations previously considered inventoried roadless areas. In many cases the Forest Service anticipates this increased activity to adversely affect ESA-listed species and their designated critical habitats. However, all future projects in these areas will be required to comply with environmental law, regulation, and policy, including the Endangered Species Act. Projects that are determined that they “may affect” ESA-listed species or designated critical habitats will undergo consultation under section 7(a)(2) of the ESA. Through consultation with the Services at the project level, jeopardizing listed species or adversely modifying critical habitat should be avoided.

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APPENDICES

Appendix A. Maps

Species critical habitat maps compared to the affected environment (action area) with a 2-mile buffer.

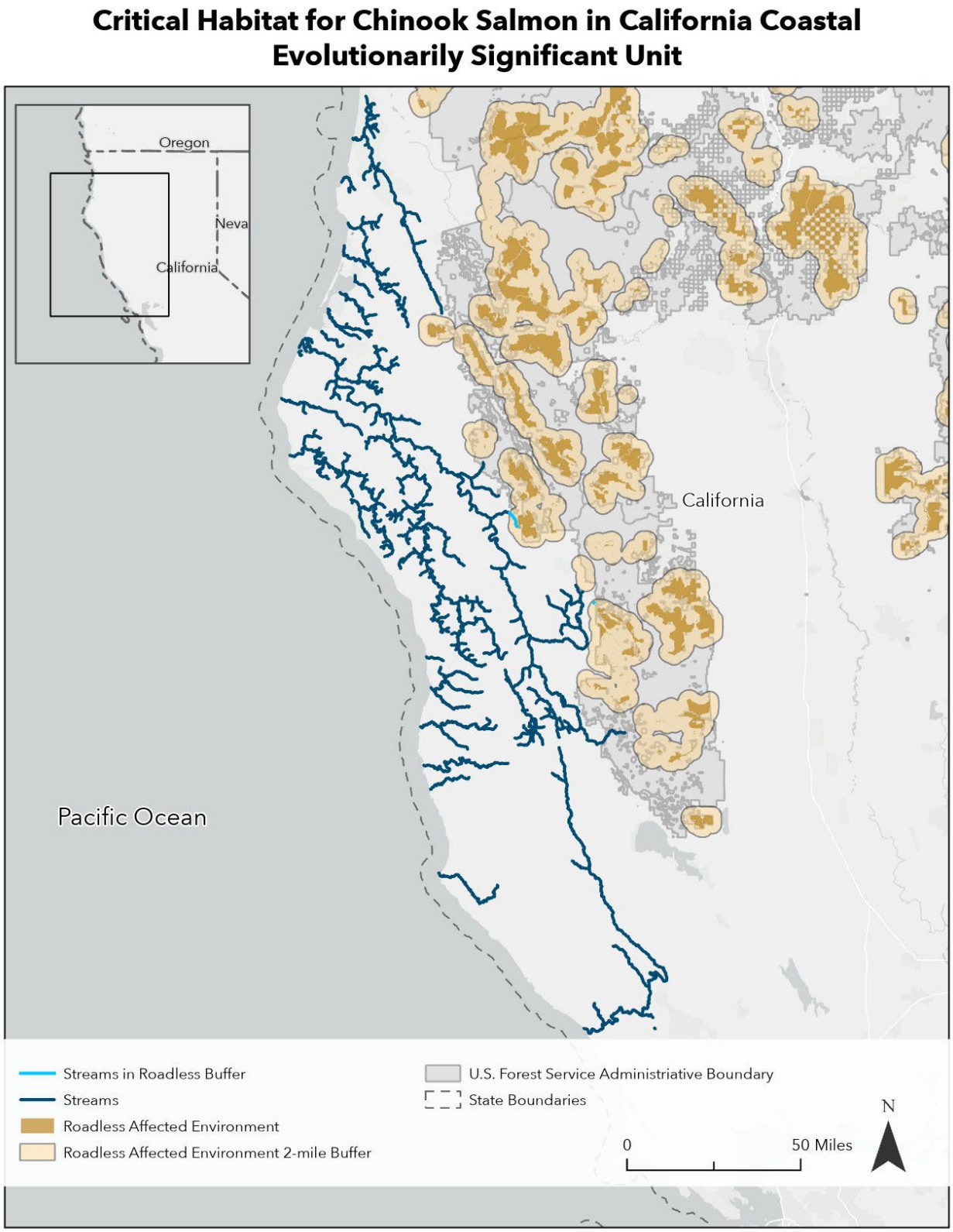


Figure 1. California Coastal Chinook salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

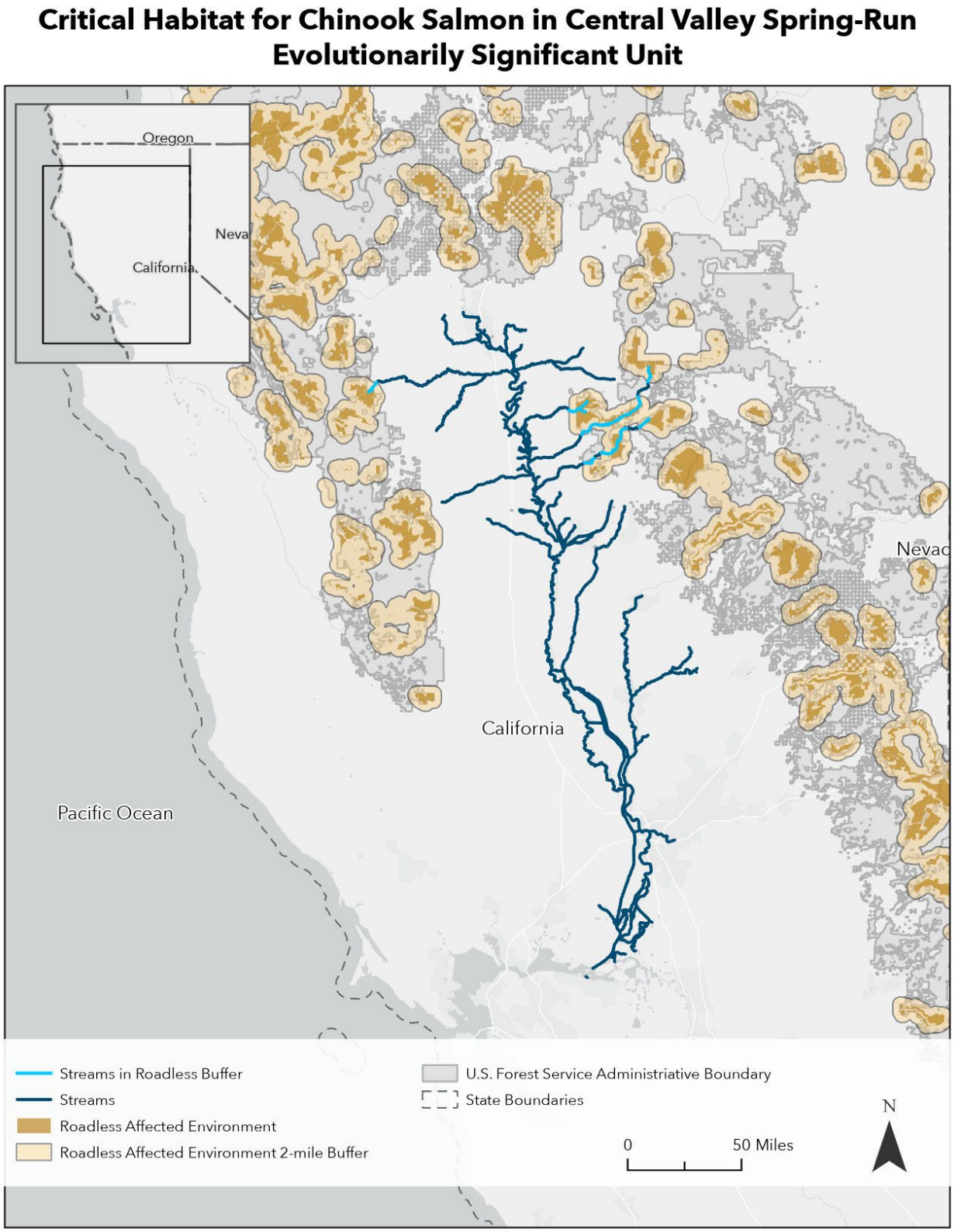


Figure 2. Central Valley Chinook salmon, spring-run ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

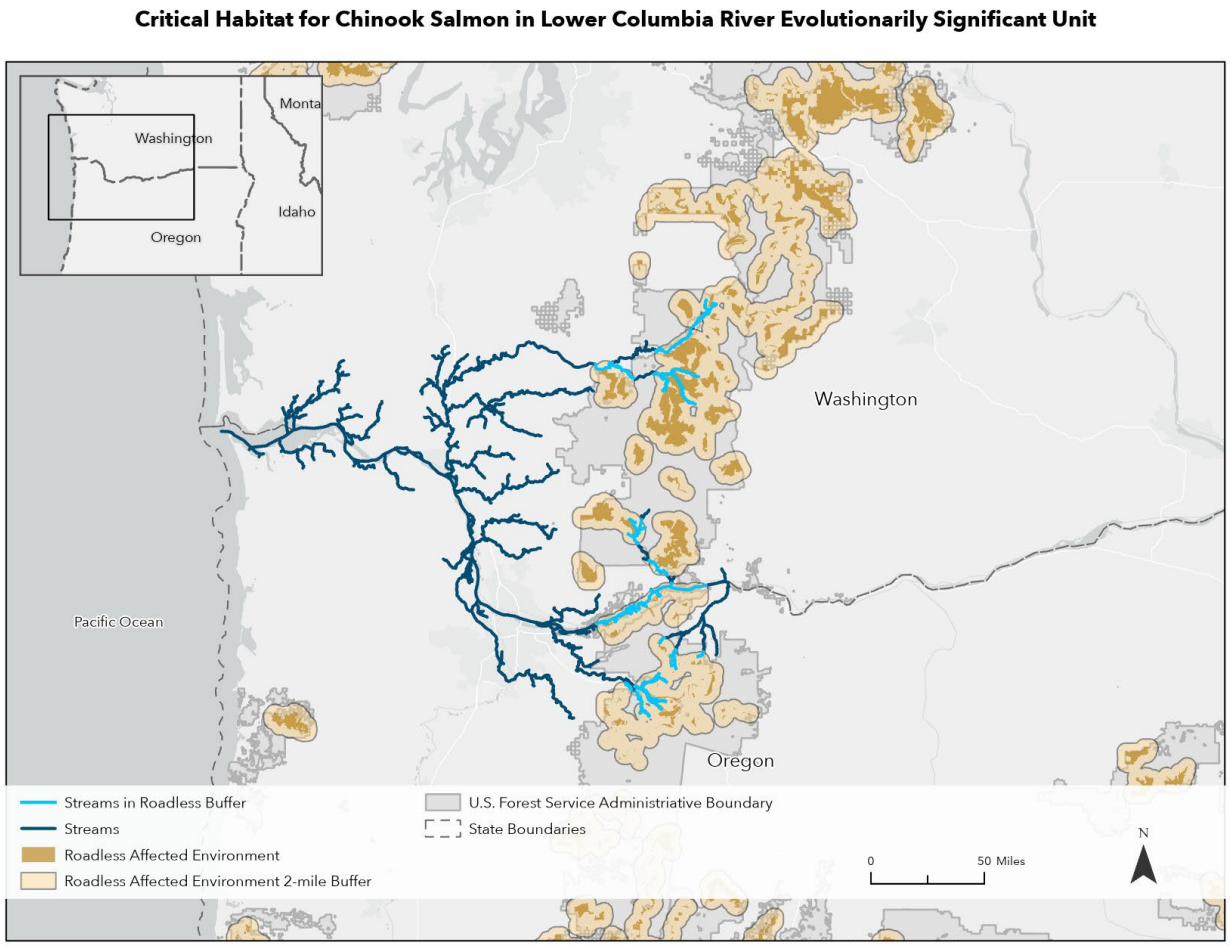


Figure 3. Lower Columbia River Chinook salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

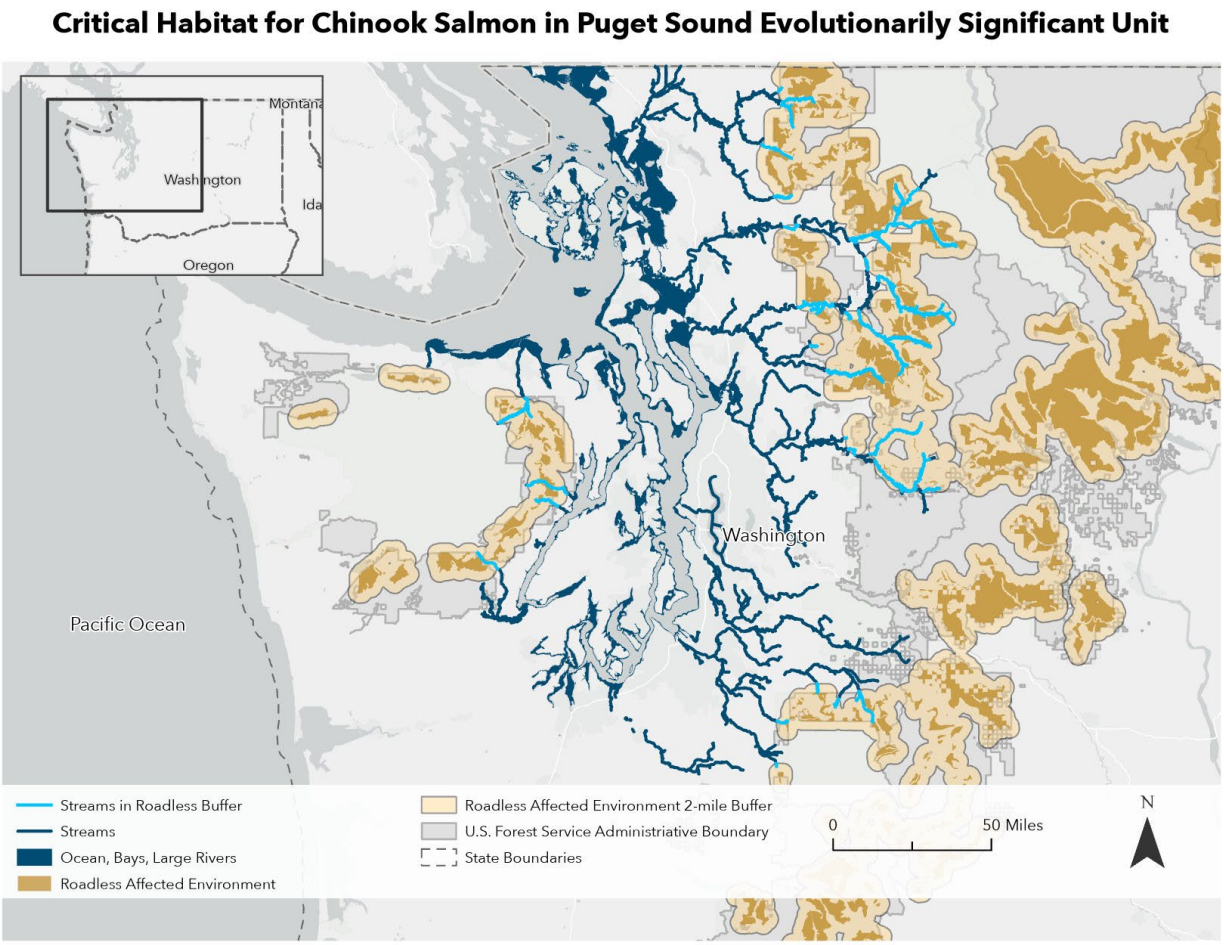


Figure 4. Puget Sound Chinook salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

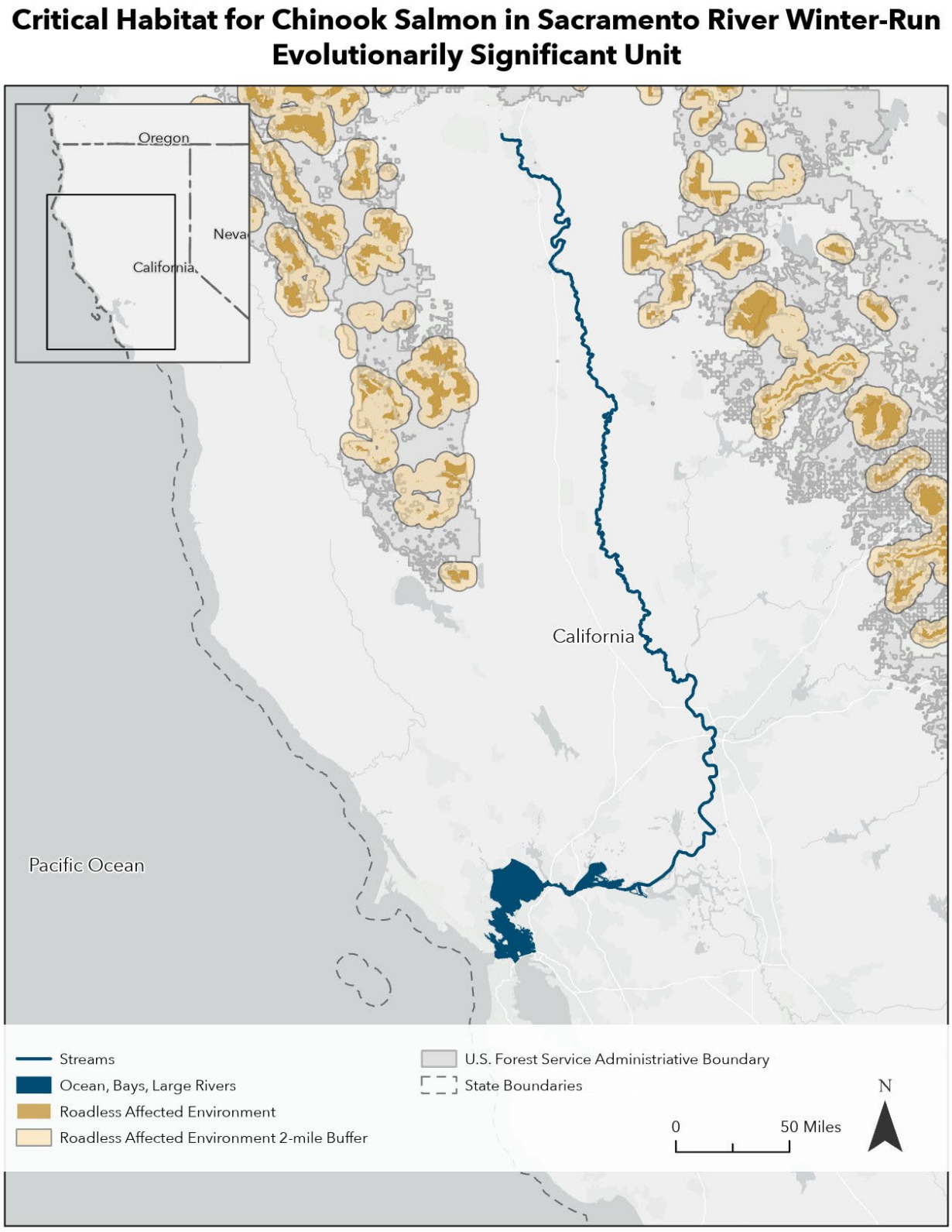


Figure 5. Sacramento River Chinook salmon winter-run ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

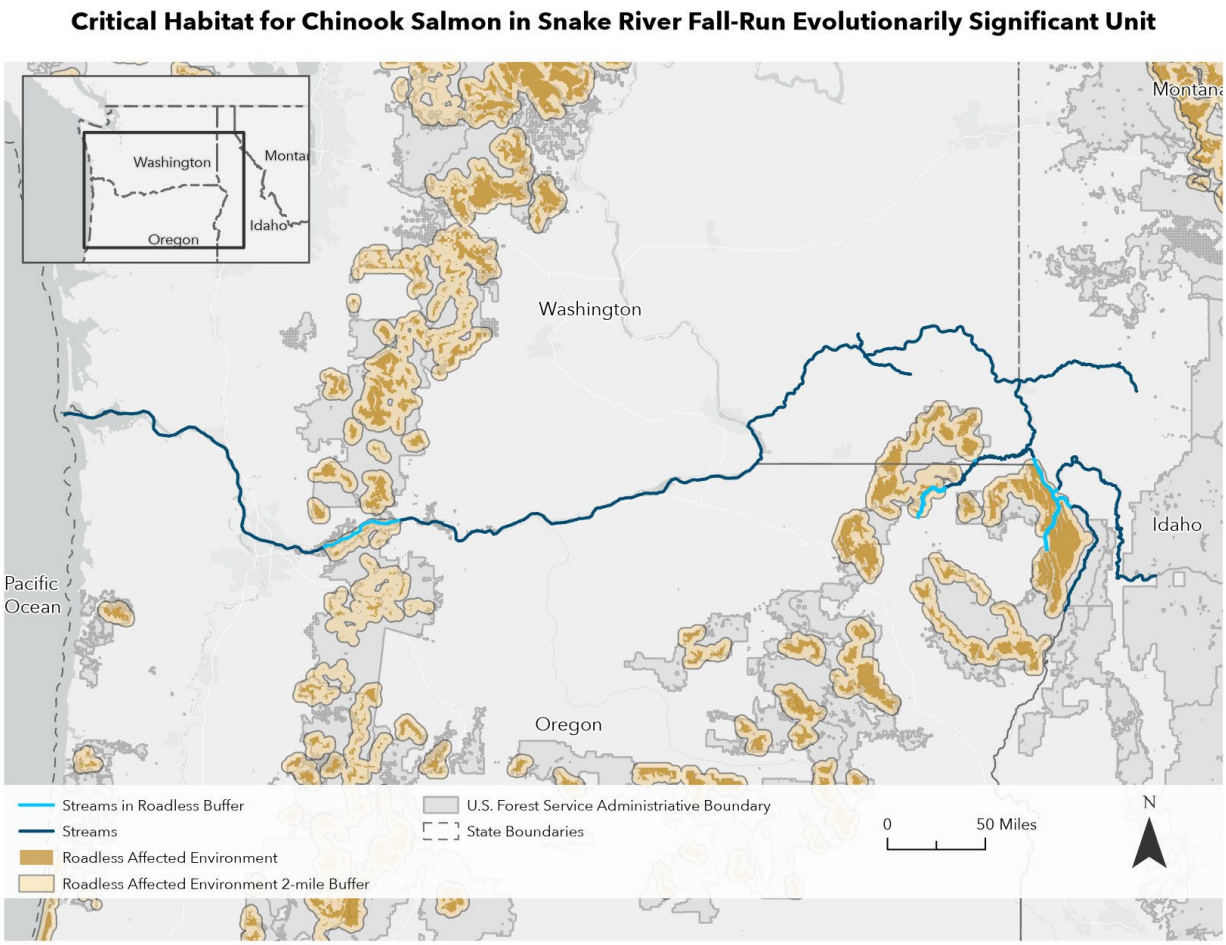


Figure 6. Snake River Chinook salmon fall-run ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

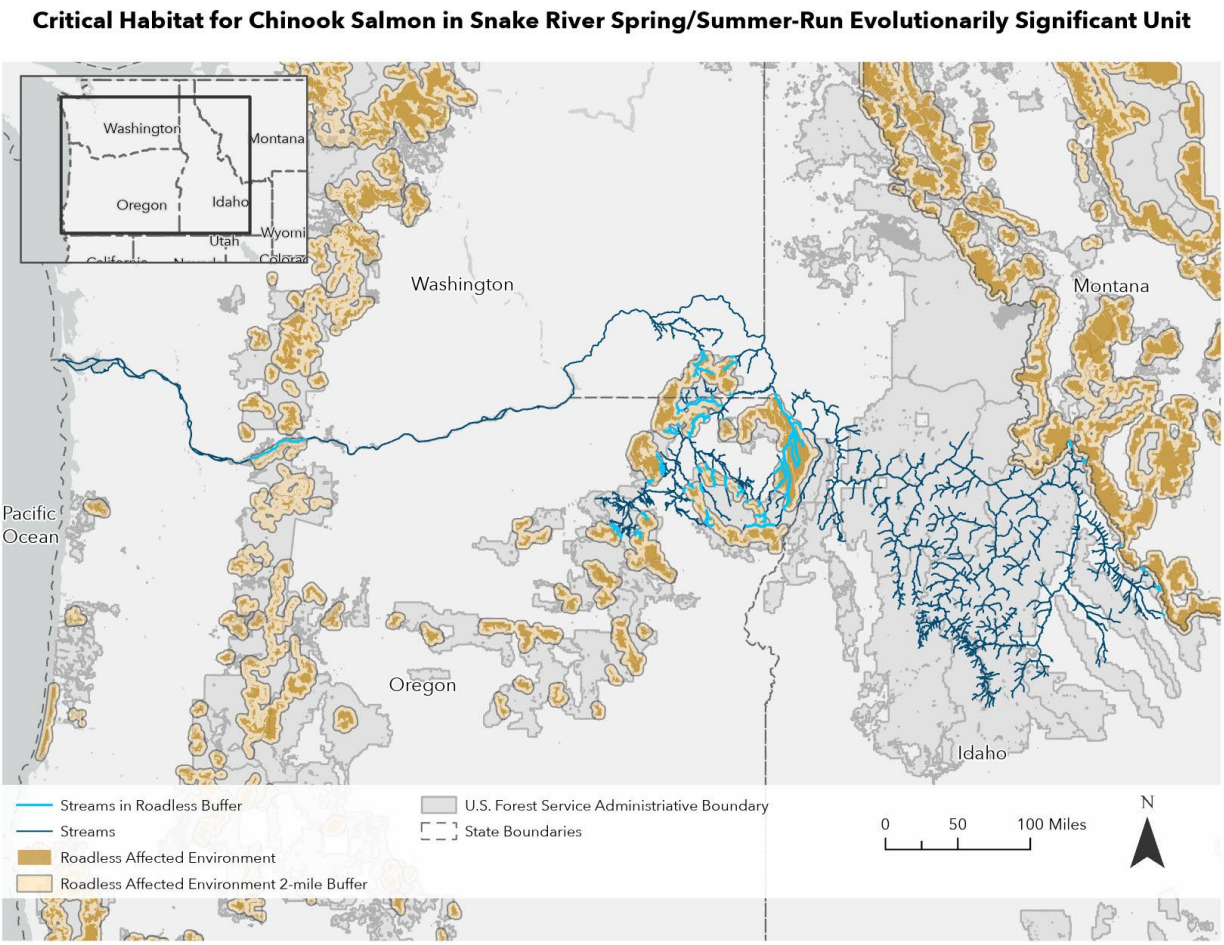


Figure 7. Snake River Chinook salmon spring/summer-run ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

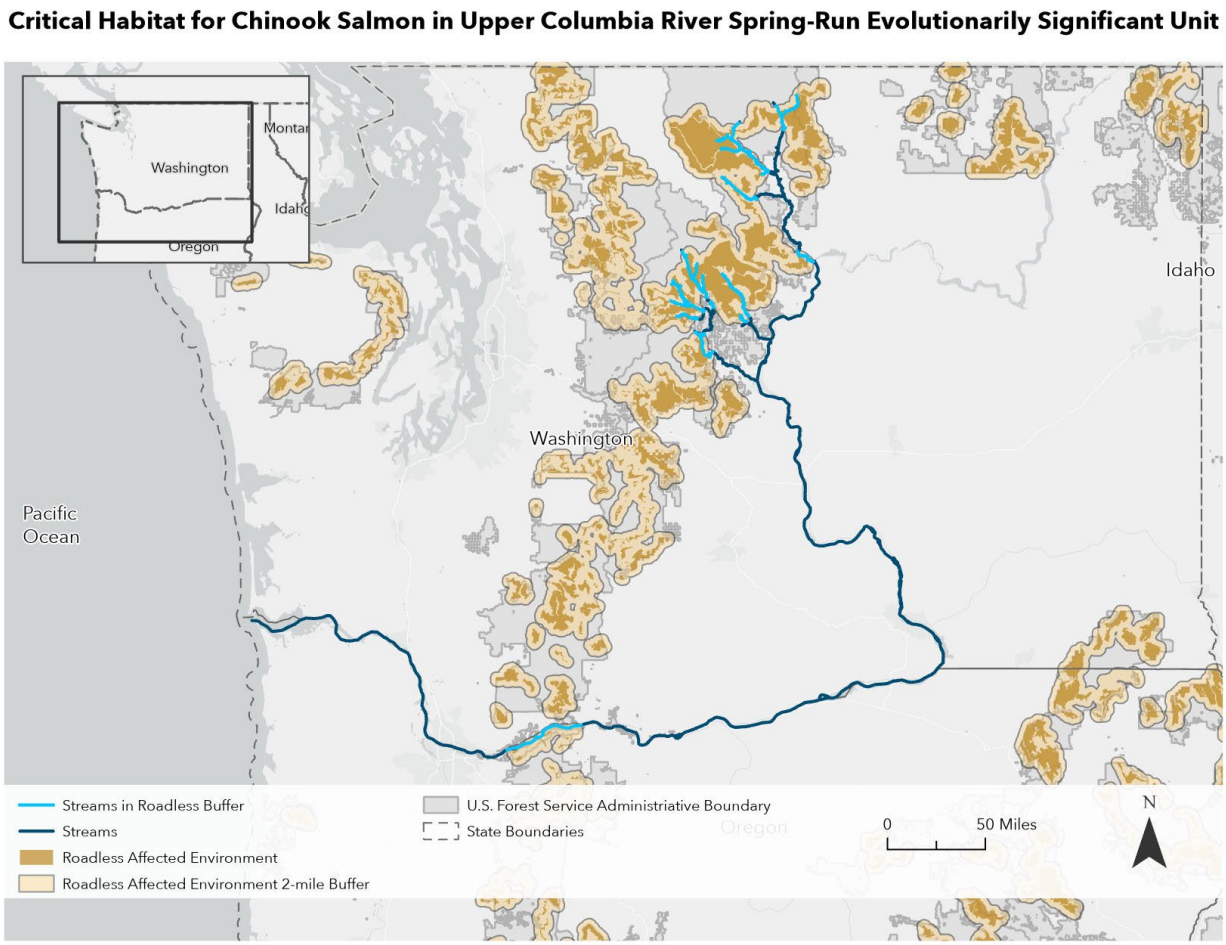


Figure 8. Upper Columbia River Chinook salmon spring/summer-run ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

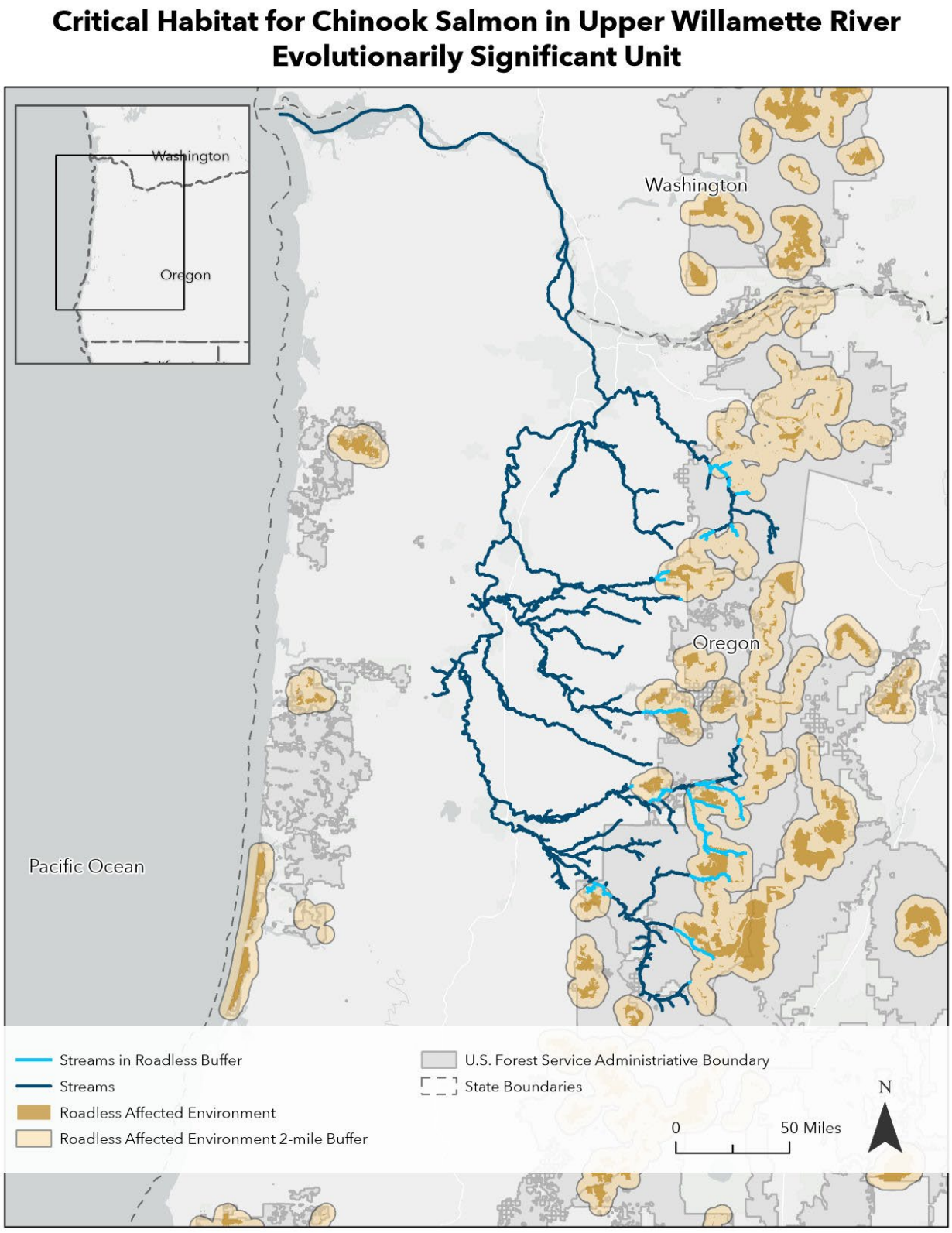


Figure 9. Upper Willamette River Chinook salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

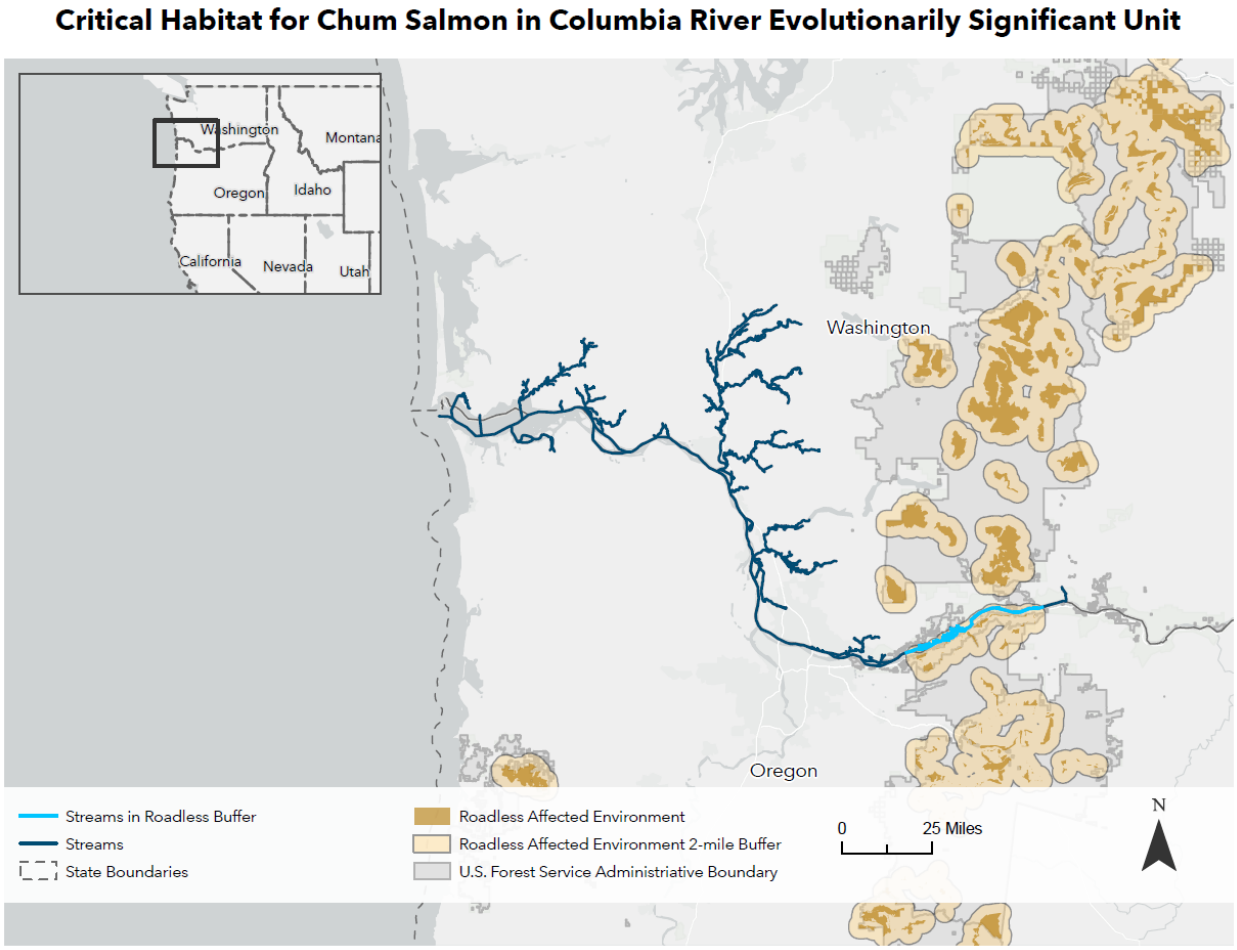


Figure 10. Columbia River chum salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

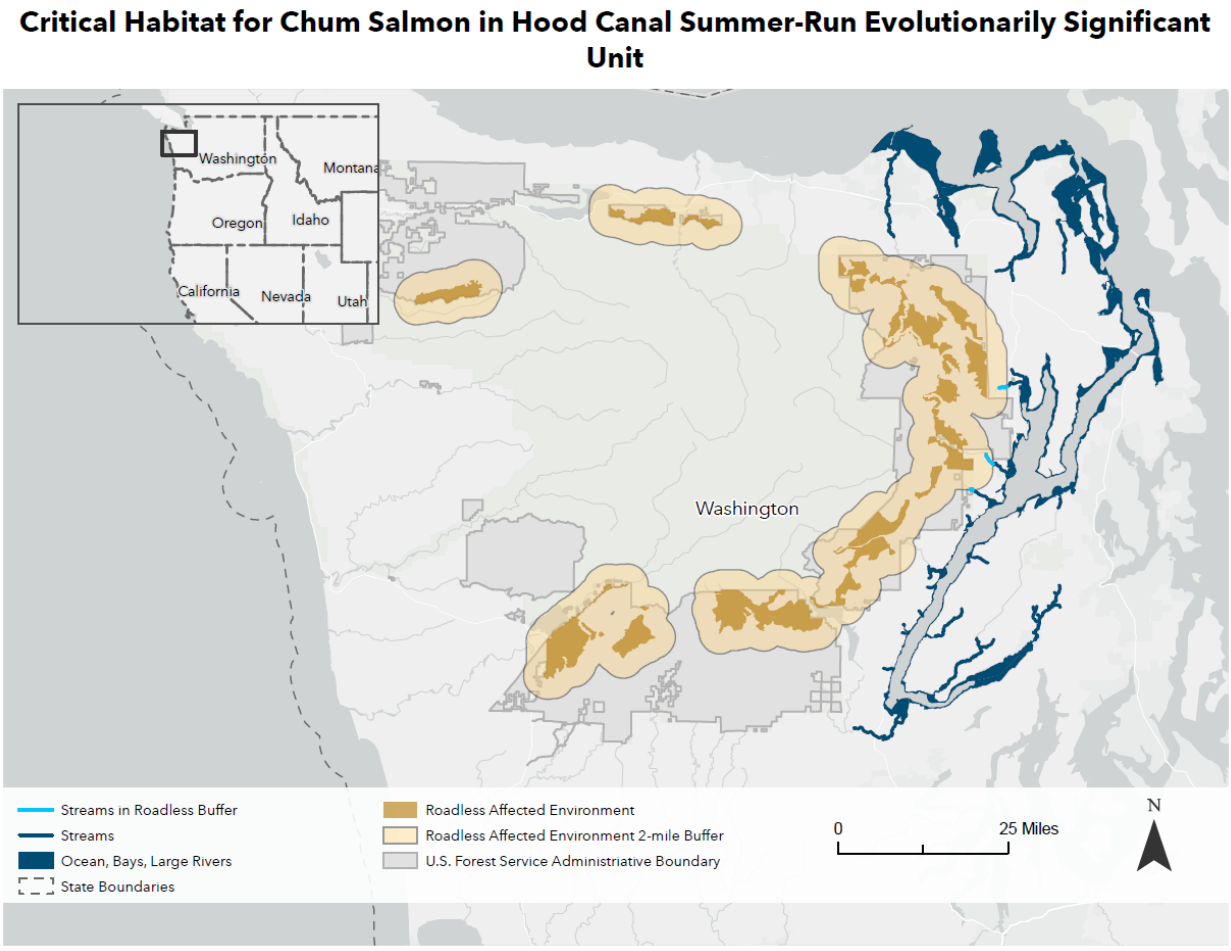


Figure 11. Hood Canal chum salmon summer-run ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

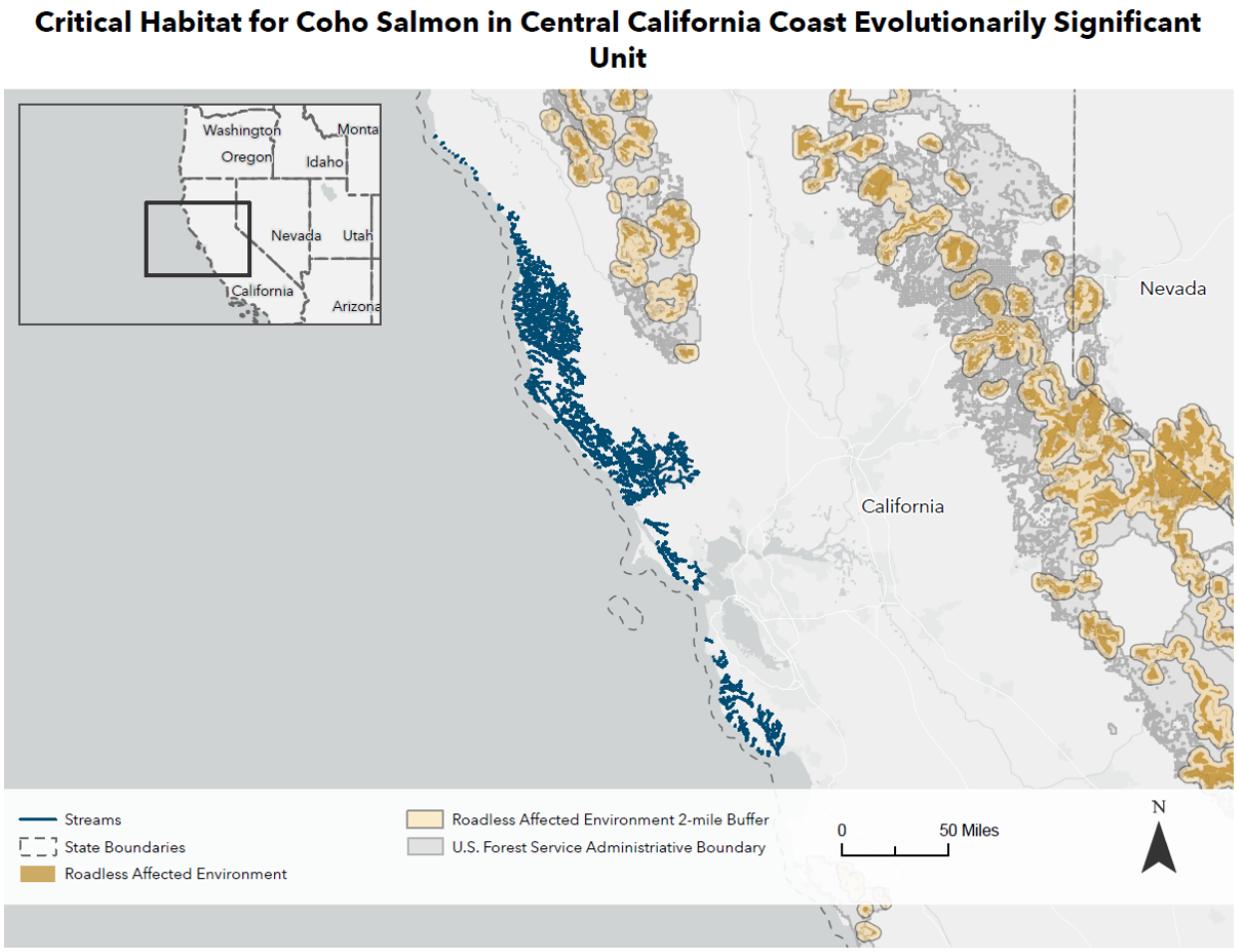


Figure 12. Central California Coast coho salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

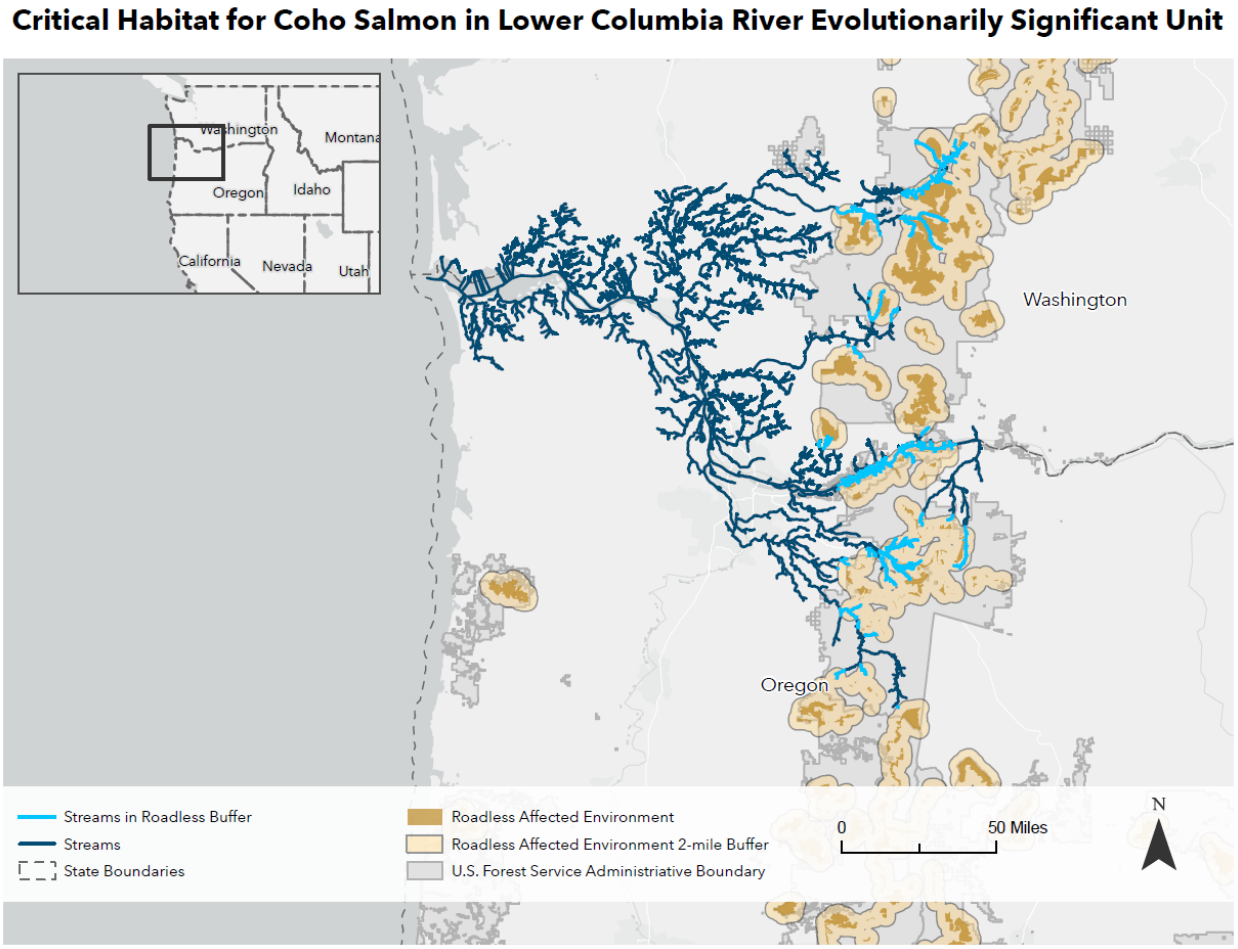


Figure 13. Lower Columbia River coho salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

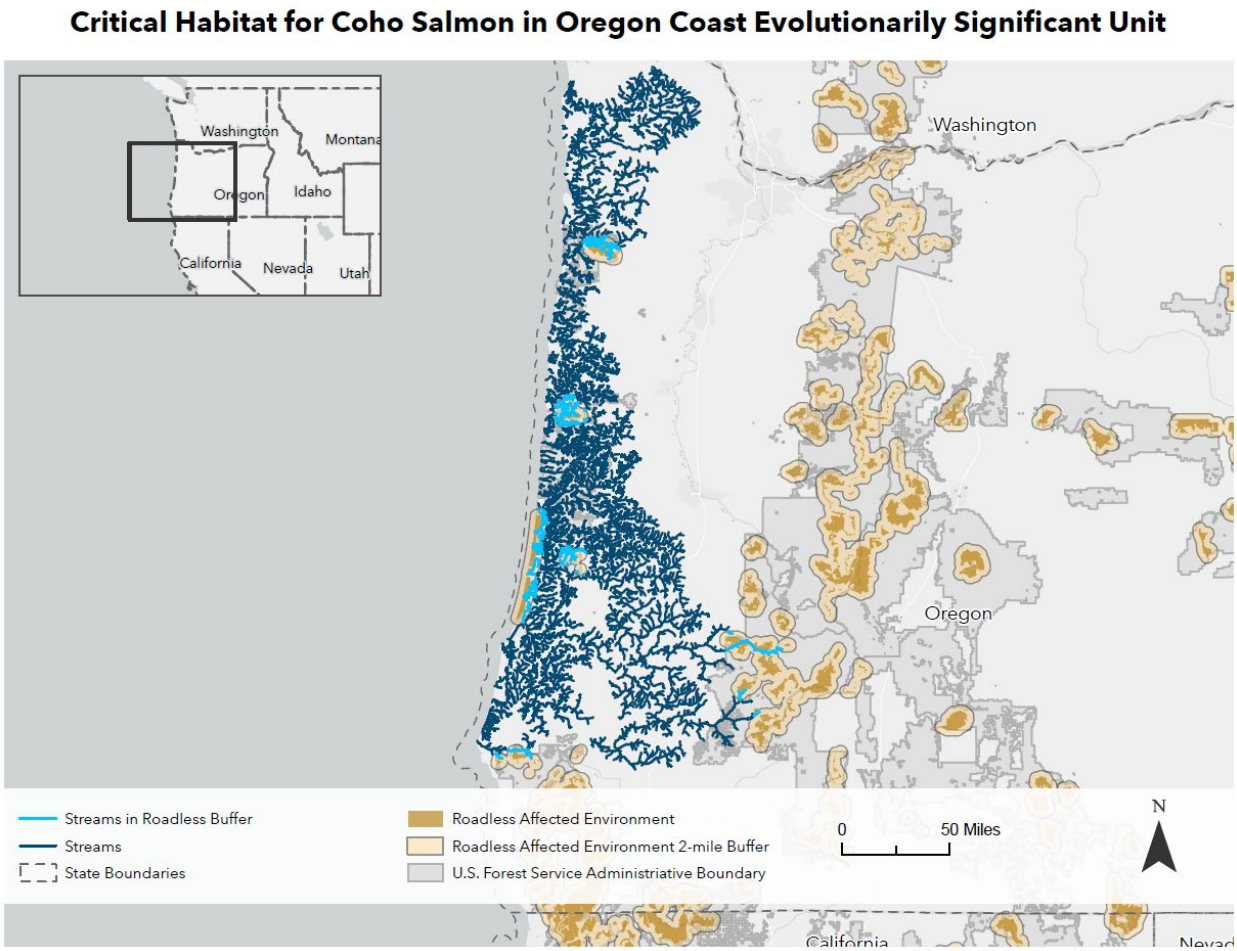


Figure 14. Oregon Coast coho salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

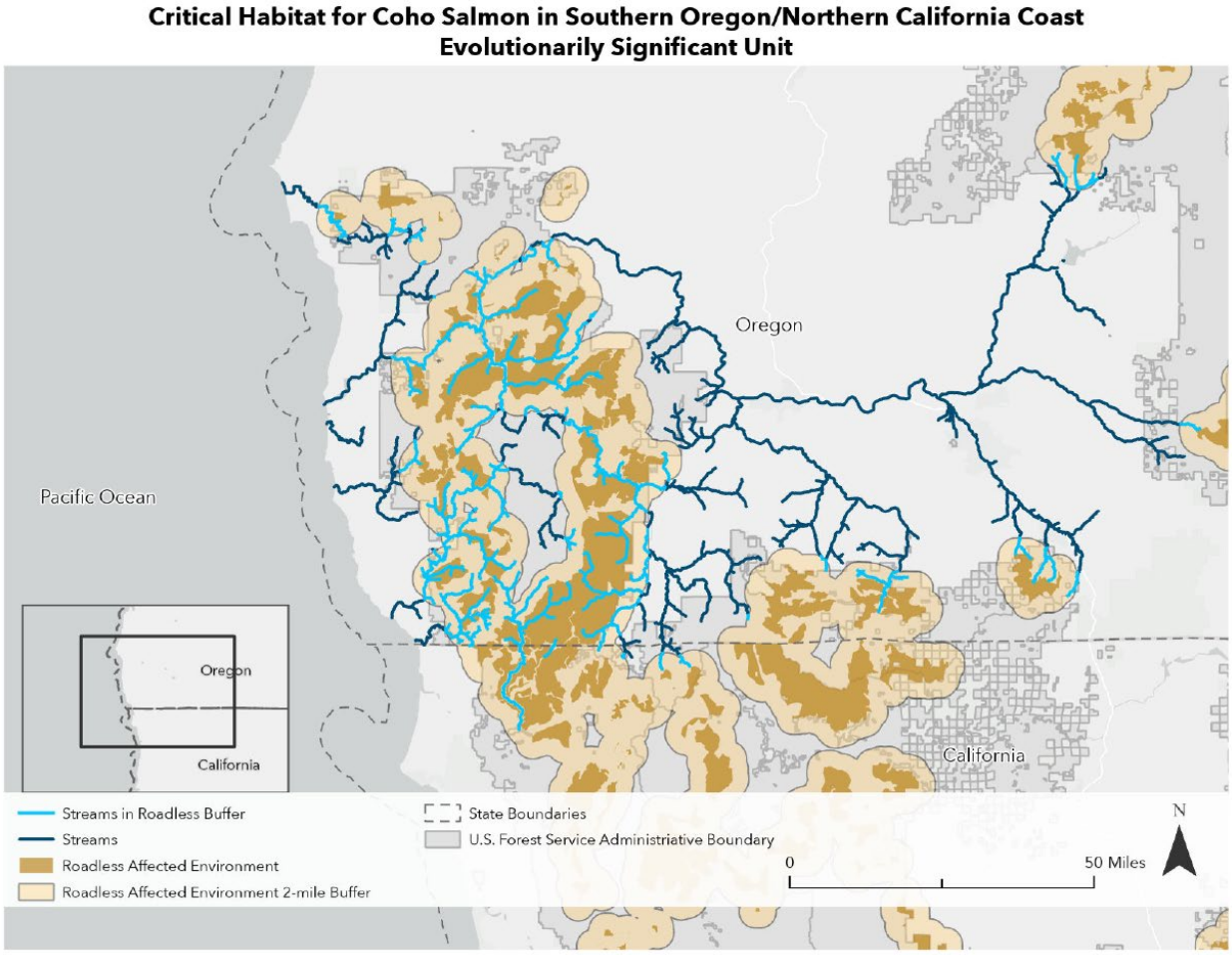


Figure 15. Southern Oregon/Northern California Coast coho salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

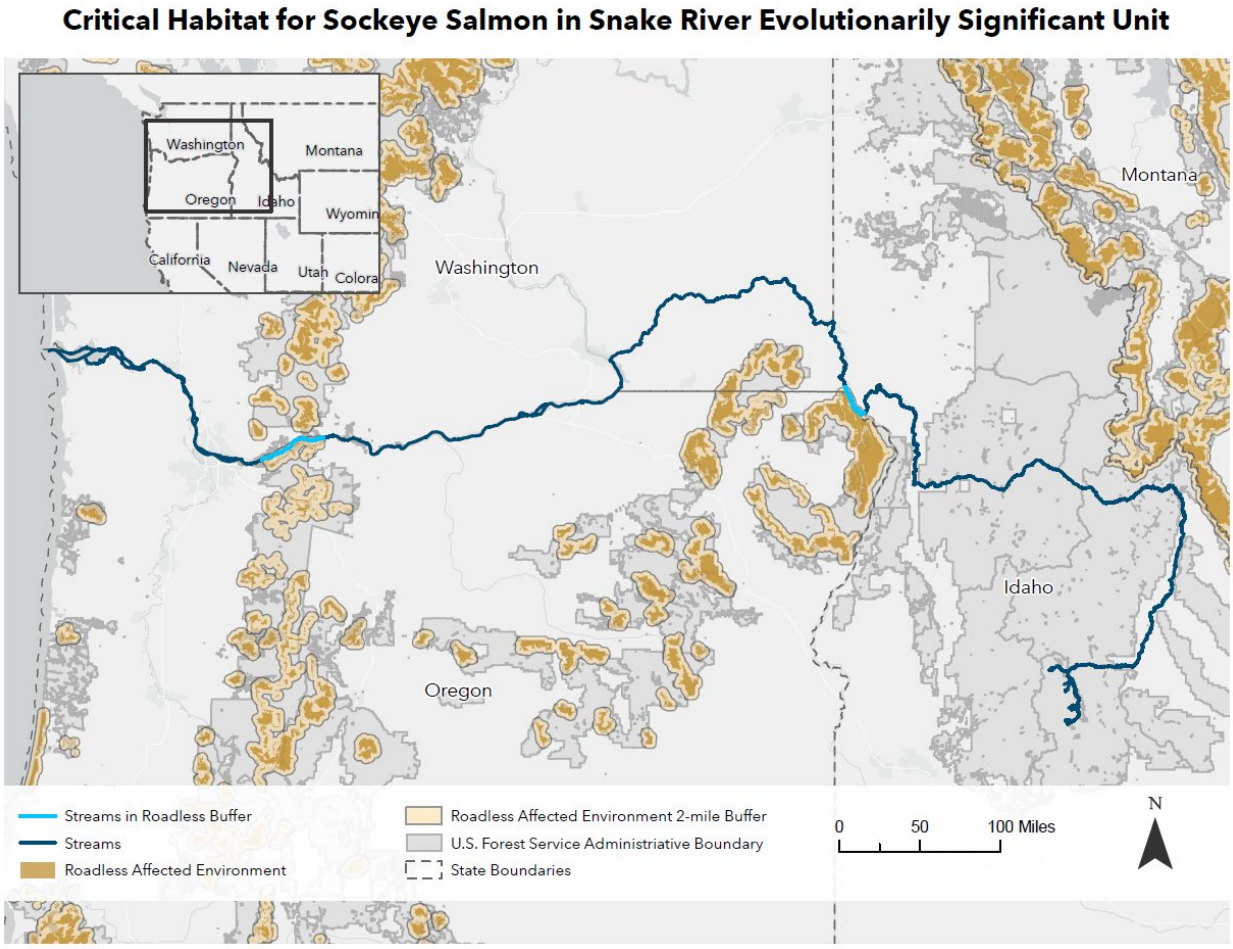


Figure 16. Snake River sockeye salmon ESU critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

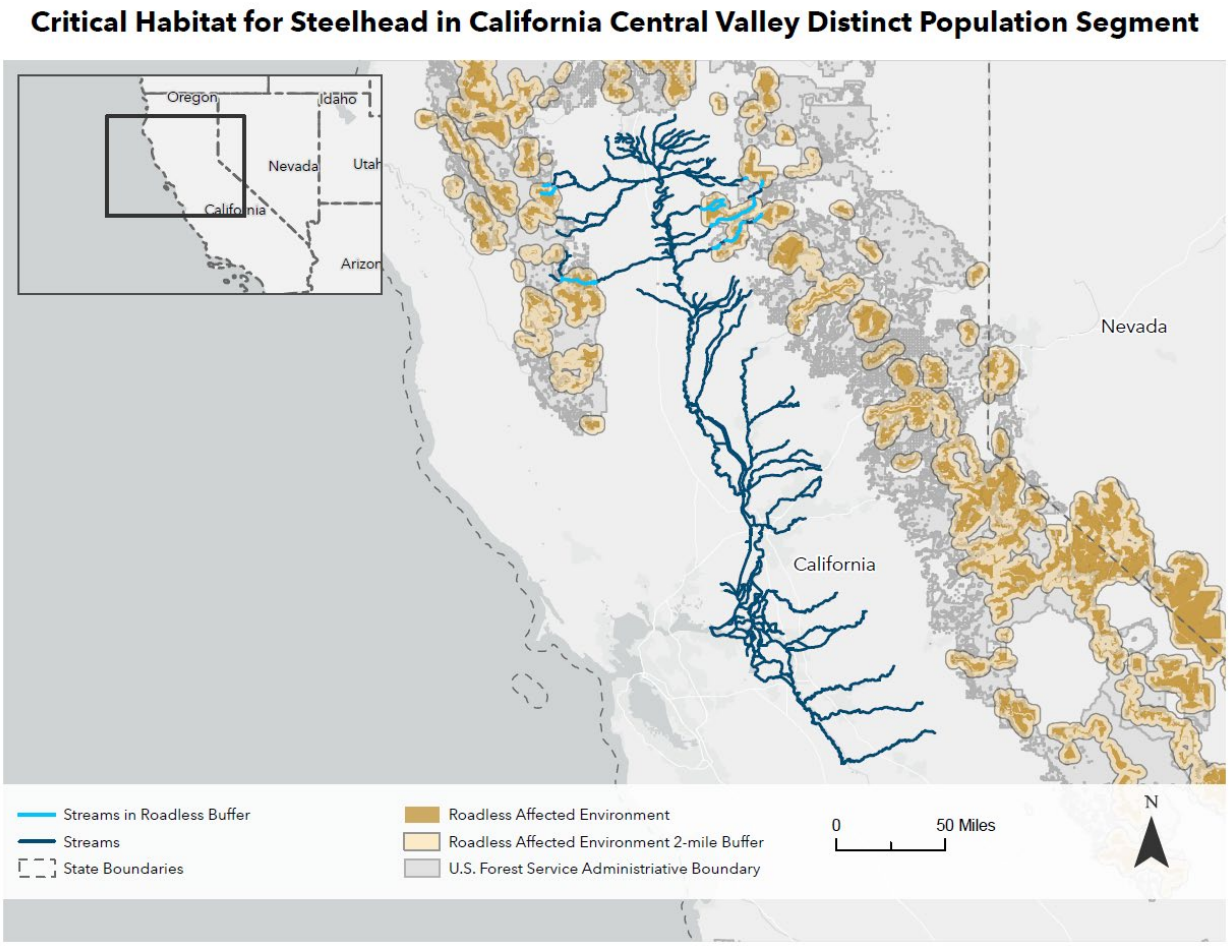


Figure 17. California Central Valley steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

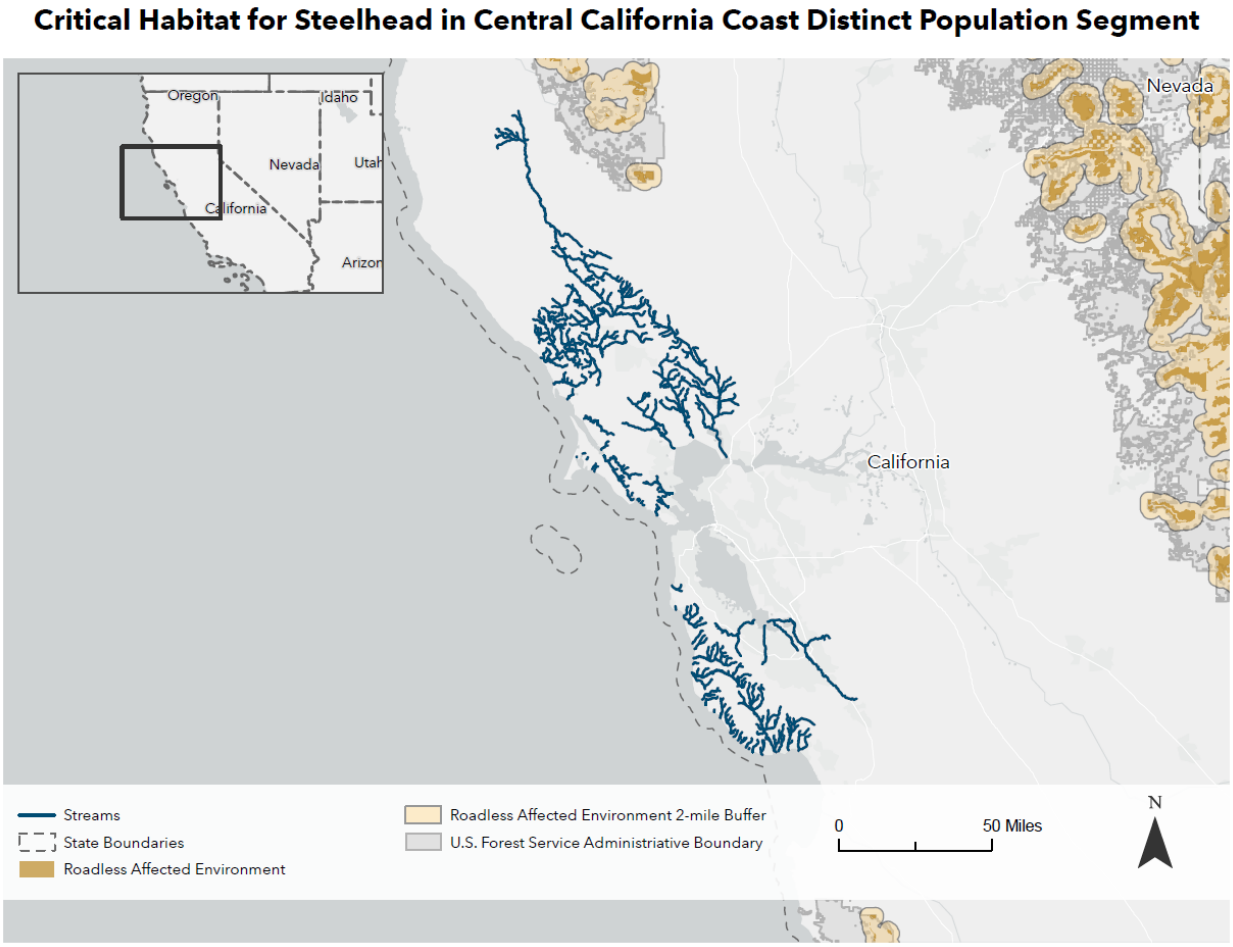


Figure 18. Central California coast steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

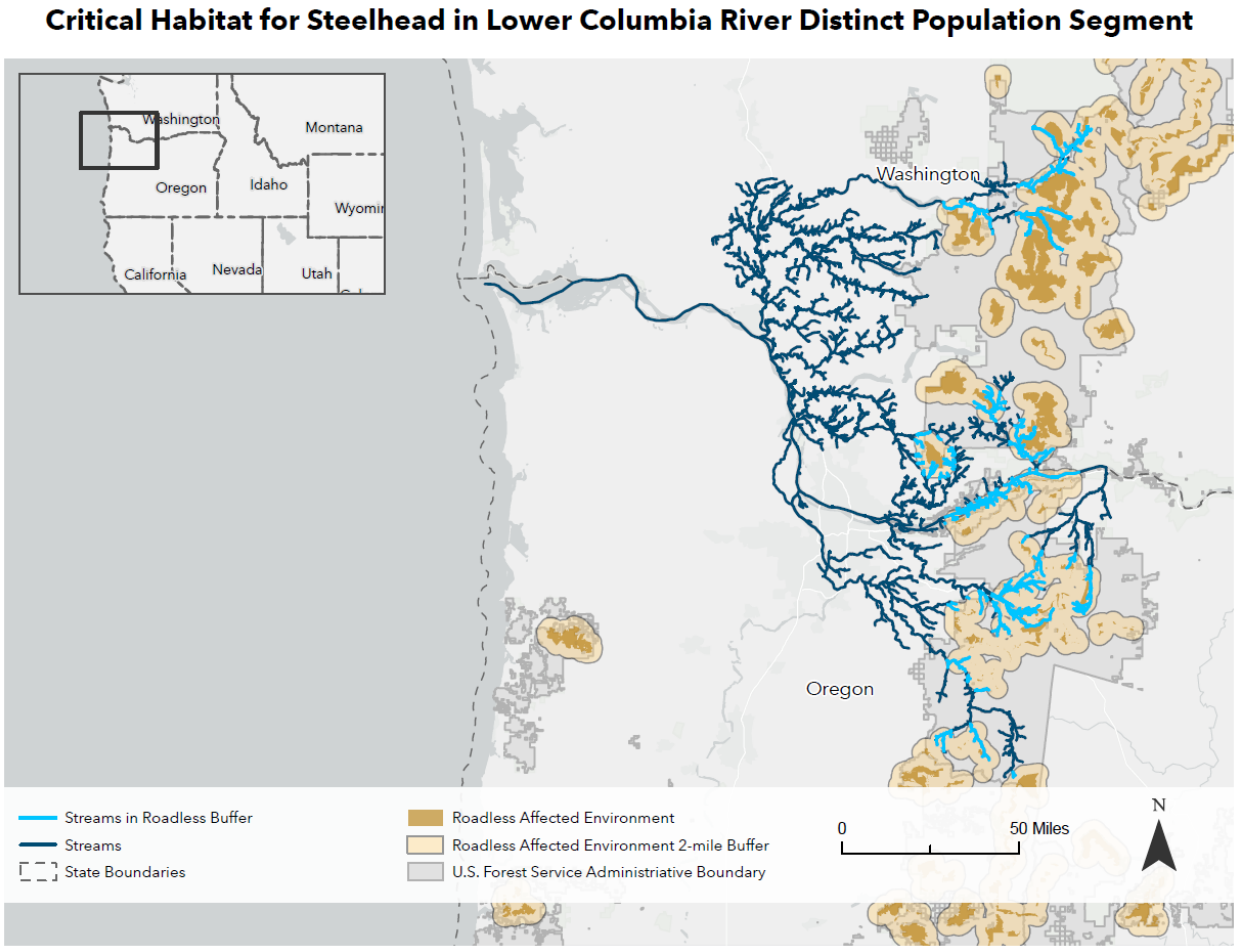


Figure 19. Lower Columbia River steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

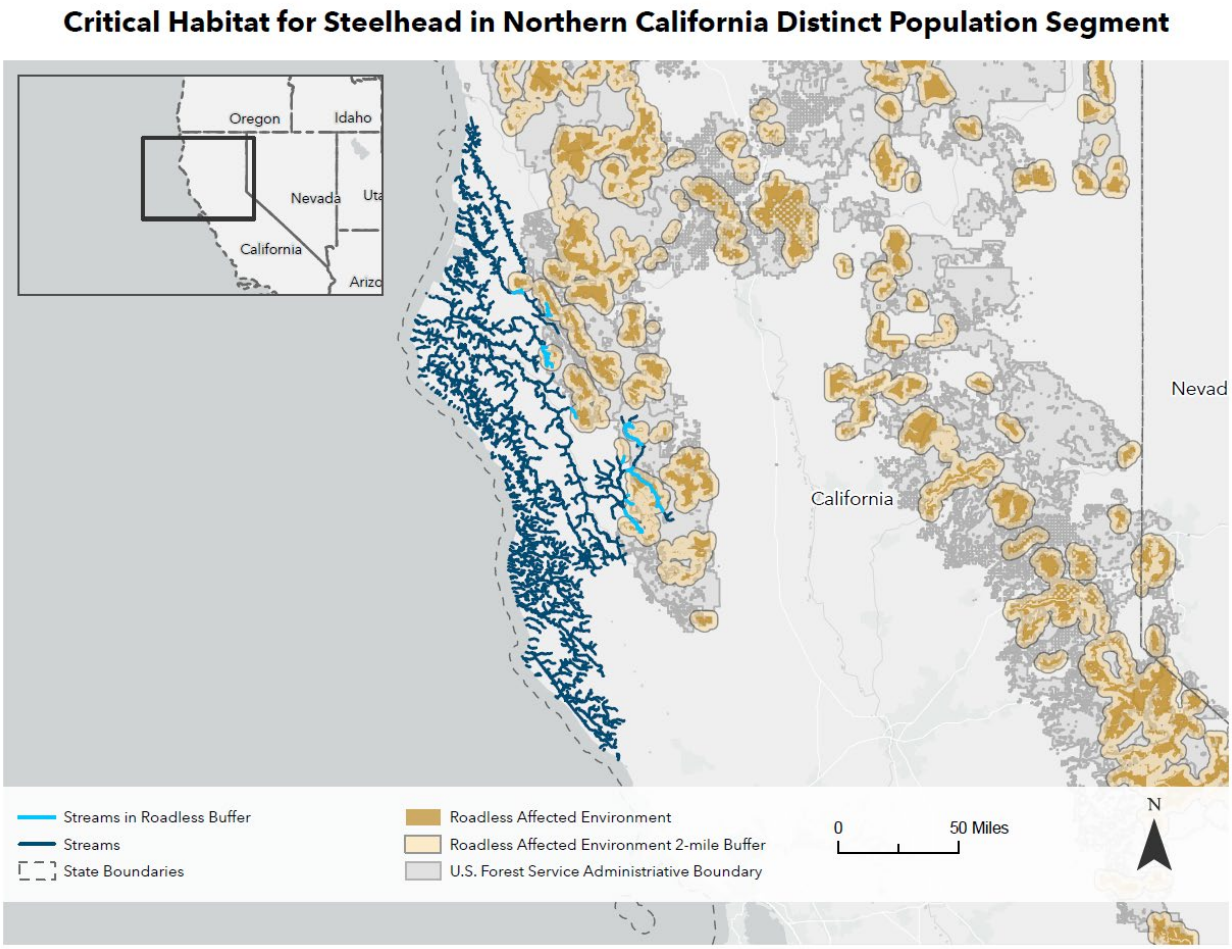


Figure 20. Northern California steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

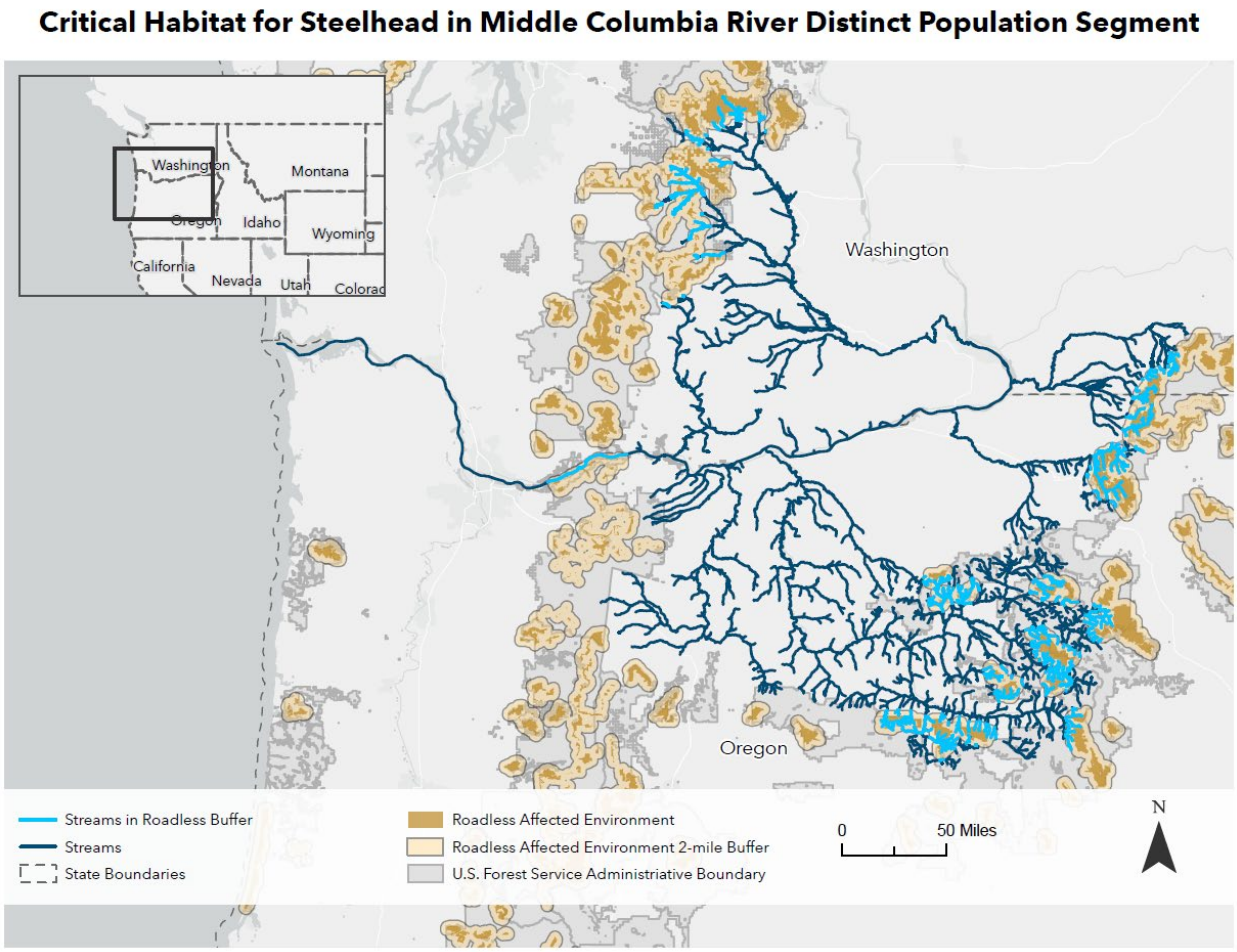


Figure 21.Middle Columbia River steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

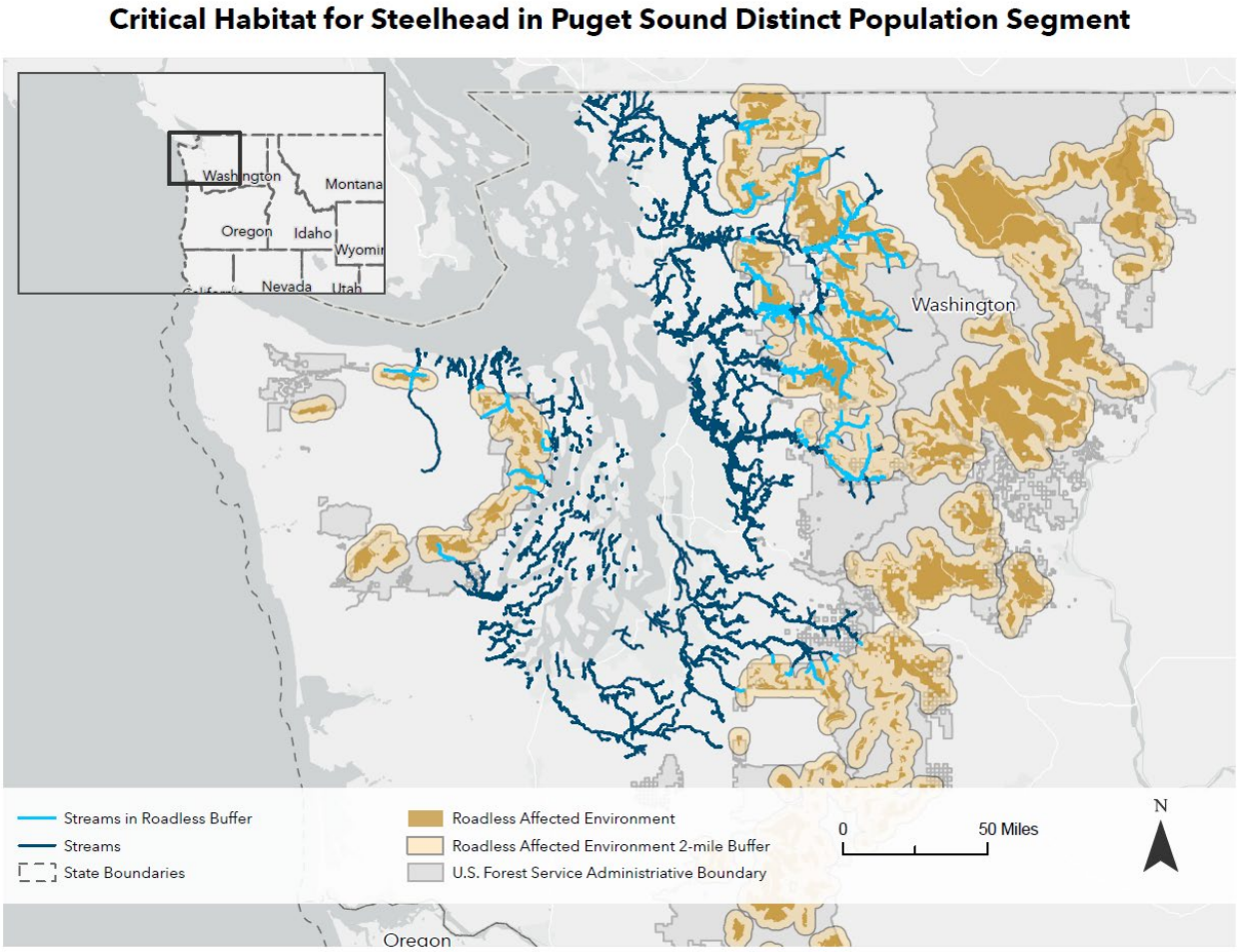


Figure 22. Puget Sound steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

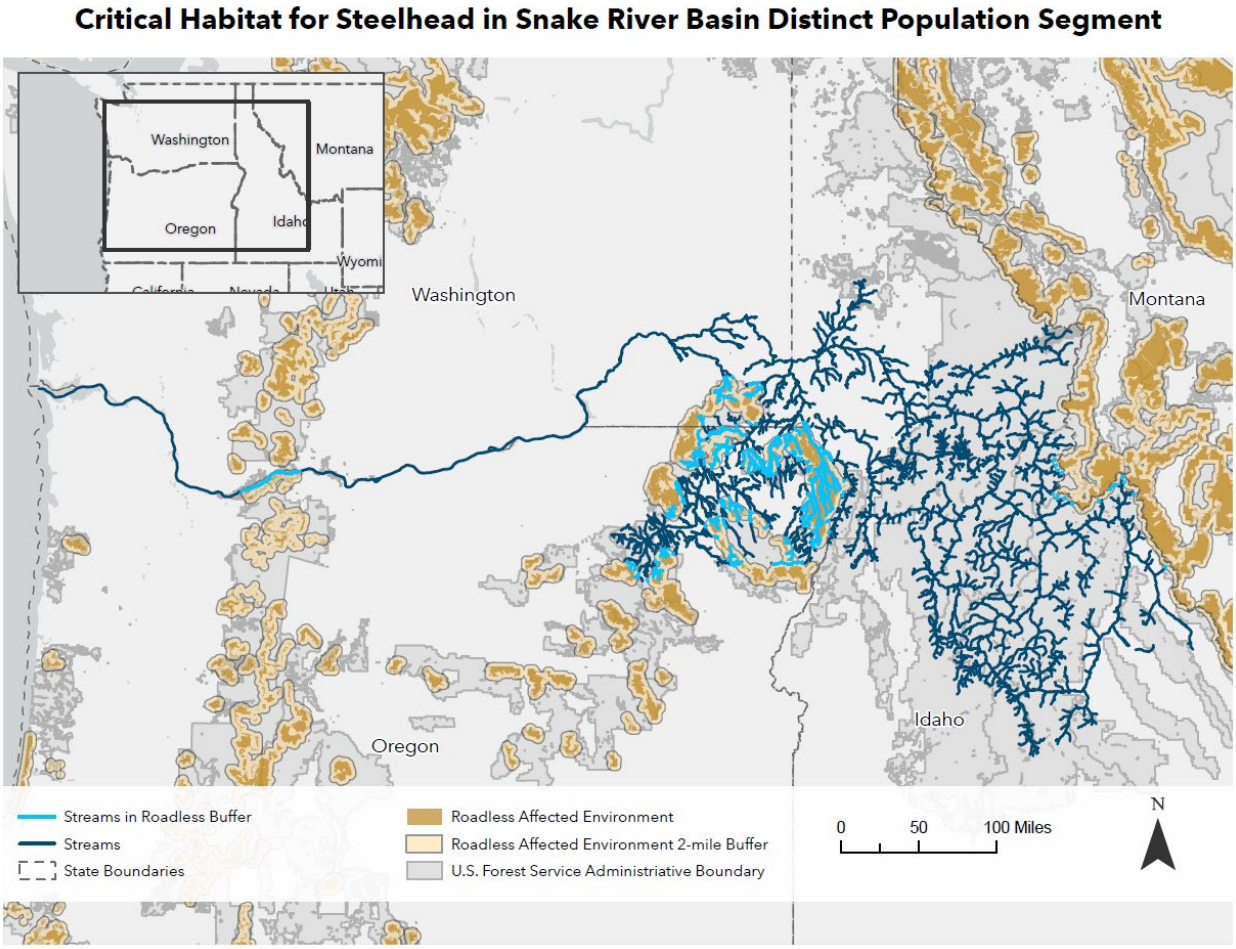


Figure 23. Snake River basin steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

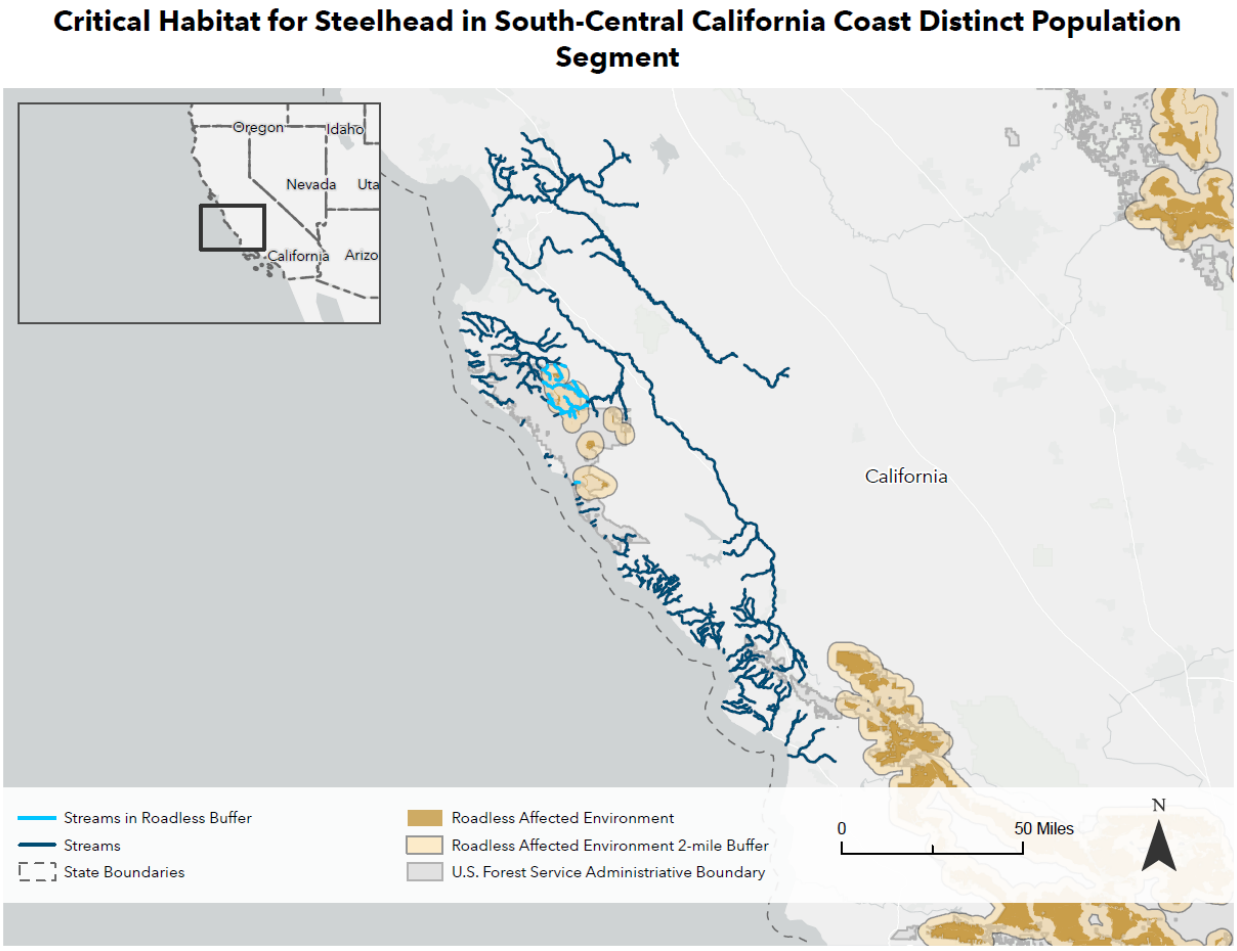


Figure 24. South-Central California coast steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

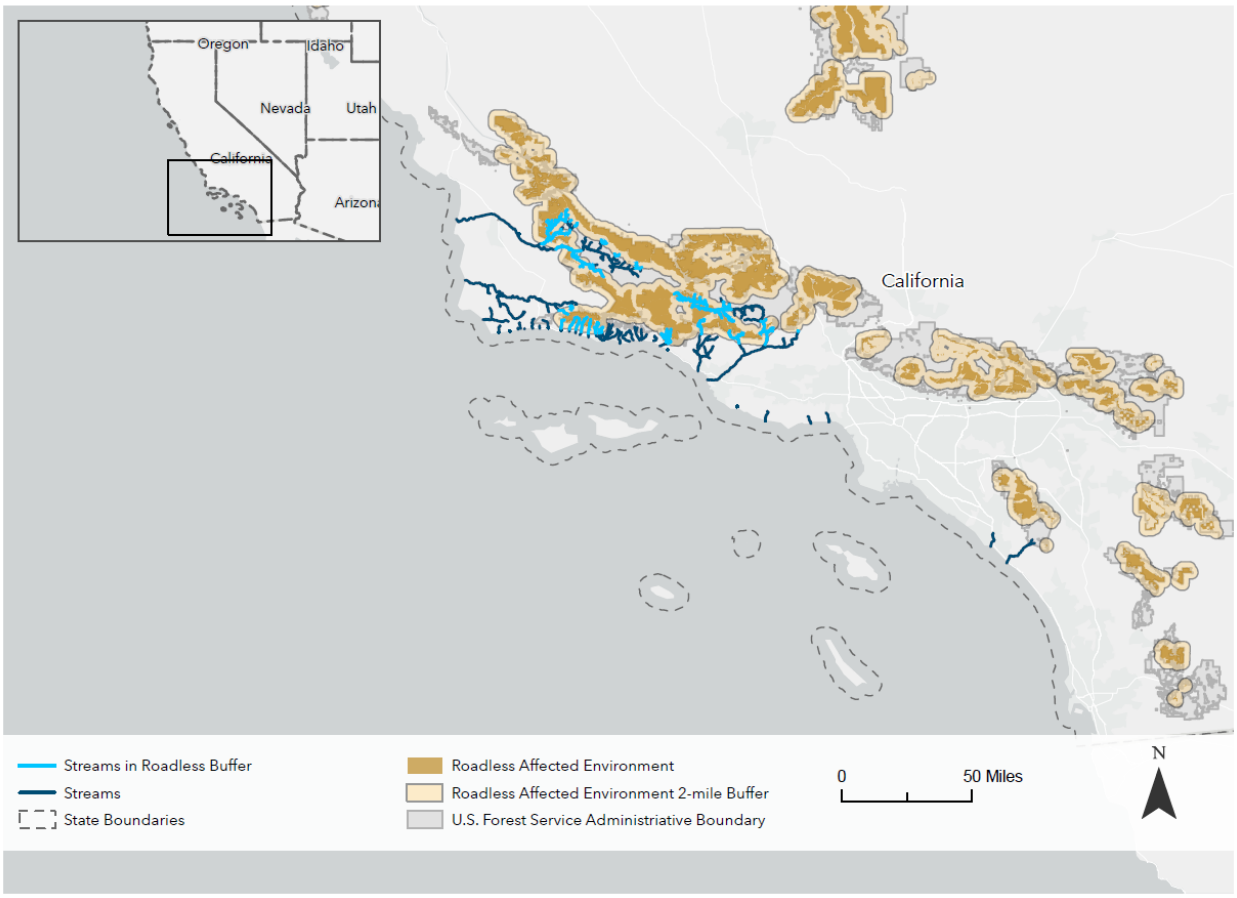


Figure 25. Southern California steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

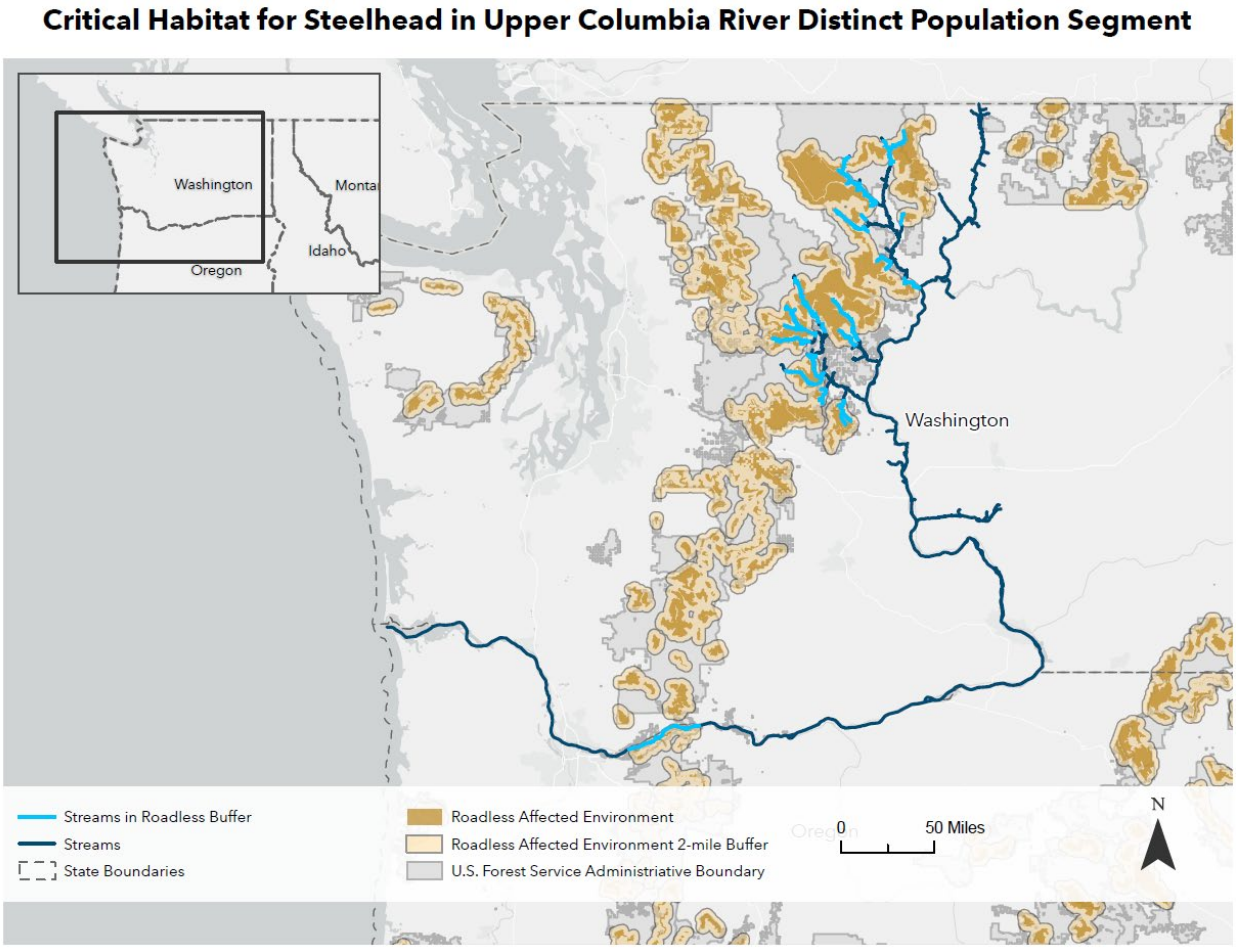


Figure 26. Upper Columbia River steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

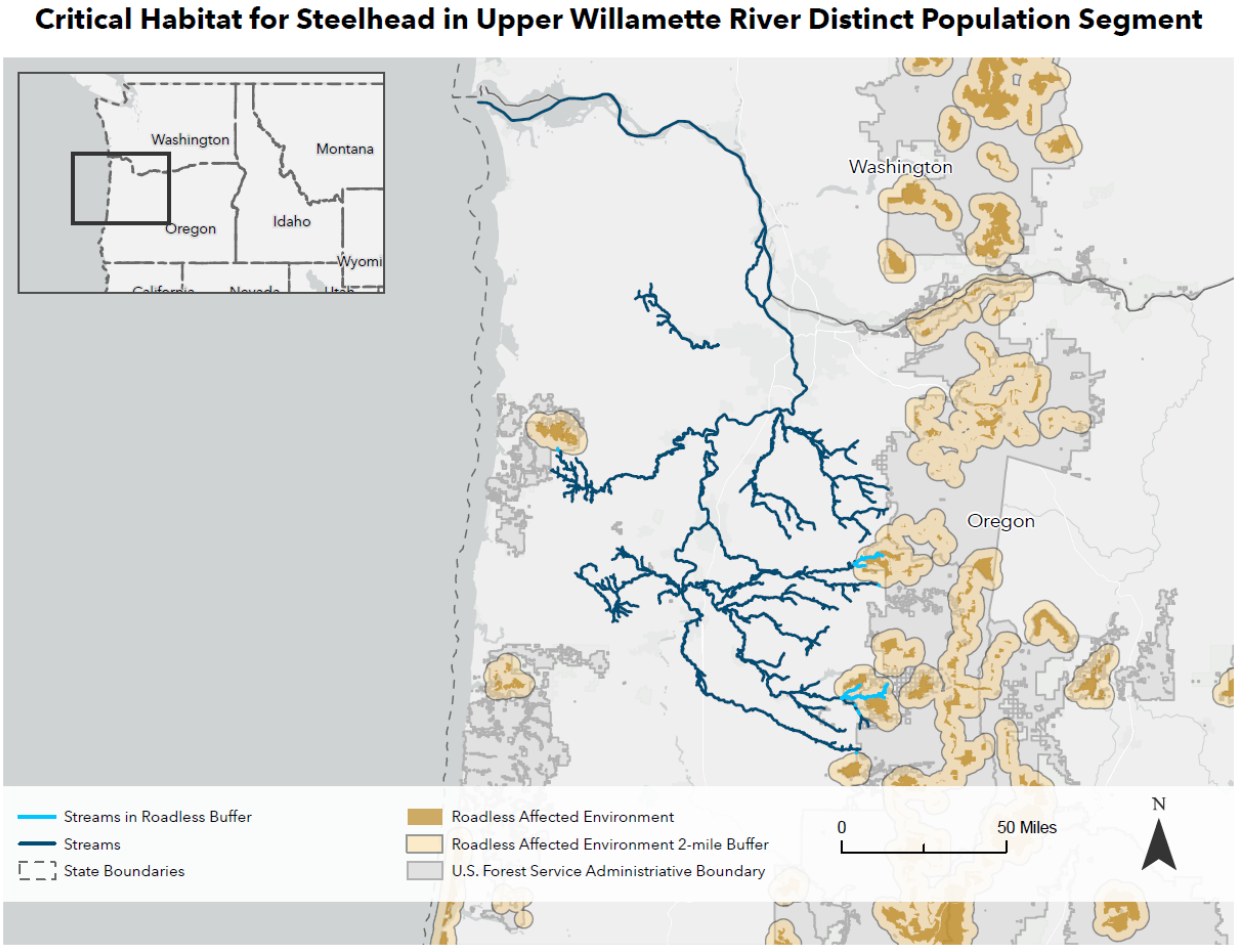


Figure 27. Upper Willamette River steelhead DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

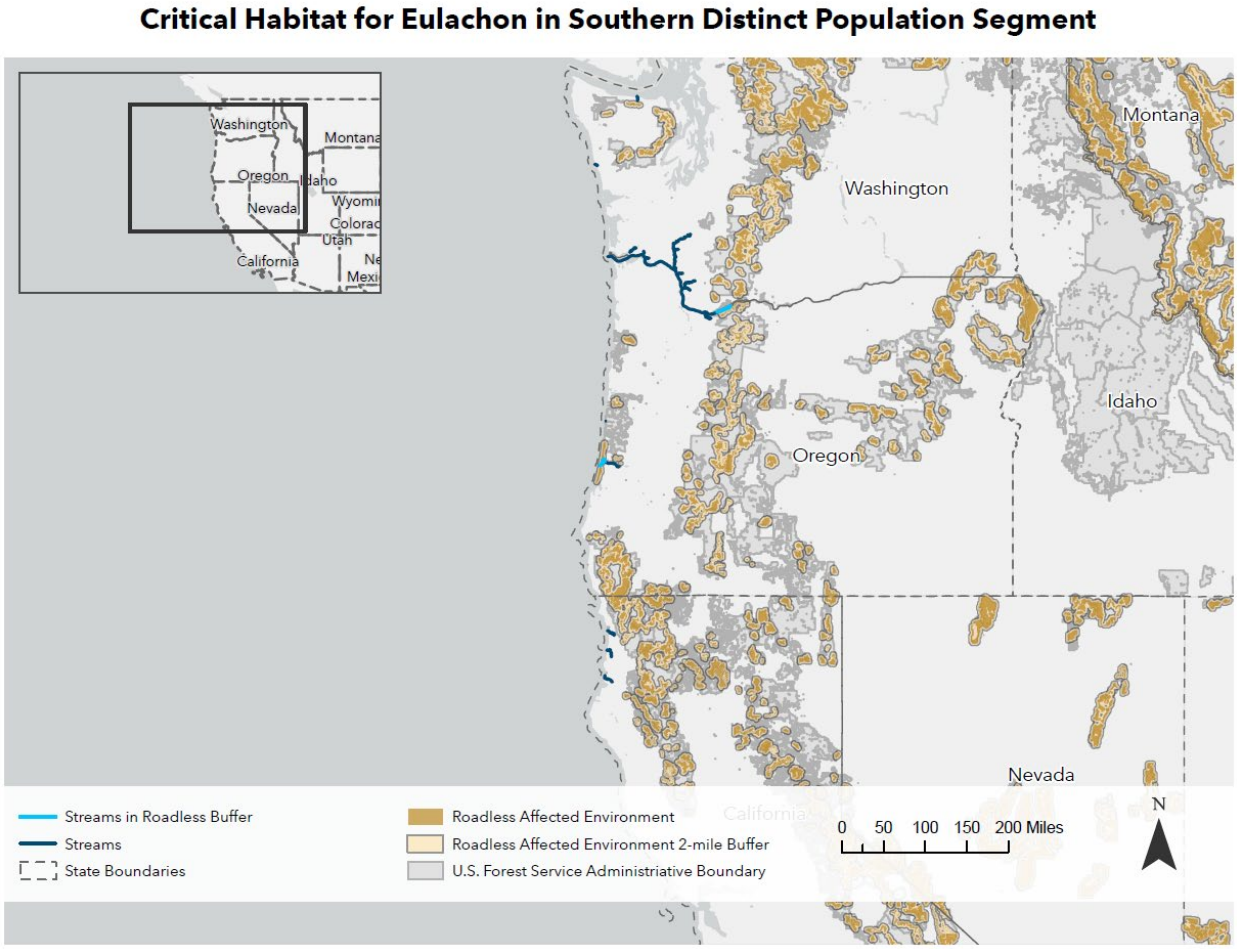


Figure 28. Eulachon southern DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

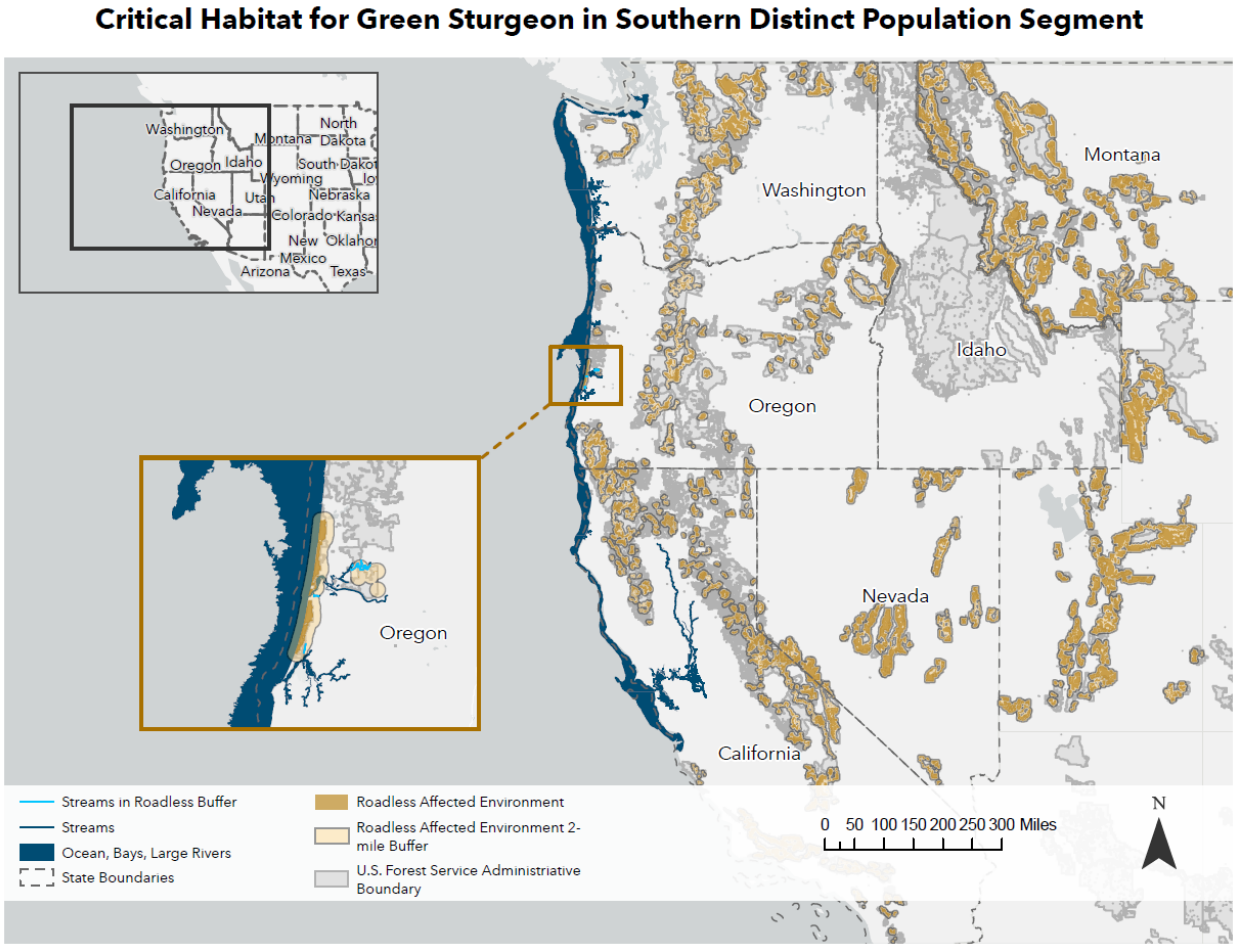


Figure 29. Green sturgeon southern DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

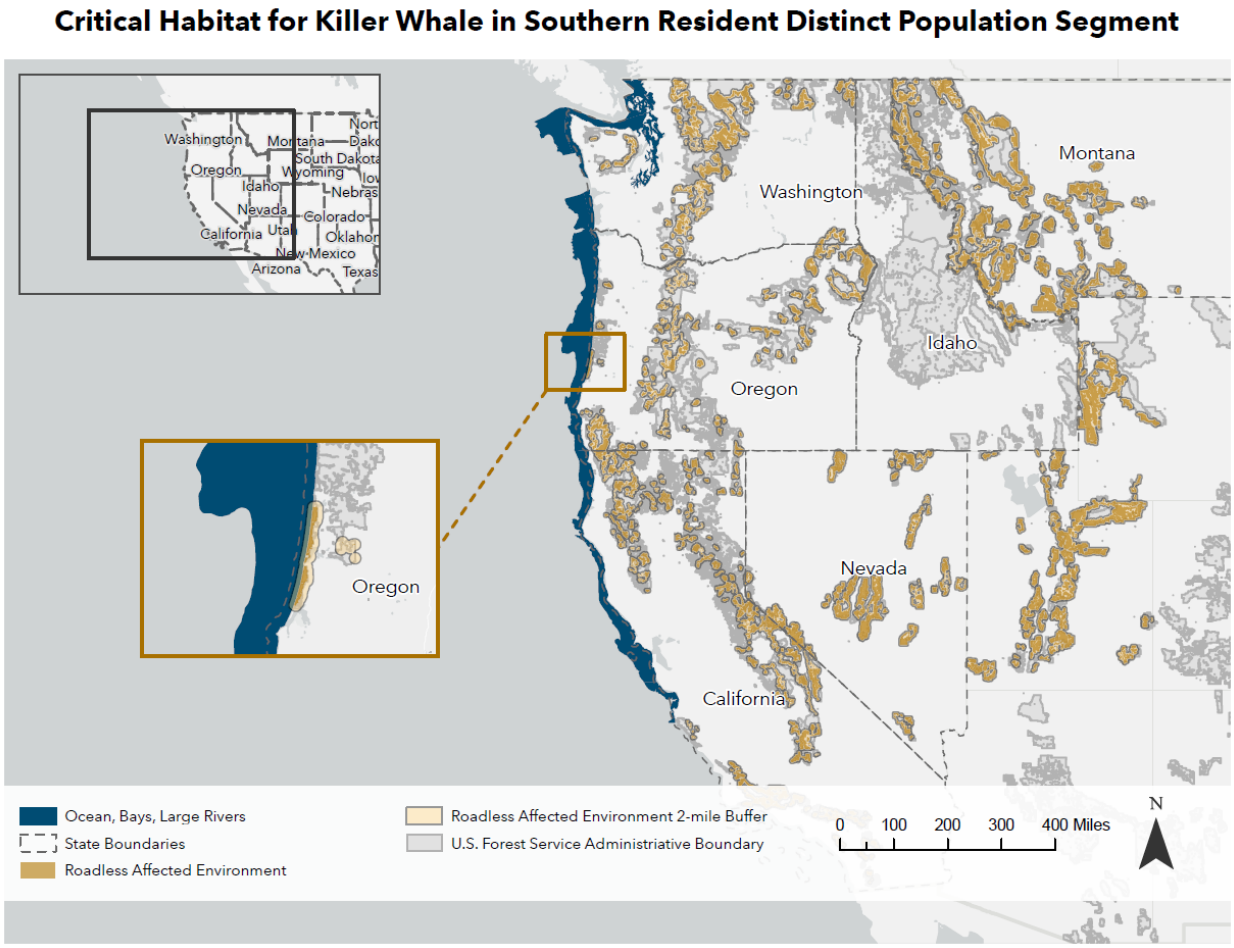


Figure 30. Killer whale southern resident DPS critical habitat compared to Roadless Rule Rescission affected environment with a 2 mile buffer.

Appendix B. Forest Service Road Standards

The Forest Service Standards are described in several different Forest Service publications. Primary references are available in FS directives such as Forest Service Handbook (FSH) 7709.56 Chapter 60 for typical road maintenance, FSH2409.15 Chapter 50 for timber related road maintenance, and FSH7709.56 Chapter 40 for development and construction requirements. In addition, performance of the work is guided by established regional specifications like Performance Based Maintenance Specifications and Forest Service Specifications for Maintenance of Roads in Timber Sales.

Transportation Facilities - Forest Service Handbook 2409.15 Chapter 50 Transportation Facilities. Web Link: https://cms.fs.usda.gov/directives/chapter-50-transportation-facilities

Road Design - Forest Service Handbook 7709.56 - Road Pre-Construction Handbook Chapter 40 – Design Subchapter 44 – Road Drainage. Web Link: http://www.fs.fed.us/im/directives/dughtml/fsh7000.html

Road Maintenance - Forest Service Handbook 7709.59 Chapter 60. Web Link: https://cms.fs.usda.gov/directives/chapter-60-road-maintenance

Guidelines for Road Maintenance Levels. Web Link: https://www.fs.usda.gov/Internet/FSE_DOCUMENTS/stelprd3793545.pdf

Sediment and Chemical Monitoring Protocols - National Best Management Practices for Water Quality Management on National Forest System Lands. Web Link: http://www.fs.fed.us/biology/resources/pubs/watershed/FS_National_Core_BMPs_April2012.pdf

Water/Road Interactions Technology Series Documents (18 total). Web Link: Water-Road Interaction Tech Series Documents (usda.gov) https://www.fs.usda.gov/biology/nsaec/products-publications-streamteamdocuments-writsd.html

Storm Damage Risk Reduction Guide for Low Volume Roads. Web Link: https://fsweb.mtdc.wo.fs.fed.us/pubs/pdfpubs/pdf12771814/pdf12771814dpi100.pdf

Stabilization and Rehabilitation Measures for Low Volume Forest Roads. Web Link: https://fcds.org.co/wp-content/uploads/2021/01/road-stabilization-rehab-measures-keller.pdf

Environmentally Sensitive Road Maintenance Practices for Dirt and Gravel Roads. Web Link: https://fsweb.sdtdc.wo.fs.fed.us/pubs/pdf/11771802.pdf

Culvert Scour Assessment. Web Link: https://fsweb.sdtdc.wo.fs.fed.us/pubs/pdf/35dpi/08771812.pdf

Appendix C. Definitions

Road: A motor vehicle travelway over 50 inches wide, unless designated and managed as a trail. A road may be classified, unclassified, or temporary.

(1) **Classified road:** A road wholly or partially within or adjacent to National Forest System lands that is determined to be needed for long-term motor vehicle access, including State roads, county roads, privately owned roads, National Forest System roads, and other roads authorized by the Forest Service.

(2) **Unclassified road:** A road on National Forest System lands that is not managed as part of the forest transportation system, such as unplanned roads, abandoned travelways, and off-road vehicle tracks that have not been designated and managed as a trail; and those roads that were once under permit or other authorization and were not decommissioned upon the termination of the authorization.

(3) **Temporary road.** A road authorized by contract, permit, lease, other written authorization, or emergency operation, not intended to be part of the forest transportation system and not necessary for long-term resource management.

Road construction: Activity that results in the addition of forest classified or temporary road miles.

Road maintenance: The ongoing upkeep of a road necessary to retain or restore the road to the approved road management objective.

Road reconstruction: Activity that results in improvement or realignment of an existing classified road defined as follows:

(1) **Road improvement:** Activity that results in an increase of an existing road's traffic service level, expansion of its capacity, or a change in its original design function.

(2) **Road realignment:** Activity that results in a new location of an existing road or portions of an existing road, and treatment of the old roadway.